



ADVANCED INTERACTIVE EXECUTIVE

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History of AIX

Aix stands for Advanced Interactive Executive Environment. Aix is one of the unix operating system, it is introduced by the IBM in the year of 1986. AIX OS will install only IBM power systems. It will be provided more security for the customer's data and AIX OS introduced many advanced concepts. Also, AIX OS is more expensive.

AIX Concepts:

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Advanced Concepts:

Mirroring, migrate, DLPAR, etc...

UNIX BASICS

Kernel:

Kernel is Interface between users & Hardware

Shell:

Shell is Interface between users & Kernel, Hardware.

Actually users required information will give on shell then shell talk with kernel then kernel talk with hardware then finally result will have displayed on the shell if users are executed correct information/commands in the shell.

If users are executed wrong commands/information in the shell, the shell will pass the wrong information to kernel the kernel will not understand then immediately kernel will through the error like command not found without talking with hardware.

When users asking information:

USERS --> SHELL --> KERNEL --> HARDWARE

When users are receiving information

USERS <-- SHELL <-- KERNEL <-- HARDWARE

When users asking wrong information:

USERS --> SHELL --> KERNEL

USERS <-- SHELL <-- KERNEL

#bootinfo -K -> To list the current running kernel bit

#bootinfo -y -> To list the MAX kernel bit level support

Note:

1) In aix we are having two type of kernel bits

a) 32 bit

b) 64 bit

2) In Aix 5.3 version we can configure 32 bit or 64 bit.

3) In 6.1, 7.1, 7.2 versions by default having 64 bit only.

To change the kernel bit from 32 (mp) bit to 64 bit:

SYN:

#ln -sf <sourcefile><DestinationFile>

ln -sf /usr/lib/boot/unix_64 /unix

ln -sf /usr/lib/boot/unix_64 /usr/lib/boot/unix

bosboot -ad /dev/ipldevice (rootvg active disk) --> Create Boot image.

shutdown -r

To change the kernel bit from 64 bit to 32 (mp) bit:

SYN:

```
#ln -sf <sourcefile><DestinationFile>
# ln -sf /usr/lib/boot/unix_mp /unix
# ln -sf /usr/lib/boot/unix_mp /usr/lib/boot/unix
# bosboot -ad /dev/ipldevice (rootvg active disk) --> Create Boot image.
# shutdown -r
```

To Login into the Unix Servers:

Two ways to login into the system.

- 1) Console login ---> Direct login
- 2) Remote login ---> Remotely

1) Console login steps:

- a) go to the front of the system
- b) Press the power button to power on the system
- c) provide user name & password

Note: For Console login network is not required, In console login only one person can login into the system at a time.

192.168.1.200

2) Remote login:

in the remote login we can connect the servers through network, from the laptops, or desktops via putty software.

Remote login steps:

Open putty --> provide the IP/hostname --> Select telnet or SSH ---> Click on open button --> Add user name & password.

Note: Remote login can access multi users at a time.

Unix Basics:

Unix Basics are common on all Unix Operating Systems (Aix, Linux, Solaris and HP-UX)

- 1) File Management
- 2) Directory Management

File Management:

#touch -> By using touch command, we can create empty file or files.

Syn: #touch <Filename> --> to create empty file

Ex: #touch aix --> To create aix file.

#touch <file1><file2><file3> --> To create multiple files at a time.

#vi --> vi stands for visual editor, by using vi we can add/remove/modify the text into the files.

Note: By using vi we can also create file & add text at a time.

Syn: #vi <filename>

Ex: #vi aix

we are learning Aix operating system

Esc +:wq --> To save the text

Esc +:wq! --> To save the text with force option

Esc +:q --> without saving text exit from VI editor

Esc +:q! --> without saving text exit from VI editor

Esc +:x --> To save & exit from VI editor

#cat --> By using cat command we display the content of the file.

SYN: #cat <filename>

EX: #cat aix

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Note:

1) By using cat command we can also append the text into the files.

2) By using cat command we can also create file & add text at a time.

#pg --> pg command equal to cat command, By using pg command we can display the content of file.

#ls -l <filename> -> To list the file

#touch india

#ls --> To list files & directories

aix india

#cp --> By using cp command we can copy the files/directories.

#cp <source file><destinationfile>

#cat aix

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#touch unix

```
#cp aix unix
```

```
#cat aix  
we are learning Aix operating system  
#cat unix  
we are learning Aix operating system
```

Note: cp command also will create destination file, if does not exists.

```
#cp aix aix1
```

```
#cat aix1  
we are learning Aix operating system
```

```
/home/thiru/aix/basics.txt  
/home/firoz/
```

```
cp /home/thiru/aix/basics.txt /home/firoz/  
cp <source file><destination path>
```

#ls -l --> To list file & directories with below details.
Permissions, owner, group name, Data & time of creation file etc..

```
#mv --> By using mv command we can rename the files & directories.  
#mv <oldname><newname>  
#mv <aix><linux> --To rename aix file name with Linux name  
#cat linux  
we are learning aix operating system.
```

```
#rm --> By using rm command we can remove the files & Directories.  
#rm <file name>  
#rm linux --> To remove the linux file
```

2) Directory Management:

By using directories, we can maintain the data in structure level. suppose if we need to separate the data by category level (Education, Entertainment, Music etc) we can create respected directories and store the data on them. In windows directories we can call folders.

```
#mkdir --> To create directory we can use mkdir command.
```

```
SYN: #mkdir <dir name>  
EX : #mkdir cricket
```

```
#cd --> Change directory, by using cd command we can enter into the  
directory or we can exit from the directory.
```

```
#cd <dir name> -> To enter into the directory
#cd cricket
#pwd --> To display the present working directory.
O/P: cricket
#cd .. -> To exit from the current directory
#touch sachin dhoni sehwag dravid kohli
#ls
#pwd
```

```
#cd cricket
#ls
sachi dhoni sehwag dravid etc
#mkdir india
#cd india
#pwd
/cricket/india/
```

```
#cd .. -> To exit from the current directory.
#cd -> To exit from the all directories at a time.
```

```
#pwd
/cricket
```

```
cd /cricket/india/ap/
```

```
pwd
/cricket/india/ap/
#cd --> To exit from the all directories at a time.
#rmdir --> To remove the empty directories only.
#rmdir <dirname>
#rmdir india -> To remove the india directory
#rm -rf --> To remove the directory with sub directories/inside files
```

SYN: #rm -rf <Dir Name>

EX: #rm -rf cricket

r - recursively
f - force

```
#mv <oldname><newname> --> To rename the directory.
```

Q) how to create directory and sub directories at a time.

```
#mkdir -p /cricket/india/ap/hyderabad
#rmdir <dirname>
#shutdown --> To power off/shutdown the system after 60 seconds.
```

#shutdown -F --> To power of the system immediately or with force.

#halt --> To shutdown the system

#init 0 --> To shutdown the system

#shutdown -r (or) #reboot --> To restart the system

#shutdown -Fr --> To restart system with force option.

#prtconf (or) #lsconf --> To display the system configuration information.

#prtconf -m --> To display total memory in the system

#prtconf -c --> To display the CPU type

#prtconf -k --> To display the kernel type.

#wc (word count):

By using wc command we can count the lines, words and letters of files.

#cat aix |wc -l --> To display the total lines of Aix file.

#cat aix |wc -w --> To display the total words of Aix file.

#cat aix |wc -c --> To display the total letters of Aix file.

#diff (Difference):

diff command will display the difference between files.

#cat aix

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It is one of the unix OS

#cat unix

we are learning aix operating system

Syn: #diff <source file><destination file>

#diff aix unix

o/p: it is one of the unix OS

#date --> To display system current date & time,

Syn: #date <MMDDhhmmYY> -To change the date

Ex: #date 0324093017 -> To change date to March 24th @ 9:30 AM 2017 year

#find ==> By using find command we can search files/directories.

Syn: #find <location> -name <file/dir name> -print

Ex: `#find /home/surya -name apple -print -->` To search apple file under /home/surya directory.

Ex: `#find / -name <file/dir name> -print -->` To search file/dir entire system.

Relative path & absolute path:

Relative path: if we need to access a file/dir in relative method, first enter into the target location by using `cd` command and access them this is called relative path

Ex: `/home/surya/india/ap/apple`

in above example if we need to access apple file in relative method follow the below steps.

```
#cd /home/surya/india/ap
#vi apple
apples are famous in simla
:wq
```

absolute path: if we need to access a file/dir in absolute path method, we have to specify the full path of file/dir while accessing them.

above example apple file if we need to access in absolute method follow the below steps.

```
Ex: vi /home/surya/india/ap/apple
apples are famous in simla
:wq
```

#man: Man command will provide information about specified command like use of command, how to run commands with example it will provide.

Ex: `#man <command>`

Ex: `#man touch -->` To display the information about touch command.

#more:

more command will display the file's content page by page, from top to bottom.

Syn: `#more filename (or) cat <filename> |more`

Ex: `#more apple (or) #cat apple |more`

#head & tail:

`#head -->` By using head command we can display the top lines of file

#tail --> by using tail command we can display the bottom lines of file.

syn:#cat <filename> |head -20 --> to display top 20 lines of the file.

syn:#cat <filename> |tail -20 --> To display the last 20 lines of the file.

#sort: by using sort command we can display files content in alpha order (A to Z) or alpha reverse order (Z to A).

Ex: sort filename --> To display the file content from A to Z order.

Ex: sort -r filename --> To display the file content from Z to A

Note: Sort command first preference will give numbers then Uppercase letters then small letters.

Moving Data:

By using #mv command we can move the data (files/Dirs) from one location to another location

Syn:#mv <sourcelocation><destinationlocation>

Ex:Source :/cricket/india/sachin

Destination: /mumbai/IPL/

To move the sachin file to destination location.

#mv /cricket/india/sachin /mumbai/IPL

To move the kumar madhu files into destination location by using relative method

Source :/cricket/india/kumar madhu

Destination : /mumbai/IPL/

#cd /crikcket/india/

#mv kumar madhu /mumbai/IPL --> To move the kumar, madhu files to IPL location.

#mv /cricket/india/ra* /mumbai/IPL --> To move the files/dirs which are starting letters with "ra" to IPL location.

#mv /cricket/india/*.html /sap/data/ --> To move all html files to /sap/data location from /cricket/india location.

#mv /cricket/india/* /mumbai/IPL --> To move the all files & dirs from /cricket/india/ location to /mumbai/IPL location.

#rm aix --> To remove the aix file

#rm aix linux --> To remove the aix linux files

#rm ra* --> To remove the all files/dir's which are starting letter ra
#rm -rf * --> To remove the all files and dirs from current location/directory.
#rm -i <filename> -> To remove the file with confirmation option.

#cp /cricket/india/ra* /mumbai/IPL --> To copy the all file/dirs to mumbai/IPL location which are having starting letters "ra"
#cp /cricket/india/*.html /mumbai/IPL --> To copy the all html extension files to /mumbai/IPL location
#cp /cricket/india/ra*.html /sap/data/ --> To move all files which are having starting letters ra with extension .html.

Append:

If we copy the data by using append option previous data and new data will be available on destination file.

SYN:#cat (sourcefile) >> (destination file)
EX:#cat aix
Advanced Interactive Executive Environment

#cat linux
Linux is one of the Unix OS

#cat aix >> linux -> To append aix file data into Linux file.
#cat linux
Linux is one of the Unix OS
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#grep & egrep:

#grep: By using grep command we display the particular content (names) of the file.

syn:#cat filename |grep -i <required name>

-i - Case insensitive (upper/small letters both will display)

ex: #cat cricket|grep -i sachin
sachin
SACHIN
Sachin

#egrep: To display the multiple different words in file we can use egrep command.

Ex: cat cricket |egrep "sachin|dravid"
sachin
SACHIN
Sachin
Dravid
dravid

#who:

Who command will display the current logged users information in the system with time of login, ipaddress etc.

#who am i --> To display the login user information in the system with date & time, pts.

#whoami --> To display the only user name, which is used to login into the system.

#set -o vi --> To get previous used commands one by one.

Ex: #set -o vi and Esc + k

#history --> To list last used 10 commands.

#history -100 --> To list last used 100 commands.

#uname --> To display the using OS information

#uname -L --> To display the server name & Id,

Note: if server name and ID, displayed it is logical server. if it is listed only "1-null" it is physical server.

#uname -M --> To display the server model number

#lsconf |grep -i model

#uname -m --> To display the server serial number

#lsconf |grep -i serial

#oslevel --> To list Aix version

#bc --> To use the calculator.

Note: To exit from the bc command ctrl +d or ctrl +c.

#cal --> To display the current month calendar details.

#cal <year> --> To display the total months calendar details in the specified year.

Ex: #cal 2016

#cal <month><year> --> To display the specified month of calendar details in the required year.

Ex: cal 03 2016 --> To display march month details in 2016 year.

#echo \$? --> To display the last command executed successfully or not.

if it listed "0" last command is success.

if it listed any number last command is not success.

#wall "message" --> To send message to all users, who are logged into the system.

PERMISSIONS:

#mkuser <username> -> To create normal user

#passwd <username> -> To create password for the user

#mkgroup <groupname> -> To create the group

#lsgroup group -> To list the group information

#chgroup users=user1,user2,user3 <groupname> -> To add users into the group

#mkuser -a <username> -> To create admin user

In Unix Permissions are three types read (r), write(w), execution (x).

1) Read Permission: by using read permission we can read the content of the file and we can able to enter into the directories. read permission weightage value is 4.

2) Write Permission: by using write permission we can write/modify/remove the content of the file by using write permission. write permission weightage value is 2.

3) Execution permission: By using execution permission we can execute the script files and we can enter into the directories. Execution weightage value is 1.

When we have created files/directories by default below permissions will assigned to them.

Default File permissions:

-rw-r--r-- = 644

Default Directory permissions:

`-rwxr-xr-x = 755`

The permissions letters are 10.

1 letter for to see file/directory or file type.
234 letters for Owner permissions
567 letters for Group member permissions
890 letters for Others permissions

Note: Actually default file permissions are 666, directory permissions are 777 but if 666, 777 all users (owner, group, others) will get equal permissions, then they have introduced umask value (022) the umask value is subtract the default permissions. after subtract default permissions for file & directory below permissions will be assigned to them.

$666 - 022 = 644$ (`-rw-r--r--`) File permissions
 $777 - 022 = 755$ (`drwxr-xr-x`) Directory permissions

`#umask -->` To display the umask value
`#umask <umaskvalue>` to change the umask value.
`#umask 033`

Note: Default umask value is 022

#chmod:

By using `chmod` command we can enable/disable/change the permissions for file, directory and file systems.

Syn: `#chmod <Permissions> <file/Dir>`

Permissions can be changed two ways by using `chmod` command for file/directory.

- 1) Symbolic mode
- 2) Numeric mode

1) To change the permissions for file/dire in symbolic mode:

```
$touch aix
ls -l aix
-rw-r--r-- 1 surya  unix          57 Nov 26 00:38 aix
```

Users: surya, siva, madhu, sunil

surya -- owner of the aix file
surya, siva -- part of unix group
madhu,sunil -- others

-rw-r--r-- surya unix 57 Nov 26 00:38 aix

To enable the execution permissions to group for the aix file

```
#chmod g+x aix  
-rw-r-xr-- aix
```

To disable the read permissions to others for aix file

```
#chmod o-r aix  
-rw- r-x ---
```

To disable the write permissions to owner for aix file

```
#chmod u-w aix  
-r-- r-x --- aix
```

To enable the execution permissions to owner for aix file.

```
#chmod u+x aix  
-r-x r-x --- aix
```

To enable write permissions for all (owner/group/others).

```
#chmod u+w aix  
#chmod g+w aix  
#chmod o+w aix  
(or)  
#chmod u+w,g+w,o+w aix  
(or)  
#chmod ugo+w aix  
(or)  
#chmod +w aix  
-rwx rwx -w- aix
```

To disable the write permissions for group and enable the read permissions to others.

```
#chmod g-w,o+r aix  
(or)  
#chmod g-w aix  
#chmod o+r aix
```

-rwx r-x rw-

To disable the execution permissions for all.

```
#chmod -x aix
#chmod ugo-x aix
#chmod u-x,g-x,o-x aix
-rw-r--rw-
```

To disable the read permissions to others for cricket directory

```
ls -ld cricket
-rwx-r-xr-x cricket

#chmod o-r cricket
```

To enable the write permissions to group for cricket directory and inside of all sub directories/file.

```
#chmod -R g+w cricket
```

2) Numeric Method to change the permissions:

-rw- r-- r-- aix

To enable the execution permissions to group for the aix file

```
#chmod g+x aix
#chmod 654 aix
-rw-r-xr-- aix
```

To disable the read permissions to others for aix file

```
#chmod o-r aix
#chmod 650 aix
-rw- r-x ---
```

To disable the write permissions to owner for aix file

```
#chmod u-w aix
#chmod 450 aix
```

```
-r-- r-x --- aix
4+1 5 0
```

To enable the execution permissions to owner for aix file.

```
#chmod u+x aix
#chmod 550 aix
-r-x r-x --- aix
```

To enable write permissions for all (owner/group/other).

```
#chmod u+w aix
#chmod g+w aix
#chmod o+w aix
(or)
#chmod u+w,g+w,o+w aix
(or)
#chmod ugo+w aix
(or)
#chmod +w aix
(or)
#chmod 772 aix
-rwx rwx -w- aix
```

To disable the write permissions for group and enable the read permissions to others.

```
#chmod 756 aix
#chmod g-w,o+r aix
(or)
#chmod g-w aix
#chmod o+r aix
-rwx r-x rw-
```

To disable the execution permissions for all.

```
#chmod -x aix
-rw-r--rw-
```

To disable the read permissions to others for cricket directory

```
ls -ld cricket
-rwx-r-xr-x cricket

#chmod o-r cricket
```

To enable the write permissions to group for cricket directory and inside of all sub directories/files.

```
#chmod -R g+w cricket
```

Numeric Method:

```
#chmod -R 775 cricket
```

```
#mkgroup <group name> To create group
```

```
#mkgroup unix --> To create unix group
```

```
#lsgroup <groupname> --> To list the group information
```

```
#chgroup <users=user1,user2><groupname> --> To add the users into group
```

Ex: #chgroup users=surya,siva unix --> To add the surya,siva users into unix group.

Once added users into run the below commands for the added users.

```
#chuser pgrp=unix surya
```

```
#chuser groups=unix surya
```

```
#chuser pgrp=unix siva
```

```
#chuser groups=unix siva
```

```
#id <username> --> To display user name and group name of the user.
```

```
#mkuser <username> --> To create user
```

```
#mkuser -a <username> --> To create Admin user
```

```
#lsuser <username> --> To list user attributes
```

```
#chuser pgrp=<groupname> --> To change the primary group
```

```
#chuser groups=<groupname1,groupname2> --> To change the secondary group.
```

Note: when we are adding users to the group by using chgroup command, we have to add existing users and new users to the group.

Ex: #lsgroup unix

users: kumar,sunil

above example unix group having kumar, sunil users, now add the new user called madhu

```
#chgroup users=kumar,sunil,madhu unix
```

Default Groups:

There are two default groups in Aix

- 1) Staff
- 2) System

Staff: when we create normal users they are part of staff group.

System: When we create admin users they are part of system group,

Note: root user also part of system group.

`#chown <ownername><file/dir> -->` To change the ownership of the file or dir.

`#chown <owner:groupname><file/dir> -->` To change the owner & group name of the file/dir

`#chown -R <owner><dirname> -->` To change the owner name for the directory and inside all file/sub directories.

`#chown -R <owner:groupname><dirname> -->` To change the owner & group name of directory & inside files/sub directories.

Siva:

Create BSC group and add 6 users of your friends.

Balaji

Create Mtech group and add 4 users of your friends

Kaleedh:

Create Colleague group and add 4 users of your friends

Suvarna:

Create Bcom group and add 4 users of your friends

Sticky Bit: If we enable sticky bit permissions to file/directory the file/directory will delete by only owner even though group members & others having full permissions they can't delete files/dirs.

`#chmod <Permissions><file/dir>`

`#chmod 1777 aix`

(or)

`#chmod +t aix`

Note: small t means we have enabled sticky with execution permission, upper case (T) means without execution permission.

Note: /tmp directory having sticky bit permission by default

#SUID: Set user ID, if we enable SUID permissions to the file, the file can access all users as owner.

#chmod +s aix --> To enable the SUID permissions to aix file
(or)
#chmod 4644 aix -->

Note: small (s) means owner having execution permission.
upper case (S) means owner don't have execution permission.

#SGID:SGID stands for Set group ID, if we enable SGID permissions to file,
only group members can access as owner.

#chmod g+s <filename> --> To enable SGID permissions for the file.
Ex:#chmod g+s linux --> To enable the SGID permissions to linux file.
(or)
#chmod 2644 linux --> To enable the SGID permissions to linux file in
numeric mode.

Stickey Bit

#chmod 1644 <filename/Dir>
1 - indicate to stickey bit
644 - file/dir permissions
#chmod +t <filename/Dir>
Ex: vi, /tmp

SUID: Set user ID

if you enable SUID permissions for the file, All users(Group, others) can
access the file as a owner
#chmod 4644 <filename>
#chmod u+s <filename>

Ex: /etc/passwd
s = With x permission
S = no x permission

SGID:

If you enable SGID permissions for any file... Only group members can
access the file as a owner

#chmod 2644 <filename>
#chmod g+s <filename>
**touch, mkdir, mkdir -p, cp, mv, date, mkgroup, mkuser, mkuser -a,
passwd,pwdadm, wc, banner, diff, permissions, chmod, chown,
stickeybit, SUID, SGID. prtconf or lsconf, prtconf -k, -m, c, head, tail,
grep, egrep, more, sort, vi editor, man, who, who am i, whoami,**

**uname, uname -L, uname -m, uname -M, oslevel, history, bc, cal,
echo \$?, find, wall, set -o vi**

shutdown -> To power off the system

Note: system will down after 60 secons

shutdown -F > to power off the system force

#halt or init 0-> To power off the server

#shudown -r -> normal reboot

#shudown -Fr -> force reboot

#reboot or init 6 -> reboot the system

Basic Questions:

- 1) How to login into the server.
- 2) How to create file
- 3) How many ways we create file
- 4) How to create multiple empty files at a time
- 5) How to copy files from one location to another location
- 6) How to move the files from one location to another location
- 7) How to remove files
- 8) How to remove empty directory
- 9) How to remove content directory
- 10) Default file & directory permissions?
- 11) How to change permissions?
- 12) How many ways we can change the permissions?
- 13) What is umask value?
- 14) How to change umask value?
- 15) What is default umask value?
- 13) How to find the word in vi editor?
- 14) How to replace word in vi editor?
- 15) How to delete line?
- 16) How to delete 5 lines?
- 17) How to copy & paste line?
- 18) How to copy & paste 8 lines?
- 19) How to move the cursor to top, bottom, left and right
- 20) How to replace letter
- 15) How to set numbers in vi editor?
- 16) VI editor modes?
- 17) What is stickey bit
- 18) What is SUID (Set User ID)?
- 19) What is SGID (Set group ID)?
- 20) How to remove/copy/move only .html file/dirs?
- 21) What is date format to change it?
- 22) Types of files in Unix?
- 23) How to link files?
- 24) Difference between softlink & hardlink?
- 25) How to delete line?

Vi Editor:

By using vi editor, we can add/remove/modify the data into files.

Vi editor modes:

There are three modes

1) Command mode: To executive vi commands

2) Insert mode: We can append the data

3) Last line mode / Exit mode: We can exit from the vi editor by saving the data or not saving the data

Short cut keys:

i -> To enter into Insert mode for add/remove data

Esc + /word -> to search word

Note: n to search word next time

ESC + Shift + H -> To move cursor first line of the page

Esc + shift + G -> To move cursor last line of the page

Esc + Shift + M -> To move cursor middle line of the page

Esc + w -> To move cursor next word

Esc + x -> To remove the letters

Esc + Shift + A -> To move the cursor end of the line

Esc + r + Type required word -> To replace letter

Esc + dd -> To remove the line

Esc + 5 + dd --> To remove the 5 lines from the cursor point

Esc + u -> To revert back / undo

Esc + 10 + dd -> To delete the 10 lines from the cursor point

Esc + yy -> To copy the line

Esc + 10 + yy --> To copy the 10 lines from the cursor point

Esc + p -> To paste the line

Esc + :%s/oldname/newname/g -> To replace the word

ESc + :set number or :set nu-> To set the line numbers in the file

Esc + :set nonumber or :set nonu -> To remove the numbers in the file

To move the cursor

we can use left (h), right (l), up (k) and down (j) arrows

or h, l, k and j also we can use to move the cursor

Daily/User Management

This chapter describe the Users and group creation/remove/list users/change the users attributes and password control management etc.

Commands:

mkuser, mkuser -a, lsuser, passwd, pwddadm, chuser, rmuser, userdel, mkggroup, chgroup, lsgroup, id, chsec etc...

#mkuser <username> -> To create normal users
#mkuser -a <username> -> To create Admin user
#lsuser ALL -> To list all users with respected attributes
#lsuser username -> To list user attributes
#lsuser -f username -> To list user attributes in vertical mode
#passwd username -> To create password of the user
#pwddadm username -> To create password of the user
#pwddadm -c username -> To clear the first login change password option for the user after created new password.

User Attributes:

In user Management we have few **important configuration** files. such as.

/etc/passwd
/etc/utmp
/var/adm/wtmp
/etc/nologin
/etc/motd (Message of the day)
/etc/environment
/etc/profile
\$HOME/.profile
/etc/group

Below are **Security Configuration** Files:

/usr/lib/security/mkuser.default
/usr/lib/security/mkuser.sys
/etc/security/envIRON
/etc/security/lastlog
/etc/security/limits
/etc/security/user
/etc/security/passwd
/etc/security/login.cfg
/etc/security/failedlogin
/etc/security/group

Creating Users:

#mkuser <username> --> To create normal user
#mkuser -a <username --> To create admin user
#mkuser <attribute=value><username>
#passwd <username --> To create/change the password of users.
(or)
#pwdadm <username --> To create/change the password of users.

Note: root is default user of the system.
once create password for any user except root user, first login of the users,
new password change option will be available.

To change the root user passwd:

#passwd
Or
#passwd root
#pwdadm
#pwdadm root

Note: once install OS there is no default password for the root user, without password we can login into the system.

#pwdadm -c sudheer -> It will not ask to change the password in the time
of first login of users after changed the password.

#passwd -f <username> --> To change the geccos of the user.

listing the users:

#lsuser ALL --> To list the total users with their attributes in the system.

#lsuser <username> --> To list the particular user with respected
attributes.

#lsuser -f <username> --> To list the particular user with respected
attributes in vertical mode

#lsuser -a <attribute><username> --> To list the particular attribute
information of the user.

#Chuser:

By using chuser command we can change the User's attributes:

SYN: #chuser <attribute=value><username>

EX: #chuser id=500 suresh

#rmuser:

We can remove/delete users from the system by using rmuser command

#rmuser <username> --> To delete the User.

Note: This will not delete user home directory under /home file system.

#rm -rf /home/userhomedir --> To delete user home directory.

(or)

#userdel -r <username> --> To delete the user with home directory.

Note:

1) userdel command will delete user and user's home directory under /home file system.

2) when we are deleting user from the system we should delete home directory also under /home file system.

Group Management:

By using group management, we can separate users depends on their roles.

Note: In system default groups are two.

1) staff --> All normal users are part of staff group.

2) system --> All admin users and root user part of system group.

#mkgroup <groupname> --> To create normal group

#mkgroup -a <groupname> --> To create admin group.

#lsgroup <groupname> --> To list particular group information.

#chgroup <attribute=value><groupname> --> To change the group attributes.

#rmgroup <groupname> --> To remove the group from the system.

To assign users to the groups:

if we need to add multiple users into particular group we can use chgroup command.

#chgroup <users=user1,user2,user3><groupname>

if we need to add multiple groups to the particular user we can use the #chuser command.

#chuser <pgrp=primarygrp><username>

#chuser <groups=grp1,grp2,grp3><username>

Note: In primary group we can add only one group

In secondary (groups) group we can add more than one group for the user

lsgroup ALL -> To list all groups with respected information.

```
# chgroup user=ravi,sai,meerza sap^C
# chgroup users=ravi,sai,meerza sap
# lsgroup sap
sap id=201 admin=false users=ravi,sai,meerza registry=files
# lsuser -a pgrp ravi
ravi pgrp=staff
# lsuser -a pgrp groups ravi
ravi pgrp=staff groups=staff,sap
# chuser pgrp=sap ravi
# chuser groups=sap ravi
# lsuser -a pgrp groups ravi
ravi pgrp=sap groups=sap
# chuser groups=sap,linux ravi

# lsuser ravi
ravi id=202 pgrp=sap groups=sap,linux home=/home/ravi
shell=/usr/bin/ksh login=true su=true rlogin=true daemon=true
admin=false sugroups=ALL admgroups= tpath=nosak ttys=ALL expires=0
auth1=SYSTEM auth2=NONE umask=22 registry=files SYSTEM=compat
logintimes= loginretries=0 pldwarntime=0 account_locked=false
minage=0 maxage=0 maxexpired=-1 minalpha=0 minother=0 mindiff=0
maxrepeats=8 minlen=0 histexpire=0 histsize=0 pldchecks= dictionlist=
fsize=2097151 cpu=-1 data=262144 stack=65536 core=2097151
rss=65536 nofiles=2000 time_last_login=1470002067
time_last_unsuccessful_login=1469899810 tty_last_login=/dev/pts/2
tty_last_unsuccessful_login=/dev/pts/2 host_last_login=192.168.1.55
host_last_unsuccessful_login=192.168.1.55 unsuccessful_login_count=0
roles=
# lsuser -a pgrp groups ravi
ravi pgrp=sap groups=sap,linux
# lsgroup linux
linux id=202 admin=false users=ravi registry=files

# chgroup id=600 linux
3004-719 Warning: /usr/bin/chgroup does not update /etc/passwd with the
new gid.
# cat /etc/passwd |grep linux
# lsgroup linux
linux id=600 admin=false users=ravi registry=files

# rmgroup linux
# lsgroup linux
```

3004-686 Group "linux" does not exist.

mkuser shankar

mkuser ravi

3004-689 User "ravi" exists.

mkuser sai

3004-689 User "sai" exists.

mkuser bala

mkuser upendra

mkuser balaji

passwd ravi

Changing password for "ravi"

ravi's New password:

Enter the new password again:

mkuser -a venkat

3004-689 User "venkat" exists.

mkuser -a ramesh

pwddadm balaji

Changing password for "balaji"

balaji's New password:

Enter the new password again:

passwd ravi

Changing password for "ravi"

ravi's New password:

Enter the new password again:

passswd root

ksh: passswd: not found.

passwd root

Changing password for "root"

root's New password:

Enter the new password again:

3004-600 The password entry does not match, please try again.

root's New password:

Enter the new password again:

passwd

Changing password for "root"

root's New password:

Enter the new password again:

mkuser aixadmin

mkuser admin

passwd aixadmin

Changing password for "aixadmin"

aixadmin's New password:

```
Enter the new password again:
# passwd admin
Changing password for "admin"
admin's New password:
Enter the new password again:
# pwddadm -c aixadmin
# pwddadm -c admin
# mkuser sudheer
# passwd sudheer
Changing password for "sudheer"
sudheer's New password:
Enter the new password again:
# pwddadm -c sudheer
```

```
# lsuser ALL
root id=0 pgrp=system groups=system,bin,sys,security,cron,audit,lp
home=/ shell=/usr/bin/ksh auditclasses=general login=true su=true
rlogin=true daemon=true admin=true sugroups=ALL admgroups=
tpath=nosak ttys=ALL expires=0 auth1=SYSTEM auth2=NONE umask=22
registry=files SYSTEM=compat logintimes= loginretries=0
pwdwarntime=0 account_locked=false minage=0 maxage=0 maxexpired=-
1 minalpha=0 minother=0 mindiff=0 maxrepeats=8 minlen=0 histexpire=0
histsize=0 pwdchecks= dictionlist= fsize=2097151 cpu=-1 data=262144
stack=65536 core=2097151 rss=65536 nofiles=2000
time_last_login=1470001642 time_last_unsuccessful_login=1469899804
tty_last_login=/dev/pts/0 tty_last_unsuccessful_login=/dev/pts/1
host_last_login=192.168.1.55 host_last_unsuccessful_login=192.168.1.55
unsuccessful_login_count=0 roles=
daemon id=1 pgrp=staff groups=staff home=/etc login=true su=true
rlogin=true daemon=true admin=true sugroups=ALL admgroups=
tpath=nosak ttys=ALL expires=0101000070 auth1=SYSTEM auth2=NONE
umask=22 registry=files SYSTEM=compat logintimes= loginretries=0
pwdwarntime=0 account_locked=false minage=0 maxage=0 maxexpired=-
1 minalpha=0 minother=0 mindiff=0 maxrepeats=8 minlen=0 histexpire=0
histsize=0 pwdchecks= dictionlist= fsize=2097151 cpu=-1 data=262144
stack=65536 core=2097151 rss=65536 nofiles=2000 roles=
bin id=2 pgrp=bin groups=bin,sys,adm home=/bin login=true su=true
rlogin=true daemon=true admin=true sugroups=ALL admgroups=
tpath=nosak ttys=ALL expires=0101000070 auth1=SYSTEM auth2=NONE
umask=22 registry=files SYSTEM=compat logintimes= loginretries=0
pwdwarntime=0 account_locked=false minage=0 maxage=0 maxexpired=-
1 minalpha=0 minother=0 mindiff=0 maxrepeats=8 minlen=0 histexpire=0
histsize=0 pwdchecks= dictionlist= fsize=2097151 cpu=-1 data=262144
stack=65536 core=2097151 rss=65536 nofiles=2000 roles=
sys id=3 pgrp=sys groups=sys home=/usr/sys login=true su=true
rlogin=true daemon=true admin=true sugroups=ALL admgroups=
```

```

tpath=nosak ttys=ALL expires=0101000070 auth1=SYSTEM auth2=NONE
umask=22 registry=files SYSTEM=compat logintimes= loginretries=0
pwdwarntime=0 account_locked=false minage=0 maxage=0 maxexpired=-
1 minalpha=0 minother=0 mindiff=0 maxrepeats=8 minlen=0 histexpire=0
histsize=0 pwdchecks= dictionlist= fsize=2097151 cpu=-1 data=262144
stack=65536 core=2097151 rss=65536 nofiles=2000 roles=
adm id=4 pgrp=adm groups=adm home=/var/adm login=true su=true
rlogin=true daemon=true admin=true sugroups=ALL admgroups=
tpath=nosak ttys=ALL expires=0 auth1=SYSTEM auth2=NONE umask=22
registry=files SYSTEM=compat logintimes= loginretries=0
pwdwarntime=0 account_locked=false minage=0 maxage=0 maxexpired=-
1 minalpha=0 minother=0 mindiff=0 maxrepeats=8 minlen=0 histexpire=0
histsize=0 pwdchecks= dictionlist= fsize=2097151 cpu=-1 data=262144
stack=65536 core=2097151 rss=65536 nofiles=2000 roles=
uucp id=5 pgrp=uucp groups=uucp home=/usr/lib/uucp login=false
su=true rlogin=false daemon=true admin=true sugroups=ALL
admgroups= tpath=nosak ttys=ALL expires=0 auth1=SYSTEM
auth2=NONE umask=22 registry=files SYSTEM=compat logintimes=
loginretries=0 pwdwarntime=0 account_locked=false minage=0 maxage=0
maxexpired=-1 minalpha=0 minother=0 mindiff=0 maxrepeats=8
minlen=0 histexpire=0 histsize=0 pwdchecks= dictionlist= fsize=2097151
cpu=-1 data=262144 stack=65536 core=2097151 rss=65536 nofiles=2000
roles=
guest id=100 pgrp=usr groups=usr home=/home/guest login=true su=true
rlogin=true daemon=true admin=false sugroups=ALL admgroups=
tpath=nosak ttys=ALL expires=0 auth1=SYSTEM auth2=NONE umask=22
registry=files SYSTEM=compat logintimes= loginretries=0
pwdwarntime=0 account_locked=false minage=0 maxage=0 maxexpired=-
1 minalpha=0 minother=0 mindiff=0 maxrepeats=8 minlen=0 histexpire=0
histsize=0 pwdchecks= dictionlist= fsize=2097151 cpu=-1 data=262144
stack=65536 core=2097151 rss=65536 nofiles=2000 roles=
nobody id=-2 pgrp=nobody groups=nobody home=/ login=true su=true
rlogin=true daemon=true admin=true sugroups=ALL admgroups=
tpath=nosak ttys=ALL expires=0101000070 auth1=SYSTEM auth2=NONE
umask=22 registry=files SYSTEM=compat logintimes= loginretries=0
pwdwarntime=0 account_locked=false minage=0 maxage=0 maxexpired=-
1 minalpha=0 minother=0 mindiff=0 maxrepeats=8 minlen=0 histexpire=0
histsize=0 pwdchecks= dictionlist= fsize=2097151 cpu=-1 data=262144
stack=65536 core=2097151 rss=65536 nofiles=2000 roles=
lpd id=9 pgrp=nobody groups=nobody home=/ login=true su=true
rlogin=true daemon=true admin=true sugroups=ALL admgroups=
tpath=nosak ttys=ALL expires=0101000070 auth1=SYSTEM auth2=NONE
umask=22 registry=files SYSTEM=compat logintimes= loginretries=0
pwdwarntime=0 account_locked=false minage=0 maxage=0 maxexpired=-
1 minalpha=0 minother=0 mindiff=0 maxrepeats=8 minlen=0 histexpire=0

```

```

histsize=0 pwdchecks= dictionlist= fsize=2097151 cpu=-1 data=262144
stack=65536 core=2097151 rss=65536 nofiles=2000 roles=
lp id=11 pgrp=lp groups=lp,printq home=/var/spool/lp shell=/bin/false
login=true su=true rlogin=true daemon=true admin=false sugroups=ALL
admgroups= tpath=nosak ttys=ALL expires=0 auth1=SYSTEM
auth2=NONE umask=22 registry=files SYSTEM=compat logintimes=
loginretries=0 pldwarntime=0 account_locked=false minage=0 maxage=0
maxexpired=-1 minalpha=0 minother=0 mindiff=0 maxrepeats=8
minlen=0 histexpire=0 histsize=0 pwdchecks= dictionlist= fsize=2097151
cpu=-1 data=262144 stack=65536 core=2097151 rss=65536 nofiles=2000
roles=
invscout id=6 pgrp=invscout groups=invscout home=/var/adm/invscout
shell=/usr/bin/ksh login=true su=true rlogin=true daemon=true
admin=true sugroups=ALL admgroups= tpath=nosak ttys=ALL expires=0
auth1=SYSTEM auth2=NONE umask=22 registry=files SYSTEM=compat
logintimes= loginretries=0 pldwarntime=0 account_locked=false
minage=0 maxage=0 maxexpired=-1 minalpha=0 minother=0 mindiff=0
maxrepeats=8 minlen=0 histexpire=0 histsize=0 pwdchecks= dictionlist=
fsize=2097151 cpu=-1 data=262144 stack=65536 core=2097151
rss=65536 nofiles=2000 roles=
snapp id=200 pgrp=snapp groups=snapp home=/usr/sbin/snapp
shell=/usr/sbin/snappd gecol=snapp login user login=true su=false
rlogin=false daemon=true admin=false sugroups=ALL admgroups=
tpath=nosak ttys=/dev/tty0 expires=0 auth1=SYSTEM auth2=NONE
umask=22 registry=files SYSTEM=NONE logintimes= loginretries=0
pldwarntime=0 account_locked=false minage=0 maxage=0 maxexpired=-
1 minalpha=0 minother=0 mindiff=0 maxrepeats=8 minlen=0 histexpire=0
histsize=0 pwdchecks= dictionlist= dce_export=false fsize=2097151 cpu=-
1 data=262144 stack=65536 core=2097151 rss=65536 nofiles=2000
roles=
ipsec id=201 pgrp=staff groups=staff home=/etc/ipsec shell=/usr/bin/ksh
login=true su=true rlogin=true daemon=true admin=false sugroups=ALL
admgroups= tpath=nosak ttys=ALL expires=0 auth1=SYSTEM
auth2=NONE umask=22 registry=files SYSTEM=compat logintimes=
loginretries=0 pldwarntime=0 account_locked=false minage=0 maxage=0
maxexpired=-1 minalpha=0 minother=0 mindiff=0 maxrepeats=8
minlen=0 histexpire=0 histsize=0 pwdchecks= dictionlist= fsize=2097151
cpu=-1 data=262144 stack=65536 core=2097151 rss=65536 nofiles=2000
roles=
nuucp id=7 pgrp=uucp groups=uucp home=/var/spool/uucppublic
shell=/usr/sbin/uucp/uucico gecol=uucp login user login=true su=true
rlogin=true daemon=true admin=false sugroups=ALL admgroups=
tpath=nosak ttys=ALL expires=0 auth1=SYSTEM auth2=NONE umask=22
registry=files SYSTEM=compat logintimes= loginretries=0
pldwarntime=0 account_locked=false minage=0 maxage=0 maxexpired=-
1 minalpha=0 minother=0 mindiff=0 maxrepeats=8 minlen=0 histexpire=0

```

```

histsize=0 pwdchecks= dictionlist= fsize=2097151 cpu=-1 data=262144
stack=65536 core=2097151 rss=65536 nofiles=2000 roles=
ravi id=202 pgrp=staff groups=staff home=/home/ravi shell=/usr/bin/ksh
login=true su=true rlogin=true daemon=true admin=false sugroups=ALL
admgroups= tpath=nosak ttys=ALL expires=0 auth1=SYSTEM
auth2=NONE umask=22 registry=files SYSTEM=compat logintimes=
loginretries=0 pldwarntime=0 account_locked=false minage=0 maxage=0
maxexpired=-1 minalpha=0 minother=0 mindiff=0 maxrepeats=8
minlen=0 histexpire=0 histsize=0 pwdchecks= dictionlist= fsize=2097151
cpu=-1 data=262144 stack=65536 core=2097151 rss=65536 nofiles=2000
time_last_login=1470002067 time_last_unsuccessful_login=1469899810
tty_last_login=/dev/pts/2 tty_last_unsuccessful_login=/dev/pts/2
host_last_login=192.168.1.55 host_last_unsuccessful_login=192.168.1.55
unsuccessful_login_count=0 roles=
sai id=203 pgrp=staff groups=staff home=/home/sai shell=/usr/bin/ksh
login=true su=true rlogin=true daemon=true admin=false sugroups=ALL
admgroups= tpath=nosak ttys=ALL expires=0 auth1=SYSTEM
auth2=NONE umask=22 registry=files SYSTEM=compat logintimes=
loginretries=0 pldwarntime=0 account_locked=false minage=0 maxage=0
maxexpired=-1 minalpha=0 minother=0 mindiff=0 maxrepeats=8
minlen=0 histexpire=0 histsize=0 pwdchecks= dictionlist= fsize=2097151
cpu=-1 data=262144 stack=65536 core=2097151 rss=65536 nofiles=2000
roles=
meerza id=204 pgrp=staff groups=staff home=/home/meerza
shell=/usr/bin/ksh login=true su=true rlogin=true daemon=true
admin=false sugroups=ALL admgroups= tpath=nosak ttys=ALL expires=0
auth1=SYSTEM auth2=NONE umask=22 registry=files SYSTEM=compat
logintimes= loginretries=0 pldwarntime=0 account_locked=false
minage=0 maxage=0 maxexpired=-1 minalpha=0 minother=0 mindiff=0
maxrepeats=8 minlen=0 histexpire=0 histsize=0 pwdchecks= dictionlist=
fsize=2097151 cpu=-1 data=262144 stack=65536 core=2097151
rss=65536 nofiles=2000 roles=
amar id=205 pgrp=staff groups=staff home=/home/amar shell=/usr/bin/ksh
login=true su=true rlogin=true daemon=true admin=false sugroups=ALL
admgroups= tpath=nosak ttys=ALL expires=0 auth1=SYSTEM
auth2=NONE umask=22 registry=files SYSTEM=compat logintimes=
loginretries=0 pldwarntime=0 account_locked=false minage=0 maxage=0
maxexpired=-1 minalpha=0 minother=0 mindiff=0 maxrepeats=8
minlen=0 histexpire=0 histsize=0 pwdchecks= dictionlist= fsize=2097151
cpu=-1 data=262144 stack=65536 core=2097151 rss=65536 nofiles=2000
roles=
ram id=206 pgrp=staff groups=staff home=/home/ram shell=/usr/bin/ksh
login=true su=true rlogin=true daemon=true admin=false sugroups=ALL
admgroups= tpath=nosak ttys=ALL expires=0 auth1=SYSTEM
auth2=NONE umask=22 registry=files SYSTEM=compat logintimes=
loginretries=0 pldwarntime=0 account_locked=false minage=0 maxage=0

```

```

maxexpired=-1 minalpha=0 minother=0 mindiff=0 maxrepeats=8
minlen=0 histexpire=0 histsize=0 pwdchecks= dictionlist= fsize=2097151
cpu=-1 data=262144 stack=65536 core=2097151 rss=65536 nofiles=2000
roles=
venkat id=207 pgrp=staff groups=staff home=/home/venkat
shell=/usr/bin/ksh login=true su=true rlogin=true daemon=true
admin=false sugroups=ALL admgroups= tpath=nosak ttys=ALL expires=0
auth1=SYSTEM auth2=NONE umask=22 registry=files SYSTEM=compat
logintimes= loginretries=0 pwdwarntime=0 account_locked=false
minage=0 maxage=0 maxexpired=-1 minalpha=0 minother=0 mindiff=0
maxrepeats=8 minlen=0 histexpire=0 histsize=0 pwdchecks= dictionlist=
fsize=2097151 cpu=-1 data=262144 stack=65536 core=2097151
rss=65536 nofiles=2000 roles=
shankar id=208 pgrp=staff groups=staff home=/home/shankar
shell=/usr/bin/ksh login=true su=true rlogin=true daemon=true
admin=false sugroups=ALL admgroups= tpath=nosak ttys=ALL expires=0
auth1=SYSTEM auth2=NONE umask=22 registry=files SYSTEM=compat
logintimes= loginretries=0 pwdwarntime=0 account_locked=false
minage=0 maxage=0 maxexpired=-1 minalpha=0 minother=0 mindiff=0
maxrepeats=8 minlen=0 histexpire=0 histsize=0 pwdchecks= dictionlist=
fsize=2097151 cpu=-1 data=262144 stack=65536 core=2097151
rss=65536 nofiles=2000 roles=
bala id=209 pgrp=staff groups=staff home=/home/bala shell=/usr/bin/ksh
login=true su=true rlogin=true daemon=true admin=false sugroups=ALL
admgroups= tpath=nosak ttys=ALL expires=0 auth1=SYSTEM
auth2=NONE umask=22 registry=files SYSTEM=compat logintimes=
loginretries=0 pwdwarntime=0 account_locked=false minage=0 maxage=0
maxexpired=-1 minalpha=0 minother=0 mindiff=0 maxrepeats=8
minlen=0 histexpire=0 histsize=0 pwdchecks= dictionlist= fsize=2097151
cpu=-1 data=262144 stack=65536 core=2097151 rss=65536 nofiles=2000
roles=
upendra id=210 pgrp=staff groups=staff home=/home/upendra
shell=/usr/bin/ksh login=true su=true rlogin=true daemon=true
admin=false sugroups=ALL admgroups= tpath=nosak ttys=ALL expires=0
auth1=SYSTEM auth2=NONE umask=22 registry=files SYSTEM=compat
logintimes= loginretries=0 pwdwarntime=0 account_locked=false
minage=0 maxage=0 maxexpired=-1 minalpha=0 minother=0 mindiff=0
maxrepeats=8 minlen=0 histexpire=0 histsize=0 pwdchecks= dictionlist=
fsize=2097151 cpu=-1 data=262144 stack=65536 core=2097151
rss=65536 nofiles=2000 roles=
balaji id=211 pgrp=staff groups=staff home=/home/balaji
shell=/usr/bin/ksh login=true su=true rlogin=true daemon=true
admin=false sugroups=ALL admgroups= tpath=nosak ttys=ALL expires=0
auth1=SYSTEM auth2=NONE umask=22 registry=files SYSTEM=compat
logintimes= loginretries=0 pwdwarntime=0 account_locked=false
minage=0 maxage=0 maxexpired=-1 minalpha=0 minother=0 mindiff=0

```

```

maxrepeats=8 minlen=0 histexpire=0 histsize=0 pwdchecks= dictionlist=
fsize=2097151 cpu=-1 data=262144 stack=65536 core=2097151
rss=65536 nofiles=2000 roles=
ramesh id=8 pgrp=system groups=system home=/home/ramesh
shell=/usr/bin/ksh login=true su=true rlogin=true daemon=true
admin=true sugroups=ALL admgroups= tpath=nosak ttys=ALL expires=0
auth1=SYSTEM auth2=NONE umask=22 registry=files SYSTEM=compat
logintimes= loginretries=0 pwdwarntime=0 account_locked=false
minage=0 maxage=0 maxexpired=-1 minalpha=0 minother=0 mindiff=0
maxrepeats=8 minlen=0 histexpire=0 histsize=0 pwdchecks= dictionlist=
fsize=2097151 cpu=-1 data=262144 stack=65536 core=2097151
rss=65536 nofiles=2000 roles=
aixadmin id=212 pgrp=staff groups=staff home=/home/aixadmin
shell=/usr/bin/ksh login=true su=true rlogin=true daemon=true
admin=false sugroups=ALL admgroups= tpath=nosak ttys=ALL expires=0
auth1=SYSTEM auth2=NONE umask=22 registry=files SYSTEM=compat
logintimes= loginretries=0 pwdwarntime=0 account_locked=false
minage=0 maxage=0 maxexpired=-1 minalpha=0 minother=0 mindiff=0
maxrepeats=8 minlen=0 histexpire=0 histsize=0 pwdchecks= dictionlist=
fsize=2097151 cpu=-1 data=262144 stack=65536 core=2097151
rss=65536 nofiles=2000 roles=
admin id=213 pgrp=staff groups=staff home=/home/admin
shell=/usr/bin/ksh login=true su=true rlogin=true daemon=true
admin=false sugroups=ALL admgroups= tpath=nosak ttys=ALL expires=0
auth1=SYSTEM auth2=NONE umask=22 registry=files SYSTEM=compat
logintimes= loginretries=0 pwdwarntime=0 account_locked=false
minage=0 maxage=0 maxexpired=-1 minalpha=0 minother=0 mindiff=0
maxrepeats=8 minlen=0 histexpire=0 histsize=0 pwdchecks= dictionlist=
fsize=2097151 cpu=-1 data=262144 stack=65536 core=2097151
rss=65536 nofiles=2000 roles=
sudheer id=214 pgrp=staff groups=staff home=/home/sudheer
shell=/usr/bin/ksh login=true su=true rlogin=true daemon=true
admin=false sugroups=ALL admgroups= tpath=nosak ttys=ALL expires=0
auth1=SYSTEM auth2=NONE umask=22 registry=files SYSTEM=compat
logintimes= loginretries=0 pwdwarntime=0 account_locked=false
minage=0 maxage=0 maxexpired=-1 minalpha=0 minother=0 mindiff=0
maxrepeats=8 minlen=0 histexpire=0 histsize=0 pwdchecks= dictionlist=
fsize=2097151 cpu=-1 data=262144 stack=65536 core=2097151
rss=65536 nofiles=2000 time_last_login=1470002577
tty_last_login=/dev/pts/3 host_last_login=192.168.1.55
unsuccessful_login_count=0 roles=

# lsuser sudheer
sudheer id=214 pgrp=staff groups=staff home=/home/sudheer
shell=/usr/bin/ksh login=true su=true rlogin=true daemon=true
admin=false sugroups=ALL admgroups= tpath=nosak ttys=ALL expires=0

```

```
auth1=SYSTEM auth2=NONE umask=22 registry=files SYSTEM=compat
logintimes= loginretries=0 pldwarntime=0 account_locked=false
minage=0 maxage=0 maxexpired=-1 minalpha=0 minother=0 mindiff=0
maxrepeats=8 minlen=0 histexpire=0 histsize=0 pldchecks= dictionlist=
fsize=2097151 cpu=-1 data=262144 stack=65536 core=2097151
rss=65536 nofiles=2000 time_last_login=1470002577
tty_last_login=/dev/pts/3 host_last_login=192.168.1.55
unsuccessful_login_count=0 roles=
```

```
# lsuser -a id sudheer
sudheer id=214
```

```
# lsuser -a id -a mindiff sudheer
sudheer id=214 mindiff=0
# lsuser -a id mindiff sudheer
sudheer id=214 mindiff=0
# chuser id=500 sudheer
# lsuser -a id sudheer
sudheer id=500
# mkuser id=501 raju
```

```
# ls -ld /home/sudheer
drwxr-xr-x  2 214    staff      256 Jul 31 17:02 /home/sudheer
# ls -ld /home/raju
drwxr-xr-x  2 raju   staff      256 Jul 31 17:15 /home/raju
```

```
# rmuser sudheer
# lsuser sudheer
3004-687 User "sudheer" does not exist.
# ls -ld /home/sudheer
drwxr-xr-x  2 214    staff      256 Jul 31 17:02 /home/sudheer
```

```
# userdel raju
# lsuser raju
3004-687 User "raju" does not exist.
# ls -ld /home/raju
drwxr-xr-x  2 501    staff      256 Jul 31 17:15 /home/raju
```

```
# lsuser balaji
balaji id=211 pgrp=staff groups=staff home=/home/balaji
shell=/usr/bin/ksh login=true su=true rlogin=true daemon=true
admin=false sugroups=ALL admgroups= tpath=nosak ttys=ALL expires=0
auth1=SYSTEM auth2=NONE umask=22 registry=files SYSTEM=compat
logintimes= loginretries=0 pldwarntime=0 account_locked=false
minage=0 maxage=0 maxexpired=-1 minalpha=0 minother=0 mindiff=0
maxrepeats=8 minlen=0 histexpire=0 histsize=0 pldchecks= dictionlist=
```

```
fsize=2097151 cpu=-1 data=262144 stack=65536 core=2097151
rss=65536 nofiles=2000 roles=
# ls -ld /home/balaji
drwxr-xr-x  2 balaji  staff          256 Jul 31 16:48 /home/balaji
# userdel -r balaji
# lsuser balaji
3004-687 User "balaji" does not exist.
# ls -ld /home/balaji
ls: 0653-341 The file /home/balaji does not exist.
# rm -rf /home/sudheer
# ls -ld /home/sudheer
ls: 0653-341 The file /home/sudheer does not exist.
```

```
# mkuser sudheer
# mkuser raju
```

```
# mkuser -a sachin
# lsuser -a pgrp groups sudheer
sudheer pgrp=staff groups=staff
```

```
# lsuser -a pgrp groups raju
raju pgrp=staff groups=staff
# lsuser -a pgrp groups sachin
sachin pgrp=system groups=system
# lsuser -a pgrp groups root
root pgrp=system groups=system,bin,sys,security,cron,audit,lp
```

/etc/passwd

It is one of the configuration file in user management. This file contains a stanza information for each user.

Stanza format:

Username:passwdinfo:UID:GID:Gecos:homeDir:Shell

Username: The name of user

Passwd info: This field contains star (*) symbol or explanator (!) symbol.
Star means password not created for the user.

Explanator means password has been created for the user.

UID: User id of the user.

GID: Group id of the user.

Gecos: In gecos field we can specify the role/comments of the users.

Home Dir: home director of the user

Shell: User using shell info.

#Chsh: To change the shell

`passwd -f <username> ->` To change the gecos
or
`#chuser gecos="comments" <username>`

/etc/utmp

It is one of the binary format configuration files in user management. This file contains the current logged-in users' information. The `who` command will read the data from the `utmp` file.

/var/adm/wtmp:

It is one of the binary format configuration files in user management. This file contains all users' current, old logins' information with date & time. Also, this file contains system rebooted history date information. This file we can read by using `who` or `last` commands.

Ex: `who /var/adm/wtmp`
or `#last`

/etc/nologin: This is one of the user management configuration files. If this file existed in the system, it will not allow all users to login into the system except the root user.

Note: 1) This file by default will not be available in the system; we have to create it by using the `touch /etc/nologin` command.
2) If `/etc/nologin` existed in the system, this will be removed automatically while rebooting the system.

/etc/motd (Message of the day):

It is one of the ASCII format configuration files. If we need to give any information for users after login into the system, we have to update the required information on the `motd` file.

/etc/profile: and \$HOME/.profile

It is one of the configuration files in user management. In this file, we can define environment variables.

EX: Environment variables are

```
set -o vi
alias c=clear
prompt etc...
```

/etc/environment
/etc/profile
\$HOME/.profile

/etc/group

It is one of the ASCII format configuration file, this file contains all created groups information, groupname:GID:assigned users information.

Below are Security Configuration Files:

/usr/lib/security/mkuser.default:

It is one of the important configuration file in user management, It is an ASCII format file, When ever we are creating the users by using mkuser command, the mkuser command will read this file and take few default values from this file. Such as primary & Secondary groupname, home directory and shell information.

This file contains the two stanzas, first stanza for normal users, second stanza for Admin users.

/usr/lib/security/mkuser.sys:

It is one of the ASCII format user management script file, this script file will add the default attributes for the users after created users by using mkuser command.

/etc/security/envron:

It is one of the security configuration file, its contains the default user's information.

```
#cat /etc/security/envron
```

```
default:  
root:  
daemon:  
bin:  
sys:  
adm:  
uucp:  
guest:
```

/etc/security/lastlog:

It is one of the security configuration file in user management, This file contains the last login users information in the system. for each user one stanza will be created. on stanza we can find the user last login timing, ipaddress, unsuccessful login details etc.

```
time_last_unsuccessful_login = 1470436256
tty_last_unsuccessful_login = /dev/pts/4
host_last_unsuccessful_login = 192.168.1.55
unsuccessful_login_count = 0
time_last_login = 1470436266
tty_last_login = /dev/pts/4
host_last_login = 192.168.1.55
```

/etc/security/passwd:

This is one of the security configuration file, in UM,
This file contains the three fields for each user, after created password,

nikki:

```
password = f/r/QwFW8YO6Y
lastupdate = 1470087373
flags = ADMCHG
```

Password: The is encrypted password information for the user,

last update: Created Date & Time of the password,

Flags = ADMCHG,

If we create/change the password for the any users, ADMCHG will be added under flags field, Once user logged into the system flags = (empty),

#pwdadm -c <username> --> To clear the ADMCHG under flags,

Note: After created password for the user, if we clear ADMCHG, change new password option will not available for the user at the time of first login.

/etc/security/failedlogin:

It is one the security configuration file, this file contains the all users failed login information with date & time, It is binary format file, we can read this file by using who command.

/etc/security/group:

It is one of the secutity configuration file, this file contains a stanza of each group, in stanza we can find the group is admin or not.

/etc/security/limits:

It is one of the important ASCII format security configuration file, This file contains limits values of the users. also we can define limits values of the users as per client's requirement.

Note: -1 means unlimited access.

2) if adding/changing the limits of the users we need to take the backup of limits file.

3) we can modify this file by using vi or chsec command.

Syn: #chsec -f <securityfile> -a <attribute=value> -s <username>

Ex: #chsec -f /etc/security/limits -a fsize=-1 -a nofiles=600 -s 999

/etc/security/user:

It is one of the important security configuration file in user management, this file contains the password control attributes information, such as:

histsize:

It is one of the password control attribute, we can't reuse the previous passwords until reach the histsize value.

Ex: if histsize value is 4, The user should use the last 4 passwords different.

Note: default value is 0, (unlimited)

histexpire:

It is one of the password control attribute, in histexpire the value we can define in weeks, suppose 1 means one week, 2 means two weeks,

Ex: histexpire=3,

The user cannot reuse previous passwords, until completed 3 weeks,

on aug 05th password changed as 123, next 21 days we cannot reuse 123 password,

maxage:

It is one of the password control attribute, this value we can define in weeks,

We can define maximum age of the new password for the user by using maxage attribute.

Ex maxage=5

The user new password age is 14 days, before 14 days user have to change the password, if not changed, the password will be expire after 14 days.

Maxexpired=1

It is one of the password control attribute, Defines the maximum time (in weeks) beyond the maxage value that a user can change an expired password

Minage:

It is one of the password control attribute, we can define minage value in week,

ex: minage=2

the user password changed as 123 on Aug22, next 14 days he have to use 123 password only,

minage means minimum age of the password.

maxrepeats:

It is one of the password control attribute, this attribute will define a character or letter maximum how many times can repeat in new password.

Ex : maxrepeats=2

123 (yes)

banana (No) bcoz "a" letter repeated 3 times.

banan (yes)

minalpha:

It is one of the password control attribute, this attribute will define minimum alpha characters should be in new password

Ex: minalpha=2

123 (no)

123a(no)

123ab(yes)

minother:

It is one of the password control attribute, this attribute will define minimum other characters (excep alpha) should be in new password

Ex: minother=3

123 (yes)

abc1(no)

12@abc(yes)

minlen:

It is one of the password control attribute, this attribute will define minimum password length.

Ex: minlen=4

123(no)

1@a*(yes)

a@*(no)

mindiff:

It is one of the password control attribute, this attribute will indicate to the new password, how many letters should be different, if we compare with the old password.

Ex: mindiff=3

abc123 (old)

abc124 (no)

xyz123 (yes) this passwd having xyz 3 different letters if we compare with old password.

/etc/profile

/etc/environment

\$home/.profile

/etc/security/login.cfg

lsuser -f shiva

shiva:

id=226

pgrp=staff

groups=staff,aix1

home=/home/shiva

shell=/usr/bin/ksh

login=true --> Enable console access,

login=false --> Console access disabled

su=true --> Switch user access enabled

su=false --> Switch user access disabled

rlogin=true --> Remote access enabled

rlogin=false --> Remote access disabled

admin=false --> Admin access disabled

expires=0 or MMDDhhmmYY

expires Defines the expiration time for the user account.

* Possible values: a valid date in the form MMDDHHMMYY or 0.

* If 0 the account does not expire. If 0101000070 the account

* is disabled. The range for YY is:

```
*          00 - 38   years 2000 thru 2038
*          39 - 99   years 1939 thru 1999
```

```
# lsuser -a expires shiva
shiva expires=0
# chuser expires=0913173016 shiva
# set -o vi
# lsuser -a expires shiva
shiva expires=0913173016
```

logintimes=

To give particular times access Ex: 09 AM to 5PM

```
shiva logintimes=
# chuser logintimes=:0900-1700 shiva
# date
Tue Aug 23 10:13:41 CDT 2016
```

loginretries=0

It is one of the important attribute,

Ex: loginretries=3,

above example says, the user need to give correct password in 3 logins, after 3 logins faileds if user tried to login to the system the user login access will be disable even the password is correct.

Solution:

a) need to clear the unsuccessful_login_count=0 for the user

```
#chuser unsuccessful_login_count=0 shiva, -> To clear the unsuccessful login count to unlock the user account
```

(or)

b) need to change the password, if unsuccessful_login attribute not fixed issue.

```
# lsuser -a loginretries shiva
shiva loginretries=3
# lsuser -a unsuccessful_login_count shiva
shiva unsuccessful_login_count=5
# chuser unsuccessful_login_count=0 shiva
# lsuser -a loginretries shiva
shiva loginretries=3
# lsuser -a unsuccessful_login_count shiva
shiva unsuccessful_login_count=0
```

Note: loginretries and unsuccessful_login_count both are related attributes, without changing password if need to give login access to users, need to

clear the unsuccessful_login_count=0, if users asking to change the password we can change it by using passwd command.

```
unsuccessful_login_count=0
account_locked=false --> account unlocked
account_locked=true --> account locked
minage=0
maxage=0
maxexpired=-1
minalpha=0
minother=0
mindiff=0
maxrepeats=8
minlen=0
histexpire=0
histsize=0
fsize=2097151
cpu=-1
data=262144
stack=65536
core=2097151
rss=65536
nofiles=2000
pwdwarntime=0
time_last_login=1470255207
tty_last_login=/dev/pts/1
host_last_login=192.168.1.55
pwdchecks=
dictionlist=
roles=
sugroups=ALL
admggroups=
tpath=nosak
ttys=ALL
auth1=SYSTEM
auth2=NONE
umask=22
registry=files
SYSTEM=compate
```

Q) The user not able to login to the system what are the reasons?

ANS:

- 0) Maybe he reached login attempts morethan loginretries value.
- 1) Password may be forgot,
- 2) if he tried console access may be login=false,

- 3) if he tried remoted access may be rlogin=false,
- 4) may be expires date completed
- 5) may be login timings are different.
- 6) May be /home,/(root), /usr, /tmp or /var 100% full.
- 7) May be CPU or Memory 100% full.
- 8) May be /etc/nologin file existed.
- 9) May be maxage value completed.

Q) User unable to change the password what is the reason?

Ans:minage=2
 minalpha=2
 minother=3
 mindiff=2
 maxrepeats=1
 minlen=7
 histexpire=4
 histsize=10

CRON & AT JOBS:

By using cron & at jobs we can schedule the jobs (commands or script files), the scheduled jobs will be implement automatically as per scheduled date & timings.

Cron jobs: if any job we need to schedule morethan one time or multiple times we can use cronjobs method,

At Jobs: if any job we need to schedule the only one time then we can use the at job method.

CRON JOBS:

Format:

minute --> 0 through 59
 hour --> 0 through 23
 day_of_month --> 1 through 31
 month --> 1 through 12
 weekday --> 0 through 6 for Sunday through Saturday
 command or Scriptfile--> A shell command

Ex:

Every Tuesday at 02:00 /appl/program

Every night at 02:00

Every night at midnight

On the 7th, 14th, and 21st day at 4:20 pm

on every two days implement job at 1:30 PM

from 10 to 25 the schedule job on March at 10:30 AM

Monday through Friday at 4:20 pm
1st January, 1st July, and 1st December
at 4:20 pm
To schedule the job every 2 hours
Every 10 minutes

1) We need to reboot the system on Sunday at 12:30 AM Aug 22nd.

30 00 22 08 0 /usr/sbin/shutdown -Fr &

2) We need to reboot the system today at 1:30 PM

30 13 06 08 6 /usr/sbin/shutdown -Fr

3) /sapfs/fico/script1 script file need run every day at 5:45 AM.

45 05 * * * /sapfs/fico/script1)

4) /sapfs/fico/script1 Every Tuesday at 02:00 AM

00 02 * * 2 /sapfs/fico/script1

5) Every night at 02:00 AM

00 02 * * * /sapfs/fico/script1 & (background of the system).

6) Every night at midnight

00 00 * * *

7) On the 7th, 14th, and 21st day at 4:20 pm

20 16 7,14,21 * * /sapfs/fico/script1

8) Monday through Friday at 4:20 pm

20 16 * * 1-5 /sapfs/fico/script1

9) 1st January, 1st July, and 1st December at 4:20 pm

20 16 01 01,07,12 * /sapfs/fico/script1

10) /sapfs/fico/script1 file need to run every 10 minutes.

*/10 * * * * /sapfs/fico/script1

11) /sapfs/fico/script1 file need to run every 2 hours.

```
*/2 * * * /sapfs/fico/script1
```

```
# date
```

```
Tue Aug 23 11:40:45 CDT 2016
```

```
40 11 23 08 02 /usr/sbin/mkvg -f -y sapvg -s 64 hdisk0 &
```

```
#crontab -l --> To list the scheduled cron jobs
```

```
#crontab -e --> To create/remove/modify the cron jobs.
```

```
# crontab -l |grep -i mkvg
```

```
40 11 23 08 02 /usr/sbin/mkvg -f -y sapvg -s 64 hdisk0 &
```

```
#crontab -r --> To remove the cron jobs
```

```
#ps -ef |grep -i cron --> To verify the cron daemon status.
```

```
#kill -9 cronid --> To stop the cron daemon
```

```
#ps -ef |grep cron --> after 60 seconds, cron daemon automatically will  
start, once killed the daemon.
```

```
# ps -ef |grep -i cron
```

```
root 307362    1  0 09:39:25    - 0:00 /usr/sbin/cron
```

```
# kill -9 307362
```

```
# ps -ef |grep -i cron
```

```
root 307364    1  0 11:48:45    - 0:00 /usr/sbin/cron
```

```
#mail --> To check the cron jobs status.
```

```
#tail -f /var/adm/cron/log --> To check the cron jobs status.
```

/var/adm/cron/log --> cron log location.

Crontabs or Jobs Location:

/var/spool/cron/crontabs --> Conjobs location or cron job users.

AT JOBS:

If we need to schedule any job at a time we can schdule the job method as at.

SYN: #at -f filename/cmd -t CCYYMMDDhhmmSS

Ex: at now -f appl/program

```
#atq --> To list the at jobs
```

```
#at -r <jobname> To remove the at job
```

#at -r --> To remove all at jobs.

/var/spool/cron/atjobs --> All at jobs location

Location: /var/adm/cron

This location contains the cronlog, cron & at deny, allow files and queuedefs file.

```
# cd /var/adm/cron
```

```
# ls
```

```
FIFO    at.deny  cron.deny log      queuedefs
```

allow and deny files:

if we add any users in deny files, the users will not be able to schedule the jobs. if allow files existed in the system, which are the users we have added into allow files, only these users can schedule the jobs.

Note: by default, deny files available in the system, allow files will not be available, if allow files required we can create by using #touch command.

If allow files existed in the system, we need to add the users on these files, who ever want to schedule the jobs (cron/at).

If a user presented on allow and deny files. then system will give first priority for the allow file.

USER ATTRIBUTES:

```
# lsuser -f sudheer
```

```
sudheer:
```

```
id=214
```

```
pgrp=staff
```

```
groups=staff
```

```
home=/home/sudheer
```

```
shell=/usr/bin/ksh
```

```
login=true
```

```
su=true
```

```
rlogin=true
```

```
daemon=true
```

```
admin=false
```

```
sugroups=ALL
```

```
admgroups=
```

```
tpath=nosak
```

```
ttys=ALL
```

```
expires=0
auth1=SYSTEM
auth2=NONE
umask=22
registry=files
SYSTEM=compat
logintimes=
loginretries=0
pwdwarntime=0
account_locked=false
minage=0
maxage=0
maxexpired=-1
minalpha=0
minother=0
mindiff=0
maxrepeats=8
minlen=0
histexpire=0
histsize=0
pwdchecks=
dictionlist=
fsize=2097151
cpu=-1
data=262144
stack=65536
core=2097151
rss=65536
nofiles=2000
time_last_login=1470002577
tty_last_login=/dev/pts/3
host_last_login=192.168.1.55
unsuccessful_login_count=0
roles=
```

1) id: Each user having unique ID in AIX operating system, User ID will start from 200.

#chuser id=value username -> To change user id

2) pgrp: Primary group

it is primary group of the user, in primary group we can add only one group.

#chuser pgrp=groupname username -> to change the primary group.

3) groups: secondary group

It is secondary group of the users, in groups we can add more than one group if required.

```
#chuser groups=group1,group2 username
```

4) home:

It is home directory of the user, once created user by using mkuser command for each user one default home directory will create under /home file system with username.

5) Shell:/usr/bin/ksh

In Aix we are using ksh by default (Korn Shell). if we need to change shell we can change by using #chsh command.

6) login=true

if login=true means console access enabled
if login=false means console access disabled

```
#chuser login=false username -> To disable the console access to the user.
```

7) rlogin:

remote login access

if rlogin=true means user having remote login access
if rlogin=false means remote login access disable to user

```
#chuser rlogin=true username --> To enable the remote access to the user.
```

8) su = switch user

su=false --> to other users su has been disabled
su=true --> To other users su has been enabled

```
#chuser su=false username --> To disable the su access
```

su - username --> To switch to the another user with variable values.
su username -> To switch to the another user without variable values.

8) expires:

To give the user account expired date & time.

0 = unlimited access

```
# lsuser -a expires shiva
shiva expires=0
#chuser expires=MMDDhhmmYY username
# chuser expires=0913173016 shiva
# set -o vi
# lsuser -a expires shiva
shiva expires=0913173016 -> Shiva user account will be expired on sep
13th 2016 @ 05PM 30 Minutes.
```

9) admin:

admin=true -> To enable admin access
admin=false -> To disable admin access

10) umask=022

#chuser umask=values username --> To change umask value for the user.

11) logintimes:

To specify the login timings of the user.

Ex: siva user need to login into the system from 9:00AM to 5PM.

```
#chuser logintimes=:0900-1700 siva
```

12) loginretries:

It is one of the attribute for the users, by default loginretries value is 0 means user can attempt wrong passwords unlimited.

suppose if loginretries=3, means the user need to enter correct password in 3 attempts if three attempts are wrong passwords his account will be expired/disabled.

Solution:

1) Clear the unsuccessful count to as zero, to give again 3 chances to siva user

```
#chuser unsuccessful_login_count=0 siva
```

2) Change the password if siva forgot the password.

#passwd siva

11) account_locked:

if account_locked=false --> The user account unlocked & he can able to login to the system.

if account_locked=true --> The user account locked & he can not login into the system even he is having correct username, password.

USER ATTRIBUTES:

id -> ID is unique number for each user it will generate by the OS.

pgrp -> It is primary group for the user. In primary group we can add only one group

groups -> its secondary group for the users, in secondary group we can add more then one group.

su (switch user) -> By using su attribute we can jump to from one user to another user.

#Ex: su=true

rlogin:

rlogin=true -> remote access enabled for the user

rlogin=false -> remote access disabled for the user

login:

login=true -> To enable the console access for the user

login=false -> To disable the console access.

LVM

What is LVM

LVM Stands for Logical Volume Management, By using LVM we can manage/stored the data in servers. LVM terminologies are VGs, PVs, PPs, LVs, LPs and File systems. In Aix LVM we have Advanced concepts like mirroring, export, import, migration etc...

LVM Structure1:

1) Take Hard disk

2) Scan new disk

#cfgmgr

#lspv -> To list the disks

#bootinfo -s <diskname> --> To check disk size

Create PVID for the new disk

#chdev -l <pvname> -a pv=yes -> To create PVID

3) We have to create the VG on new disk

Ex: hdisk1 1GB Disk

SYN: #mkvg -y <vgname> -s 64 hdisk1

EX: #mkvg -y appvg -s 64 hdisk1

#lsvg -> To list the VGs

4) To create Logical Volume

SYN: #mklv -y <lvname> -t <typeoflv> <vgname> <NOofLPs> <pvname>

Ex: #mklv -y saplv -t jfs2 sapvg 4 hdisk1

5) To create File system with created Logical volume (saplv)

#crfs -v <typeoffs> -d /dev/lvname -m </fsname>

Ex: #crfs -v jfs2 -d /dev/saplv -m /sapfs

6) To open the FS
#mount </fsname>
#mount /sapfs

7) To enter into the FS
#cd /sapfs
#pwd
/sapfs

8) To store the data (#touch, mkdir , cp, rm)

PVID

#chdev -l <pvname> -a pv=yes --> To create PVID
#chdev -l <pvname> -a pv=clear --> To remove PVID
#chdev -l hdisk5 -a pv=clear -> To remove the pvid
#chdev -l hdisk5 -a pv=yes -> To create the pvid

Note:1)The PVID will be 16 digits

2)) PVID stands for Physical volume Identifier

3) PVID will be generate depends on system current date& time and with combination system serial number

4) If clear PVID we will be losing the total disk data

5) If disk is assigned any VG, The disk PVID will not be clear.

#lsvg <vgname> --> To list VG attributes
#lsvg -l <vgname> --> To list the created lvs and file systems in the VG.
#lspv -l <pvname> --> To list the LVs/FSs of particular PV.
#lsvg -p <vgname> --> To list the total hard disks of the VG
#rmlv <lvname> --> To remove LV
#rmfs </fsname> --> To remove File system
#reducevg <vgname><pvname> --> To remove PV from VG

In Above picture we can see below things

- 1) VG Name is SAPVG
- 2) SAP VG contains the two hard disks (hdisk1, hdisk2)
- 3) PP/LP Size is 64 MB
- 4) Each pv contains only LV/FS (/myfs and /sapfs)

LVM Conditions/LVM Rules/Key points:

- 1) To create VG minimum one empty disk required
- 2) A disk can be assigned to only one VG,
- 3) A disk can not possible to shared on different VGs.

- 4) VG stands for volume group, Data can be stored on VGs with the help of LVs, LPs, PVs and PPs. VG contains the PVs, PPs, LVs and LPs.
- 5) Hard disk we can call as Physical Volume (PV) in LVM the data can be stored on PVs. PV contains the PPs. All PPs are same size on hard disk.
- 6) PP - Physical partion it is part of PV. The data can be stored on PPs.
- 7) LP - Logical partion it is part of LV. The data can be stored on PPs with help of LPs. Minimum 1LP can map with one PP and Maximum 3 PPs it can be map.
- 8) LV - Logical volume, by using logical volume we can store the data in disks. The data will go through the LVs to store on PVs.
- 9) To access logical volume, we should create respected file system. with help of file system, we can open/close the LVs and store the data in PVs.
- 10) ALL PPs & LPs size is always same on VG.
- 11) Different types model disks we can be add to same VG.

LVM Concepts:

LVM Structure1 and 2
 Extend/Reduce VG
 mount/umount FS/fuser/rmfs
 Remove LV/FS/VG
 VG Types/Differences
 Activate/Deactivate VG
 Df -gt/mount/lsfs
 VG/PV/LV Attributes
 Disk availability/un availability
 Migrating data
 Convert VGs
 Export/Import VG
 Rename VG/FS/LV
 Default VG/FSs/LVs in Aix
 Mirroring/Unmirroring and Mklvcopy
 Synchronization
 VGDA/VGSA/LVCB and Quorum
 Intra policy and Inter Policy
 Hot spare
 T-Factor
 LTG size,

ADD/Remove disks to Volume Groups:

Add Disks to VG:

Note: If we don't have free space (PPs) in volume group we can add new disks to VG to create new LVs/FSs or To Increase the existing FSs/LVs

Steps to add disks:

- 1) Take empty disk
- 2) Create PVID if not created
- 3) Add the disk to VGs by using below command.

Syn: `#extendvg <VGNAME><New PV>`

EX: `# extendvg sapvg hdisk1`

`#lsvg -p sapvg ->` To list the total PVs of sapvg

Remove/reduce the Hard disk from the VG:

If disk is failed or disk performance is slow, we will remove the disk from VG.

Note: Before remove the hard disk from the VG, the disk should not contain any LVs, File systems, the disk should be empty.

Steps:

- 1) Check LVs/FSs in the PV. (`#lspv -l hdisk0`)
- 2) Remove LVs/FSs if existed. if file systems are mounted we have to unmount them and remove it.
- 3) Finally remove the disk from the VG by using below command.

SYN: `#reducevg <vgname><pvname>`

Ex: `#reducevg sapvg hdisk2 -->` To remove the hdisk2 from sapvg.

Remove LV/FS/VG

Removing LV:

Note: Before removing LV from the VG/server the LV should be closed/umount.

`#rmlv <lvname> ->` To remove LV

`#rmlv -f <lvname> ->` To remove lv without confirmation

Removing FS:

Note: Before removing File System from the VG/server the FS should be closed/umount.

`#umount </fsname> ->` If file system is mounted, we have to umount

`#rmfs </fsname> ->` To remove FS

Removing VG:

Note: There is no separate command to remove the VG, if we remove/reduce all PVs from the VG, The VG will be delete from the server. if we want delete VG from server, just we have to reduce the all PVs from the VG then VG will be delete automatically.

Types of VGs:

In Aix There are 3 types of VGs

- 1) Normal VG
- 2) Big VG
- 3) Scalable VG

To create Normal VG

```
#mkvg -y <vgname> -s <ppsize><pvname pvname>
```

To create Big VG

```
#mkvg -B -y <vgname> -s <ppsize><pvname pvname>
```

To Create Scalable VG

```
#mkvg -S -y <vgname> -s <ppsize><pvname pvname>
```

3 types of VGs Differences/limitations:

VG Attributes:

```
#lsvg <VG Name> --> To display the VG attributes
```

O/P: VG Name, VGID, VG State, VG Permissions, PP Size, Total PPs, Free PPs, Used PPs, Max LV, Total LV, Opened LV, Total PVs, Active PVs, Stale PVs, AUTO ON, AUTO SYNC, STALE PPs, VGDA/VG Descriptor, QUORUM,

Convert VGs:

1)To convert normal VG to Big VG.

```
#chvg -B <vgname>
```

Note: This is online activity, no need to umount respected file systems in VG, varyoffvg not required

2) To convert the VG normal to Scalable VG or

To convert the Big VG to Scalable VG

Note: This is offline activity we need to umount the file systems and need to varyoff the VG before converting to Scalable VG.

Steps:

- 1) unmount the file systems
 - 2) varyoff the VG
 - 3) convert the VG to Scalable
- ```
#chvg -G <vgname>
```

**Note:** Each PV should have the at least one free PP to convert VG (normal to Big/Scalable, Big to Scalable)

**Migrating:**

By using migrating, we can transfer the data from one PV to another PV within VG only.

**Note:** In migrating the source and destination disks should be in Same VG

**SYN:** # migratepv <sourcepv><Destinationpv> To migrate the data

EX: # migratepv <hdisk1><hdisk2>

or nohup migratepv <sourcepv><Destinationpv>

```
lsvg -p sai
```

sai:

| PV_NAME      | PV STATE |     | TOTAL PPs | FREE PPs           | FREE |
|--------------|----------|-----|-----------|--------------------|------|
| DISTRIBUTION |          |     |           |                    |      |
| hdisk0       | active   | 39  | 38        | 08..07..07..08..08 |      |
| hdisk2       | active   | 39  | 15        | 08..01..00..00..06 |      |
| hdisk3       | active   | 399 | 399       | 80..80..79..80..80 |      |

```
lspv -l hdisk2
```

hdisk2:

| LV NAME | LPs | PPs | DISTRIBUTION       | MOUNT POINT |
|---------|-----|-----|--------------------|-------------|
| sailv   | 10  | 10  | 00..03..07..00..00 | N/A         |
| fslv00  | 4   | 4   | 00..04..00..00..00 | /saifs      |
| javav   | 10  | 10  | 00..00..00..08..02 | /javafs     |

```
lspv -l hdisk3
```

```
migratepv hdisk2 hdisk3
```

Ex: #migratelp <lvname/lpnumber><destdisk/ppnumber>

EX: #migratelp fslv00/1 hdisk4/5

```
lspv -l hdisk3
hdisk3:
LV NAME LPs PPs DISTRIBUTION MOUNT POINT
sailv 10 10 00..10..00..00 N/A
fslv00 4 4 00..04..00..00 /saifs
javalv 10 10 00..10..00..00 /javafs
lspv -l hdisk2
```

**SYN:** #migratepv -l <lvname><sourcepv><destinationpv> --> To transfer particular lv from one PV to another PV in same VG.

EX: # migratepv -l fslv00 hdisk3 hdisk2

```
lspv -l hdisk3
hdisk3:
LV NAME LPs PPs DISTRIBUTION MOUNT POINT
sailv 10 10 00..10..00..00 N/A
javalv 10 10 00..10..00..00 /javafs
lspv -l hdisk2
hdisk2:
LV NAME LPs PPs DISTRIBUTION MOUNT POINT
fslv00 4 4 00..04..00..00 /saifs
```

```
lspv -M hdisk2
hdisk2:1-8
hdisk2:9 fslv00:1
hdisk2:10 fslv00:2
hdisk2:11 fslv00:3
hdisk2:12 fslv00:4
hdisk2:13-39
lspv -M hdisk0
hdisk0:1-8
hdisk0:9 loglv00:1
hdisk0:10-39
```

# migratepv fslv00/1 hdisk0/12  
migratepv: Mirror copy 1 of logical partition 1 of logical volume  
fslv00 migrated to physical partition 12 of hdisk0.

```
lspv -M hdisk0
hdisk0:1-8
hdisk0:9 loglv00:1
hdisk0:10-11
hdisk0:12 fslv00:1
hdisk0:13-39
lspv -M hdisk2
hdisk2:1-9
hdisk2:10 fslv00:2
hdisk2:11 fslv00:3
```

hdisk2:12 fslv00:4  
hdisk2:13-39

## **Activate:Deactivate /Varyoff:Varyon VGs:**

### **Deactivate VG:**

#### **Note:**

- 1) Before deactivate VG, we should close all file systems in the VG.
- 2) If we deactivate VG, LVM operations not possible on inactive VGs

Ex: lsvg vname, lsvg -l vname etc...

- 3) if we deactivate VG the VG's data cannot accessble

#lsvg -o --> To list online/active VGs

```
lsvg
rootvg
sai
saimgrt
```

```
lspv
hdisk0 00cb9dbe18a78c3d sai active
hdisk1 00cb9dbe137eba2c saimgrt active
hdisk2 00cb9dbe13a16525 sai active
hdisk3 00cb9dbef0487445 sai active
hdisk4 00cb9dbe13623391 None
hdisk5 00cb9dbeb3812c53 rootvg active
hdisk6 00cb9dbe136c60a9 None
```

**SYN:** # varyoffvg <vname> To deactivate VG

EX: # varyoffvg saivg

0516-012 lvaryoffvg: Logical volume must be closed. If the logical volume contains a filesystem, the umount command will close the LV device.

0516-942 varyoffvg: Unable to vary off volume group sai.

# lsvg -l sai

sai:

| LV NAME | TYPE    | LPs | PPs | PVs | LV STATE     | MOUNT POINT |
|---------|---------|-----|-----|-----|--------------|-------------|
| loglv00 | jfs2log | 1   | 1   | 1   | open/syncd   | N/A         |
| fslv00  | jfs2    | 4   | 4   | 2   | open/syncd   | /saifs      |
| sailv   | jfs2    | 10  | 10  | 1   | closed/syncd | N/A         |
| javallv | jfs2    | 10  | 10  | 1   | open/syncd   | /javafs     |

**Note:** In VG all File systems/LVs should closed before deactivate VG

# umount /saifs

```
umount /javafs
umount: 0506-349 Cannot unmount /dev/javalv: The requested resource is
busy.
cd
umount /javafs
umount: 0506-349 Cannot unmount /dev/javalv: The requested resource is
busy.
pwd
/
fuser -u /javafs
/javafs: 307370c(root)
pwd
/
fuser -cuk /javafs
/javafs: 307370c(root)
fuser -u /javafs
/javafs:
```

```
umount /javafs
```

```
lsvg -l sai
```

```
sai:
```

| LV NAME | TYPE    | LPs | PPs | PVs | LV STATE             | MOUNT POINT |
|---------|---------|-----|-----|-----|----------------------|-------------|
| loglv00 | jfs2log | 1   | 1   | 1   | closed/syncd N/A     |             |
| fslv00  | jfs2    | 4   | 4   | 2   | closed/syncd /saifs  |             |
| sailv   | jfs2    | 10  | 10  | 1   | closed/syncd N/A     |             |
| javalv  | jfs2    | 10  | 10  | 1   | closed/syncd /javafs |             |

```
varyoffvg sai
```

```
lsvg
```

```
rootvg
```

```
sai
```

```
saimgrt
```

```
lsvg -o
```

```
saimgrt
```

```
rootvg
```

```
lspv
```

|        |                  |         |        |
|--------|------------------|---------|--------|
| hdisk0 | 00cb9dbe18a78c3d | sai     |        |
| hdisk1 | 00cb9dbe137eba2c | saimgrt | active |
| hdisk2 | 00cb9dbe13a16525 | sai     |        |
| hdisk3 | 00cb9dbef0487445 | sai     |        |
| hdisk4 | 00cb9dbe13623391 | None    |        |
| hdisk5 | 00cb9dbeb3812c53 | rootvg  | active |
| hdisk6 | 00cb9dbe136c60a9 | None    |        |

```
lsvg sai
```

0516-010 : Volume group must be varied on; use varyonvg command.

```
varyonvg sai
lsvg -o
sai
saimgrt
rootvg
lspv
hdisk0 00cb9dbe18a78c3d sai active
hdisk1 00cb9dbe137eba2c saimgart active
hdisk2 00cb9dbe13a16525 sai active
hdisk3 00cb9dbef0487445 sai active
hdisk4 00cb9dbe13623391 None
hdisk5 00cb9dbeb3812c53 rootvg active
hdisk6 00cb9dbe136c60a9 None
```

### **Activate VG:**

**SYN:** #varyonvg <VG name>

**EX:** #varyonvg saivg

**Note:** once VG is activated file systems we have to mount manually by using mount command.

### **Export/Import VGs:**

- 1) We can use export/import commands to move the VG from one server(aix1) to another server (aix2) or
- 2) By using export/import we can change/rename the VG name.

To move the VG from one server (ser1) to another (ser2)

By using export/import VG commands we can move the VG from one server to another server follow the below steps:

#### **On Source Server Steps:**

- 1) umount the all file systems of VG  
#umount /Fsname
- 2) Varyoff/deactivate VG  
#varyoffvg <Vgname>
- 3) Export VG  
#exportvg <vgname>

**Note:** Once exported VG, The VG's file systems definations will remove from the OS/ODM/System or /etc/filesystems configuration file.

4) remove the disks from the OS.

```
#rmdev -dl <diskname>
```

```
#rmdev -dl hdiskx
```

```
#rmdev -dl hdiskx
```

5) Remove the disks from the source system physically

```
lsvg -p sai
```

sai:

| PV NAME | PV STATE |     | TOTAL PPs | FREE PPs           | FREE DISTRIBUTION |
|---------|----------|-----|-----------|--------------------|-------------------|
| hdisk0  | active   | 39  | 37        | 08..06..07..08..08 |                   |
| hdisk2  | active   | 39  | 36        | 08..05..07..08..08 |                   |
| hdisk3  | active   | 399 | 379       | 80..60..79..80..80 |                   |

```
lsvg -l sai
```

sai:

| LV NAME | TYPE    | LPs | PPs | PVs | LV STATE     | MOUNT POINT |
|---------|---------|-----|-----|-----|--------------|-------------|
| loglv00 | jfs2log | 1   | 1   | 1   | open/syncd   | N/A         |
| fslv00  | jfs2    | 4   | 4   | 2   | open/syncd   | /saifs      |
| sailv   | jfs2    | 10  | 10  | 1   | closed/syncd | N/A         |
| javallv | jfs2    | 10  | 10  | 1   | open/syncd   | /javafs     |

```
umount /saifs
```

```
umount /javafs
```

```
lsvg -l sai
```

sai:

| LV NAME | TYPE    | LPs | PPs | PVs | LV STATE     | MOUNT POINT |
|---------|---------|-----|-----|-----|--------------|-------------|
| loglv00 | jfs2log | 1   | 1   | 1   | closed/syncd | N/A         |
| fslv00  | jfs2    | 4   | 4   | 2   | closed/syncd | /saifs      |
| sailv   | jfs2    | 10  | 10  | 1   | closed/syncd | N/A         |
| javallv | jfs2    | 10  | 10  | 1   | closed/syncd | /javafs     |

```
lsvg -o
```

sai

saimgrt

rootvg

```
varyoffvg sai
```

```
lsvg -o
```

saimgrt

rootvg

```
cat /etc/filesystems |grep -ip /saifs
```

/saifs:

dev = /dev/fslv00

vfs = jfs2

```

log = /dev/loglv00
mount = true
account = false

cat /etc/filesystems|grep -ip /javafs
/javafs:
dev = /dev/javalv
vfs = jfs2
log = /dev/loglv00
mount = false
account = false

exportvg sai
cat /etc/filesystems |grep -ip /saifs
cat /etc/filesystems|grep -ip /javafs
lsvg
rootvg
saimgrt
lsvg -o
saimgrt
rootvg
lspv
hdisk0 00cb9dbe18a78c3d None
hdisk1 00cb9dbe137eba2c saimgrt active
hdisk2 00cb9dbe13a16525 None
hdisk3 00cb9dbef0487445 None
hdisk4 00cb9dbe13623391 None
hdisk5 00cb9dbeb3812c53 rootvg active
hdisk6 00cb9dbe136c60a9 None

rmdev -dl hdisk0
hdisk0 deleted
rmdev -dl hdisk2
hdisk2 deleted
rmdev -dl hdisk3
hdisk3 deleted
lspv
hdisk1 00cb9dbe137eba2c saimgrt active
hdisk4 00cb9dbe13623391 None
hdisk5 00cb9dbeb3812c53 rootvg active
hdisk6 00cb9dbe136c60a9 None

Go to Destination server^C

```

**Destination server:**

1) Add the disks to the destination server physically.

2) Scan the added disks

```
#cfgmgr
```

3) Importvg the VG

```
#importvg -y <vgname> -n <pvname> (or)
```

```
#importvg -y <vgname><pvname>
```

**Note:** a) -n -> will not activate VG automatically.

b) Once VG imported, The VG file system's definations will add into the /etc/filesystems file/OS/ODM/System.

c) While importing VG if you want you can change the VG name.

4) Varyon VG if not online

```
#varyonvg <vgname>
```

5) mount the file systems.

```
#mount /fsname
```

```
#mount /fsname
```

```
lspv
```

|        |                  |         |        |
|--------|------------------|---------|--------|
| hdisk1 | 00cb9dbe137eba2c | saimgrt | active |
| hdisk4 | 00cb9dbe13623391 | None    |        |
| hdisk5 | 00cb9dbeb3812c53 | rootvg  | active |
| hdisk6 | 00cb9dbe136c60a9 | None    |        |

```
cfgmgr
```

```
lspv
```

|        |                  |         |        |
|--------|------------------|---------|--------|
| hdisk0 | 00cb9dbe18a78c3d | None    |        |
| hdisk1 | 00cb9dbe137eba2c | saimgrt | active |
| hdisk2 | 00cb9dbe13a16525 | None    |        |
| hdisk3 | 00cb9dbef0487445 | None    |        |
| hdisk4 | 00cb9dbe13623391 | None    |        |
| hdisk5 | 00cb9dbeb3812c53 | rootvg  | active |
| hdisk6 | 00cb9dbe136c60a9 | None    |        |

```
importvg -y sai -n hdisk2
```

```
sai
```

```
lspv
```

|        |                  |         |        |
|--------|------------------|---------|--------|
| hdisk0 | 00cb9dbe18a78c3d | sai     |        |
| hdisk1 | 00cb9dbe137eba2c | saimgrt | active |
| hdisk2 | 00cb9dbe13a16525 | sai     |        |
| hdisk3 | 00cb9dbef0487445 | sai     |        |
| hdisk4 | 00cb9dbe13623391 | None    |        |
| hdisk5 | 00cb9dbeb3812c53 | rootvg  | active |
| hdisk6 | 00cb9dbe136c60a9 | None    |        |

```
exportvg sai
```

```

lspv
hdisk0 00cb9dbe18a78c3d None
hdisk1 00cb9dbe137eba2c saimgirt active
hdisk2 00cb9dbe13a16525 None
hdisk3 00cb9dbef0487445 None
hdisk4 00cb9dbe13623391 None
hdisk5 00cb9dbeb3812c53 rootvg active
hdisk6 00cb9dbe136c60a9 None

importvg -y sai hdisk2
sai
lspv
hdisk0 00cb9dbe18a78c3d sai active
hdisk1 00cb9dbe137eba2c saimgirt active
hdisk2 00cb9dbe13a16525 sai active
hdisk3 00cb9dbef0487445 sai active
hdisk4 00cb9dbe13623391 None
hdisk5 00cb9dbeb3812c53 rootvg active
hdisk6 00cb9dbe136c60a9 None

lsvg -o
sai
saimgirt
rootvg
lsvg -l sai
sai:
LV NAME TYPE LPs PPs PVs LV STATE MOUNT POINT
loglv00 jfs2log 1 1 1 closed/syncd N/A
fslv00 jfs2 4 4 2 closed/syncd /saifs
sailv jfs2 10 10 1 closed/syncd N/A
javalv jfs2 10 10 1 closed/syncd /javafs
mount /saifs
mount /javafs
lsvg -l
0516-992 lsvg: Illegal combination of command line options.
Usage: lsvg [-o] [[-L] -n PVname]
 lsvg [-L] [-i] [-M | -l | -p] VGname...
Lists the characteristics of a volume group.
lsvg -l sai
sai:
LV NAME TYPE LPs PPs PVs LV STATE MOUNT POINT
loglv00 jfs2log 1 1 1 open/syncd N/A
fslv00 jfs2 4 4 2 open/syncd /saifs
sailv jfs2 10 10 1 closed/syncd N/A
javalv jfs2 10 10 1 open/syncd /javafs
cd /javafs

```

```
ls
aix hp-ux linux lost+found solaris
ls -l
total 0
-rw-r--r-- 1 root system 0 Jul 24 10:03 aix
-rw-r--r-- 1 root system 0 Jul 24 10:03 hp-ux
-rw-r--r-- 1 root system 0 Jul 24 10:03 linux
drwxr-xr-x 2 root system 256 Jul 24 10:03 lost+found
-rw-r--r-- 1 root system 0 Jul 24 10:03 solaris
```

```
cat /etc/filesystems|grep -ip /saifs
```

```
/saifs:
```

```
dev = /dev/fslv00
vfs = jfs2
log = /dev/loglv00
mount = true
check = false
account = false
```

```
cat /etc/filesystems|grep -ip /javafs
```

```
/javafs:
```

```
dev = /dev/javalv
vfs = jfs2
log = /dev/loglv00
mount = false
check = false
```

**To rename/change the VG name, FS name, LV name**

### **To Change/Rename VG Name:**

By using export/import commands we can change the VG name

1) umount Filesystems

```
#umount /fsname
```

```
#umouut /fsname
```

2) varyoffvg

```
#varyoffvg <vgname>
```

3) Exportvg

```
#exportvg <vgname>
```

4) importvg

```
#importvg -y <vgname><pvname>
```

**Note:** While importing VG we can change the VG name which is required. spicify the required VG name after -y flag.

5) Activate VG if not active

```
#Varyonvg <vgname>
```

6) mount the File systems

```
#mount /fsname
```

#mount /fsname

## How to rename the File system

# lsfs

| Name         | Nodename | Mount Pt   | VFS    | Size    | Options   | Auto |
|--------------|----------|------------|--------|---------|-----------|------|
| Accounting   |          |            |        |         |           |      |
| /dev/hd4     | --       | /          | jfs2   | 524288  | -- yes no |      |
| /dev/hd1     | --       | /home      | jfs2   | 524288  | -- yes no |      |
| /dev/hd2     | --       | /usr       | jfs2   | 2097152 | -- yes no |      |
| /dev/hd9var  | --       | /var       | jfs2   | 524288  | -- yes no |      |
| /dev/hd3     | --       | /tmp       | jfs2   | 524288  | -- yes no |      |
| /proc        | --       | /proc      | procfs | --      | -- yes no |      |
| /dev/hd10opt | --       | /opt       | jfs2   | 524288  | -- yes no |      |
| /dev/lv00    | --       | /rajani    | jfs    | --      | -- no no  |      |
| /dev/lvnrlv  | --       | /lvnfs     | jfs2   | --      | -- no no  |      |
| /dev/fslv01  | --       | /saimgrtfs | jfs2   | 4194304 | -- yes no |      |
| /dev/fslv00  | --       | /saifs     | jfs2   | 2097152 | -- yes no |      |
| /dev/javalv  | --       | /javafs    | jfs2   | 5242880 | -- no no  |      |

# df -gt -> to list the only mounted file systems and we can see the total size/used/free size of the file systems

# lsvg -l sapvg

sapvg:

| LV NAME | TYPE    | LPs | PPs | PVs | LV STATE     | MOUNT POINT |
|---------|---------|-----|-----|-----|--------------|-------------|
| loglv00 | jfs2log | 1   | 1   | 1   | open/syncd   | N/A         |
| fslv00  | jfs2    | 4   | 4   | 2   | open/syncd   | /saifs      |
| sailv   | jfs2    | 10  | 10  | 1   | closed/syncd | N/A         |
| javalv  | jfs2    | 10  | 10  | 1   | closed/syncd | /javafs     |

# df -gt

| Filesystem   | GB blocks | Used | Free | %Used | Mounted on |
|--------------|-----------|------|------|-------|------------|
| /dev/hd4     | 0.25      | 0.01 | 0.24 | 5%    | /          |
| /dev/hd2     | 1.00      | 0.55 | 0.45 | 55%   | /usr       |
| /dev/hd9var  | 0.25      | 0.01 | 0.24 | 4%    | /var       |
| /dev/hd3     | 0.25      | 0.00 | 0.25 | 2%    | /tmp       |
| /dev/hd1     | 0.25      | 0.00 | 0.25 | 1%    | /home      |
| /proc        | -         | -    | -    | -     | /proc      |
| /dev/hd10opt | 0.25      | 0.04 | 0.21 | 15%   | /opt       |
| /dev/fslv01  | 2.00      | 0.00 | 2.00 | 1%    | /saimgrtfs |
| /dev/fslv00  | 1.00      | 0.00 | 1.00 | 1%    | /saifs     |

# df

| Filesystem | 512-blocks | Free   | %Used | Iused | %Iused | Mounted on |
|------------|------------|--------|-------|-------|--------|------------|
| /dev/hd4   | 524288     | 501904 | 5%    | 1526  | 3%     | /          |
| /dev/hd2   | 2097152    | 953768 | 55%   | 17739 | 15%    | /usr       |

|              |         |         |     |     |    |            |
|--------------|---------|---------|-----|-----|----|------------|
| /dev/hd9var  | 524288  | 508320  | 4%  | 339 | 1% | /var       |
| /dev/hd3     | 524288  | 516912  | 2%  | 64  | 1% | /tmp       |
| /dev/hd1     | 524288  | 523360  | 1%  | 30  | 1% | /home      |
| /proc        | -       | -       | -   | -   | -  | /proc      |
| /dev/hd10opt | 524288  | 447464  | 15% | 700 | 2% | /opt       |
| /dev/fslv01  | 4194304 | 4193008 | 1%  | 4   | 1% | /saimgrtfs |
| /dev/fslv00  | 2097152 | 2096176 | 1%  | 7   | 1% | /saifs     |

# df -k

| Filesystem   | 1024-blocks | Free    | %Used | Iused | %Iused | Mounted on |
|--------------|-------------|---------|-------|-------|--------|------------|
| /dev/hd4     | 262144      | 250952  | 5%    | 1526  | 3%     | /          |
| /dev/hd2     | 1048576     | 476884  | 55%   | 17739 | 15%    | /usr       |
| /dev/hd9var  | 262144      | 254160  | 4%    | 339   | 1%     | /var       |
| /dev/hd3     | 262144      | 258456  | 2%    | 64    | 1%     | /tmp       |
| /dev/hd1     | 262144      | 261680  | 1%    | 30    | 1%     | /home      |
| /proc        | -           | -       | -     | -     | -      | /proc      |
| /dev/hd10opt | 262144      | 223732  | 15%   | 700   | 2%     | /opt       |
| /dev/fslv01  | 2097152     | 2096504 | 1%    | 4     | 1%     | /saimgrtfs |
| /dev/fslv00  | 1048576     | 1048088 | 1%    | 7     | 1%     | /saifs     |

# df -m

| Filesystem   | MB blocks | Free    | %Used | Iused | %Iused | Mounted on |
|--------------|-----------|---------|-------|-------|--------|------------|
| /dev/hd4     | 256.00    | 245.07  | 5%    | 1526  | 3%     | /          |
| /dev/hd2     | 1024.00   | 465.71  | 55%   | 17739 | 15%    | /usr       |
| /dev/hd9var  | 256.00    | 248.20  | 4%    | 339   | 1%     | /var       |
| /dev/hd3     | 256.00    | 252.40  | 2%    | 64    | 1%     | /tmp       |
| /dev/hd1     | 256.00    | 255.55  | 1%    | 30    | 1%     | /home      |
| /proc        | -         | -       | -     | -     | -      | /proc      |
| /dev/hd10opt | 256.00    | 218.49  | 15%   | 700   | 2%     | /opt       |
| /dev/fslv01  | 2048.00   | 2047.37 | 1%    | 4     | 1%     | /saimgrtfs |
| /dev/fslv00  | 1024.00   | 1023.52 | 1%    | 7     | 1%     | /saifs     |

# df -g

| Filesystem   | GB blocks | Free | %Used | Iused | %Iused | Mounted on |
|--------------|-----------|------|-------|-------|--------|------------|
| /dev/hd4     | 0.25      | 0.24 | 5%    | 1526  | 3%     | /          |
| /dev/hd2     | 1.00      | 0.45 | 55%   | 17739 | 15%    | /usr       |
| /dev/hd9var  | 0.25      | 0.24 | 4%    | 339   | 1%     | /var       |
| /dev/hd3     | 0.25      | 0.25 | 2%    | 64    | 1%     | /tmp       |
| /dev/hd1     | 0.25      | 0.25 | 1%    | 30    | 1%     | /home      |
| /proc        | -         | -    | -     | -     | -      | /proc      |
| /dev/hd10opt | 0.25      | 0.21 | 15%   | 700   | 2%     | /opt       |
| /dev/fslv01  | 2.00      | 2.00 | 1%    | 4     | 1%     | /saimgrtfs |
| /dev/fslv00  | 1.00      | 1.00 | 1%    | 7     | 1%     | /saifs     |

# df -gt

| Filesystem | GB blocks | Used | Free | %Used | Mounted on |
|------------|-----------|------|------|-------|------------|
| /dev/hd4   | 0.25      | 0.01 | 0.24 | 5%    | /          |

|              |      |      |      |     |            |
|--------------|------|------|------|-----|------------|
| /dev/hd2     | 1.00 | 0.55 | 0.45 | 55% | /usr       |
| /dev/hd9var  | 0.25 | 0.01 | 0.24 | 4%  | /var       |
| /dev/hd3     | 0.25 | 0.00 | 0.25 | 2%  | /tmp       |
| /dev/hd1     | 0.25 | 0.00 | 0.25 | 1%  | /home      |
| /proc        | -    | -    | -    | -   | /proc      |
| /dev/hd10opt | 0.25 | 0.04 | 0.21 | 15% | /opt       |
| /dev/fslv01  | 2.00 | 0.00 | 2.00 | 1%  | /saimgrtfs |
| /dev/fslv00  | 1.00 | 0.00 | 1.00 | 1%  | /saifs     |

# mount

| node         | mounted    | mounted over | vfs          | date                | options |
|--------------|------------|--------------|--------------|---------------------|---------|
| -----        |            |              |              |                     |         |
| /dev/hd4     | /          | jfs2         | Jul 24 09:51 | rw,log=/dev/hd8     |         |
| /dev/hd2     | /usr       | jfs2         | Jul 24 09:51 | rw,log=/dev/hd8     |         |
| /dev/hd9var  | /var       | jfs2         | Jul 24 09:51 | rw,log=/dev/hd8     |         |
| /dev/hd3     | /tmp       | jfs2         | Jul 24 09:51 | rw,log=/dev/hd8     |         |
| /dev/hd1     | /home      | jfs2         | Jul 24 09:51 | rw,log=/dev/hd8     |         |
| /proc        | /proc      | procfs       | Jul 24 09:51 | rw                  |         |
| /dev/hd10opt | /opt       | jfs2         | Jul 24 09:51 | rw,log=/dev/hd8     |         |
| /dev/fslv01  | /saimgrtfs | jfs2         | Jul 24 09:51 | rw,log=/dev/loglv01 |         |
| /dev/fslv00  | /saifs     | jfs2         | Jul 24 11:21 | rw,log=/dev/loglv00 |         |

# lsfs

| Name         | Nodename | Mount Pt   | VFS    | Size    | Options   | Auto |
|--------------|----------|------------|--------|---------|-----------|------|
| Accounting   |          |            |        |         |           |      |
| /dev/hd4     | --       | /          | jfs2   | 524288  | -- yes no |      |
| /dev/hd1     | --       | /home      | jfs2   | 524288  | -- yes no |      |
| /dev/hd2     | --       | /usr       | jfs2   | 2097152 | -- yes no |      |
| /dev/hd9var  | --       | /var       | jfs2   | 524288  | -- yes no |      |
| /dev/hd3     | --       | /tmp       | jfs2   | 524288  | -- yes no |      |
| /proc        | --       | /proc      | procfs | --      | -- yes no |      |
| /dev/hd10opt | --       | /opt       | jfs2   | 524288  | -- yes no |      |
| /dev/lv00    | --       | /rajani    | jfs    | --      | -- no no  |      |
| /dev/lvnrlv  | --       | /lvnfs     | jfs2   | --      | -- no no  |      |
| /dev/fslv01  | --       | /saimgrtfs | jfs2   | 4194304 | -- yes no |      |
| /dev/fslv00  | --       | /saifs     | jfs2   | 2097152 | -- yes no |      |

# lsvg -l sapvg

sapvg:

| LV NAME | TYPE    | LPs | PPs | PVs | LV STATE     | MOUNT POINT |
|---------|---------|-----|-----|-----|--------------|-------------|
| loglv00 | jfs2log | 1   | 1   | 1   | open/syncd   | N/A         |
| fslv00  | jfs2    | 4   | 4   | 2   | open/syncd   | /saifs      |
| sailv   | jfs2    | 10  | 10  | 1   | closed/syncd | N/A         |
| javalv  | jfs2    | 10  | 10  | 1   | closed/syncd | /javafs     |

# chfs -m /orafs /javafs

```
lsvg -l sapvg
sapvg:
LV NAME TYPE LPs PPs PVs LV STATE MOUNT POINT
loglv00 jfs2log 1 1 1 open/syncd N/A
fslv00 jfs2 4 4 2 open/syncd /saifs
sailv jfs2 10 10 1 closed/syncd N/A
javalv jfs2 10 10 1 closed/syncd /orafs
```

```
lsfs |grep -i /javafs
lsfs |grep -i /orafs
/dev/javalv -- /orafs jfs2 5242880 -- no no
```

```
chfs -m /sapfs /saifs
lsvg -l sapvg
sapvg:
LV NAME TYPE LPs PPs PVs LV STATE MOUNT POINT
loglv00 jfs2log 1 1 1 open/syncd N/A
fslv00 jfs2 4 4 2 open/syncd /sapfs
sailv jfs2 10 10 1 closed/syncd N/A
javalv jfs2 10 10 1 closed/syncd /orafs
```

```
lsfs |grep /saifs
lsfs |grep /sapfs
/dev/fslv00 -- /sapfs jfs2 2097152 -- yes no
df -gt |grep /sapfs
df -gt |grep /saifs
/dev/fslv00 1.00 0.00 1.00 1% /saifs
```

```
mount |grep /saifs
 /dev/fslv00 /saifs jfs2 Jul 24 11:21 rw,log=/dev/loglv00
mount |grep /sapfs
umount /saifs
mount /sapfs
mount |grep /saifs
df -gt |grep /saifs
df -gt |grep /sapfs
/dev/fslv00 1.00 0.00 1.00 1% /sapfs
mount |grep /sapfs
 /dev/fslv00 /sapfs jfs2 Jul 24 11:36 rw,log=/dev/loglv00
cd /sapfs
ls
bangla india lost+found westindises
```

### **To rename the File system**

```
#chfs -m <newfsname><oldfsname>
```

**Note:** If you change the name of mounted file system name, the new name will effect only in VG, it will not effect in #df and mount commands. to effect on these commands. we need to umount with old name/mount file system with new name

### To rename Logical Volume:

```
#chlv -n <newname><oldname>
```

**Note:** we need to unmount the LV before change/rename the lv name

```
lsvg -l sapvg
```

```
sapvg:
```

| LV NAME | TYPE    | LPs | PPs | PVs | LV STATE     | MOUNT POINT |
|---------|---------|-----|-----|-----|--------------|-------------|
| loglv00 | jfs2log | 1   | 1   | 1   | open/syncd   | N/A         |
| fslv00  | jfs2    | 4   | 4   | 2   | open/syncd   | /sapfs      |
| sailv   | jfs2    | 10  | 10  | 1   | closed/syncd | N/A         |
| javlv   | jfs2    | 10  | 10  | 1   | closed/syncd | /orafs      |

```
chlv -n <newname><oldname>^C
```

```
chlv -n saplv fslv00
```

0516-708 chlv: Cannot rename open logical volume. If LV contains a filesystem, unmount to close logical volume fslv00.

```
chlv -n javlv oralv
```

0516-306 getlvodm: Unable to find oralv in the Device Configuration Database.

0516-704 chlv: Unable to change logical volume oralv.

```
chlv -n oralv javlv
```

```
lsvg -l sapvg
```

```
sapvg:
```

| LV NAME | TYPE    | LPs | PPs | PVs | LV STATE     | MOUNT POINT |
|---------|---------|-----|-----|-----|--------------|-------------|
| loglv00 | jfs2log | 1   | 1   | 1   | open/syncd   | N/A         |
| fslv00  | jfs2    | 4   | 4   | 2   | open/syncd   | /sapfs      |
| sailv   | jfs2    | 10  | 10  | 1   | closed/syncd | N/A         |
| oralv   | jfs2    | 10  | 10  | 1   | closed/syncd | /orafs      |

```
umount /sapfs
```

umount: 0506-349 Cannot unmount /dev/fslv00: The requested resource is busy.

```
pwd
```

```
/sapfs
```

```
cd
```

```
umount /sapfs
```

```
chlv -n saplv fslv00
```

```
lsvg -l sapvg
```

```
sapvg:
```

| LV NAME | TYPE    | LPs | PPs | PVs | LV STATE     | MOUNT POINT |
|---------|---------|-----|-----|-----|--------------|-------------|
| loglv00 | jfs2log | 1   | 1   | 1   | closed/syncd | N/A         |
| saplv   | jfs2    | 4   | 4   | 2   | closed/syncd | /sapfs      |
| sailv   | jfs2    | 10  | 10  | 1   | closed/syncd | N/A         |
| oralv   | jfs2    | 10  | 10  | 1   | closed/syncd | /orafs      |

## VG Attributes

```
lsvg siva
VOLUME GROUP: siva VG IDENTIFIER:
00cb9dbe00004c000000015627b8c65c
VG STATE: active PP SIZE: 256 megabyte(s)
VG PERMISSION: read/write TOTAL PPs: 39 (9984
megabytes)
MAX LVs: 256 FREE PPs: 36 (9216 megabytes)
LVs: 2 USED PPs: 3 (768 megabytes)
OPEN LVs: 2 QUORUM: 2
TOTAL PVs: 1 VG DESCRIPTORS: 2
STALE PVs: 0 STALE PPs: 0
```

VOLUME GROUP The name of the volume group.

VG STATE The state of the volume group.

VG PERMISSION Access permission: read-only or read-write.

MAX LVs Maximum number of LVs allowed in the volume group.

OPEN LVs The number of logical volumes currently open.

TOTAL PVs The total number of PVs in the volume group.

STALE PVs The number of PVs that contain stale partitions.

ACTIVE PVs The number of PVs currently active.

MAX PPs per VG Maximum number of PPs allowed in the volume group.

MAX PPs per PV Maximum number of PPs per physical volume.

LTG size LTG size of the volume group.

HOT SPARE Hot spare policy of the volume group.

VG IDENTIFIER The numerical identifier of the volume group.

PP SIZE The size of physical partitions from the volume group.

TOTAL PPs The total number of PPs within the volume group.

FREE PPs The total number of PPs within the volume group that have not yet been allocated to any logical volume.

USED PPs The total number of PPs within the volume group that have been already allocated to logical volumes.

QUORUM The number of physical volumes needed for quorum.

VG DESCRIPTORS The number of VGDA's within the volume group.

STALE PPs The number of stale PPs within the volume group.

AUTO ON Determines if the volume group will automatically vary on after system reboot.

MAX PVs The maximum number of PVs allowed in the volume group.  
 AUTO SYNC Determines if stale partitions are synchronized automatically.  
 BB POLICY Bad block relocation policy.

```
ACTIVE PVs: 1 AUTO ON: yes
MAX PPs per VG: 32512
MAX PPs per PV: 1016 MAX PVs: 32
LTG size (Dynamic): 256 kilobyte(s) AUTO SYNC: no
HOT SPARE: no BB POLICY: relocatable
```

### **Auto on:**

The auto on option is available for the VGs and filesystems. if we set auto on yes for the VG/file system automatically will activate while booting the system. if we set auto on=no the VG/File system will automatically will deactivate while booting system.

### **VGs Auto on:**

**Note:** For VGs by default auto on=yes

```
#chvg -ay <vgname> = to set auto on = yes
#chvg -an <vgname> = to set auto on = no
```

### **File systems:**

```
#chfs -A yes </fsname> = To set auto on=yes for the file system
#chfs -A no </fsname> = To set auto on=no for the file system.
```

### **PV Attributes:**

```
#lspv <pvname>
```

### **ALLOCATABLE**

If you set ALLOCATABLE=NO for the pv not possible to use the free PPs to create/increase FS, LVs

if you set ALLOCATABLE=Yes for the pv we can use free PPs to create/increase FS, LVs

```
#chpv -an <pvname> To change the ALLOCATABLE=NO
#chpv -ay <pvname> To change the ALLOCATABLE=yes
```

```
#extendlv <lvname><nooflp><pvname> --> To increase the LV size.
```

if you set auto on=yes for the vg, the VG automatically activate while booting system.

if you set auto on=no for the vg, the VG automatically deactivate while booting system.

### **FSs Auto on**

#chfs -A yes </fsname> -> to change the auto on yes for the file system

#chfs -A no </fsname> -> to change the auto on no for the file system

if you set auto on=yes for the FS, the FS automatically mount while booting system.

if you set auto on=no for the FS, the FS automatically umount while booting system.

### **Default VG/FSs/LVs in Aix**

rootvg

LV     Filesystem   Use of it

/dev/hd1 -

### **Mirroring:**

By using mirroring, we can make duplicate copies of existing data. we can do mirroring on rootvg and user created VGs (ex: sapvg, javavg etc). In mirroring no of copies minimum 2, maximum three copies.

**Note:** If we mirror rootvg we have to create BLV and set the bootlist order.

### **Requirements:**

1) Take the empty hard disk equal size of existing disk or more than existing disk size.

Ex: if existing disk (hdisk1) size is 10G, empty hard disk(hdisk2) should be =>10G.

or

Ex: if existing disk (hdisk1) size is 10G but data is 5GB, empty hard disk should be =>5G or =>10G.

### **Implementation:**

#### **User Created VGs Mirroring:**

1) Take the new/empty hard disk and scan it.

#cfgmgr

**Note:** if empty disk already available scan not require

2) Add the new/empty hard disk into VG, which is you want mirror  
`#extendvg <vgname><pvname>`  
 3) Mirror the VG with new/empty disk.  
`#mirrorvg <vgname><diskname>`  
 4) Verify the mirroring  
`#lsvg -l <vgname>`

**Note:** Here we will verify the LPs: PPs ratio  
 if it is 1:1 its not mirrored  
 if it is 1:2 its mirrored, available two copies  
 if it is 1:3 its mirrored, available 3 copies

## Unmirroring of User created VGs:

### Unmirrorvg:

1) Break the Mirroring  
`#unmirrorvg <vgname><pvname>`

**Note:** pvname should be specify, Which copy we need to delete.

2) Verify the mirroring  
`#lsvg -l <vgname>`  
 3) remove the empty disk from the VG  
`#reducevg <vgname><pvname>`  
 4) remove the disk from the OS/ODM if required  
`#rmdev -dl <diskname>`  
 =====

```
lsvg -p sapvg
sapvg:
PV_NAME PV STATE TOTAL PPs FREE PPs FREE
DISTRIBUTION
hdisk0 active 159 150 32..23..31..32..32
```

```
lspv |grep -i none
hdisk1 00cb9dbe137eba2c None
hdisk2 00cb9dbe13a16525 None
```

```
bootinfo -s hdisk0
10240
```

```
bootinfo -s hdisk2
10240
```

```
extendvg sapvg hdisk2
0516-1398 extendvg: The physical volume hdisk2, appears to belong
toanother volume group. Use the force option to add this physical volumeto
a volume group.
```

0516-792 extendvg: Unable to extend volume group.

```
extendvg -f sapvg hdisk2
```

```
lsvg -p sapvg
```

sapvg:

| PV_NAME      | PV STATE | TOTAL PPs | FREE PPs | FREE               |
|--------------|----------|-----------|----------|--------------------|
| DISTRIBUTION |          |           |          |                    |
| hdisk0       | active   | 159       | 150      | 32..23..31..32..32 |
| hdisk2       | active   | 159       | 159      | 32..32..31..32..32 |

```
lsvg -l sapvg
```

sapvg:

| LV NAME | TYPE    | LPs | PPs | PVs | LV STATE   | MOUNT POINT |
|---------|---------|-----|-----|-----|------------|-------------|
| loglv00 | jfs2log | 1   | 1   | 1   | open/syncd | N/A         |
| fslv00  | jfs2    | 8   | 8   | 1   | open/syncd | /sapfs      |

```
mirrorvg sapvg hdisk2
```

0516-1125 mirrorvg: Quorum requirement turned off, varyoff and varyon volume group for this to take effect.

```
lsvg -l sapvg
```

sapvg:

| LV NAME | TYPE    | LPs | PPs | PVs | LV STATE   | MOUNT POINT |
|---------|---------|-----|-----|-----|------------|-------------|
| loglv00 | jfs2log | 1   | 2   | 2   | open/syncd | N/A         |
| fslv00  | jfs2    | 8   | 16  | 2   | open/syncd | /sapfs      |

```
lsvg -p sapvg
```

sapvg:

| PV_NAME      | PV STATE | TOTAL PPs | FREE PPs | FREE               |
|--------------|----------|-----------|----------|--------------------|
| DISTRIBUTION |          |           |          |                    |
| hdisk0       | active   | 159       | 150      | 32..23..31..32..32 |
| hdisk2       | active   | 159       | 150      | 32..23..31..32..32 |

```
lspv -l hdisk0
```

hdisk0:

| LV NAME | LPs | PPs | DISTRIBUTION       | MOUNT POINT |
|---------|-----|-----|--------------------|-------------|
| fslv00  | 8   | 8   | 00..08..00..00..00 | /sapfs      |
| loglv00 | 1   | 1   | 00..01..00..00..00 | N/A         |

```
lspv -l hdisk2
```

hdisk2:

| LV NAME | LPs | PPs | DISTRIBUTION       | MOUNT POINT |
|---------|-----|-----|--------------------|-------------|
| fslv00  | 8   | 8   | 00..08..00..00..00 | /sapfs      |
| loglv00 | 1   | 1   | 00..01..00..00..00 | N/A         |

```
unmirrorvg sapvg hdisk2
```

0516-1133 unmirrorvg: Quorum requirement turned on, varyoff and varyon volume group for this to take effect.

```
lsvg -l sapvg
```

sapvg:

| LV NAME | TYPE | LPs | PPs | PVs | LV STATE | MOUNT POINT |
|---------|------|-----|-----|-----|----------|-------------|
|---------|------|-----|-----|-----|----------|-------------|

=====

### RootVG Mirroring:

1) Take the new disk and scan it

2) Add the disk to the rootvg

3) Mirror the rootvg with new disk

# mirrorvg <vgname><pvname> -> This command will mirror VG, but # prompt will give after mirroring completed 100%

(or)

# mirrorvg -S <vgname><pvname> -> By using this command, mirroring will be happen background and # prompt will give immediately

(or)

# mirrorvg -S -c 3 <vgname> -> By using this command, we can do the mirroring at a time 3 copies.

**Note:** once mirrored VGs, we have to verify the ratio of LPs:PPs and ALL LVs/FSs should be syncd state.

4) verify the mirroring

5) Create the Boot Logical Volume (BLV) on new disk.

```
#bosboot -ad </dev/newpv>
```

6) Create the Bootlist order in the system

```
#bootlist -om normal <hdiskx hdiskx>
```

7) To list bootlist order / To verify the bootlist order

```
#bootlist -om normal
```

```
lsvg -p rootvg
```

rootvg:

| PV NAME      | PV STATE | TOTAL PPs | FREE PPs | FREE                  |
|--------------|----------|-----------|----------|-----------------------|
| DISTRIBUTION |          |           |          |                       |
| hdisk5       | active   | 546       | 501      | 109..75..99..109..109 |

```
bootinfo -s hdisk5
```

140013

```
lspv |grep -i none
```

|        |                  |      |
|--------|------------------|------|
| hdisk1 | 00cb9dbe137eba2c | None |
|--------|------------------|------|

```
bootinfo -s hdisk1
```

70006

```
lsvg rootvg
VOLUME GROUP: rootvg VG IDENTIFIER:
00cb9dbe00004c000000015618487caa
VG STATE: active PP SIZE: 256 megabyte(s)
VG PERMISSION: read/write TOTAL PPs: 546 (139776
megabytes)
MAX LVs: 256 FREE PPs: 501 (128256 megabytes)
LVs: 10 USED PPs: 45 (11520 megabytes)
OPEN LVs: 9 QUORUM: 2
TOTAL PVs: 1 VG DESCRIPTORS: 2
STALE PVs: 0 STALE PPs: 0
ACTIVE PVs: 1 AUTO ON: yes
MAX PPs per VG: 32512
MAX PPs per PV: 1016 MAX PVs: 32
LTG size (Dynamic): 256 kilobyte(s) AUTO SYNC: no
HOT SPARE: no BB POLICY: relocatable
```

```
lsvg rootvg |grep -i used
LVs: 10 USED PPs: 45 (11520 megabytes)
```

```
lspv |grep -i none
hdisk1 00cb9dbe137eba2c None
```

```
lsvg -p rootvg
rootvg:
PV_NAME PV STATE TOTAL PPs FREE PPs FREE
DISTRIBUTION
hdisk5 active 546 501 109..75..99..109..109
```

```
extendvg rootvg hdisk1
0516-1398 extendvg: The physical volume hdisk1, appears to belong
toanother volume group. Use the force option to add this physical volumeto
a volume group.
```

```
0516-792 extendvg: Unable to extend volume group.
```

```
extendvg -f rootvg hdisk1
```

```
lsvg -l rootvg
rootvg:
LV NAME TYPE LPs PPs PVs LV STATE MOUNT POINT
hd5 boot 1 1 1 closed/syncd N/A
hd6 paging 2 2 1 open/syncd N/A
hd8 jfs2log 1 1 1 open/syncd N/A
hd4 jfs2 1 1 1 open/syncd /
hd2 jfs2 4 4 1 open/syncd /usr
hd9var jfs2 1 1 1 open/syncd /var
hd3 jfs2 1 1 1 open/syncd /tmp
```

```

hd1 jfs2 1 1 1 open/syncd /home
hd10opt jfs2 1 1 1 open/syncd /opt
mydump1v dump 32 32 1 open/syncd N/A
lsvg -p rootvg
rootvg:
PV_NAME PV STATE TOTAL PPs FREE PPs FREE
DISTRIBUTION
hdisk5 active 546 501 109..75..99..109..109
hdisk1 active 273 273 55..55..54..54..55

```

```

lspv -l hdisk5
hdisk5:
LV NAME LPs PPs DISTRIBUTION MOUNT POINT
mydump1v 32 32 00..32..00..00..00 N/A
hd4 1 1 00..00..01..00..00 /
hd2 4 4 00..00..04..00..00 /usr
hd6 2 2 00..02..00..00..00 N/A
hd8 1 1 00..00..01..00..00 N/A
hd1 1 1 00..00..01..00..00 /home
hd10opt 1 1 00..00..01..00..00 /opt
hd9var 1 1 00..00..01..00..00 /var
hd3 1 1 00..00..01..00..00 /tmp
hd5 1 1 01..00..00..00..00 N/A

```

```

lspv -l hdisk1
mirrorvg <vgname><pvname>
mirrorvg -S <vgname><pvname>
mirrorvg -c 2 <vgname>
mirrorvg -c rootvg
lsvg -p rootvg
rootvg:
PV_NAME PV STATE TOTAL PPs FREE PPs FREE
DISTRIBUTION
hdisk5 active 546 501 109..75..99..109..109
hdisk1 active 273 273 55..55..54..54..55

```

```

mirrorvg -c 2 rootvg
0516-1124 mirrorvg: Quorum requirement turned off, reboot system for this
to take effect for rootvg.
0516-1126 mirrorvg: rootvg successfully mirrored, user should perform
bosboot of system to initialize boot records. Then, user must modify bootlist
to include: hdisk1 hdisk5.

```

```

bootlist -om normal
hdisk5
bosboot -ad /dev/hdisk1
bosboot: Boot image is 25166 512 byte blocks.
bootlist -om normal hdisk1 hdisk5

```

```

hdisk1 blv=hd5
hdisk5 blv=hd5
bootlist -om normal
hdisk1 blv=hd5
hdisk5 blv=hd5

lspv -l hdisk1
hdisk1:
LV NAME LPs PPs DISTRIBUTION MOUNT POINT
hd4 1 1 00..00..01..00..00 /
hd2 4 4 00..00..04..00..00 /usr
hd6 2 2 00..02..00..00..00 N/A
hd8 1 1 00..00..01..00..00 N/A
hd1 1 1 00..00..01..00..00 /home
hd10opt 1 1 00..00..01..00..00 /opt
hd9var 1 1 00..00..01..00..00 /var
hd3 1 1 00..00..01..00..00 /tmp
hd5 1 1 01..00..00..00..00 N/A

lspv -l hdisk5
hdisk5:
LV NAME LPs PPs DISTRIBUTION MOUNT POINT
mydumplv 32 32 00..32..00..00..00 N/A
hd4 1 1 00..00..01..00..00 /
hd2 4 4 00..00..04..00..00 /usr
hd6 2 2 00..02..00..00..00 N/A
hd8 1 1 00..00..01..00..00 N/A
hd1 1 1 00..00..01..00..00 /home
hd10opt 1 1 00..00..01..00..00 /opt
hd9var 1 1 00..00..01..00..00 /var
hd3 1 1 00..00..01..00..00 /tmp
hd5 1 1 01..00..00..00..00 N/A

```

### Unmirror rootvg:

```

1) Break the rootvg mirroring.
#unmirrorvg <rootvg><rmovepv>
2) Verify the mirroring
#lsvg -l rootvg
3) remove the BLV on removed pv
#chpv -c <pvname>
4) remove the removed pv from the bootlist order
#bootlist -om normal <OSpv>
5) Verify the bootlist order
#bootlist -om normal
6) Remove the empty PV from the rootvg
#reducevg rootvg <pvname>

```

7) Remove the disk from the OS/ODM, if needed.

```
#rmdev -dl <pvname>
```

**Note:**1) In rootvg mirroring dumpLv will not mirror automatically, if you want mirror dumpLv we can mirror manually by using #mklvcopy command.

2) In rootvg unmirroring dumpLv will not remove, we need to move the dumpLv lv to the existing OS disk,

```
lsvg -p rootvg
```

```
rootvg:
```

| PV_NAME | PV STATE | TOTAL PPs | FREE PPs              | FREE |
|---------|----------|-----------|-----------------------|------|
| hdisk5  | active   | 546       | 109..75..99..109..109 |      |
| hdisk1  | active   | 273       | 54..53..44..54..55    |      |

```
lsvg -l rootvg
```

```
rootvg:
```

| LV NAME  | TYPE    | LPs | PPs | PVs | LV STATE     | MOUNT POINT |
|----------|---------|-----|-----|-----|--------------|-------------|
| hd5      | boot    | 1   | 2   | 2   | closed/syncd | N/A         |
| hd6      | paging  | 2   | 4   | 2   | open/syncd   | N/A         |
| hd8      | jfs2log | 1   | 2   | 2   | open/syncd   | N/A         |
| hd4      | jfs2    | 1   | 2   | 2   | open/syncd   | /           |
| hd2      | jfs2    | 4   | 8   | 2   | open/syncd   | /usr        |
| hd9var   | jfs2    | 1   | 2   | 2   | open/syncd   | /var        |
| hd3      | jfs2    | 1   | 2   | 2   | open/syncd   | /tmp        |
| hd1      | jfs2    | 1   | 2   | 2   | open/syncd   | /home       |
| hd10opt  | jfs2    | 1   | 2   | 2   | open/syncd   | /opt        |
| mydumpLv | dump    | 32  | 32  | 1   | open/syncd   | N/A         |

```
lspv -l hdisk1
```

```
hdisk1:
```

| LV NAME | LPs | PPs | DISTRIBUTION       | MOUNT POINT |
|---------|-----|-----|--------------------|-------------|
| hd4     | 1   | 1   | 00..00..01..00..00 | /           |
| hd2     | 4   | 4   | 00..00..04..00..00 | /usr        |
| hd6     | 2   | 2   | 00..02..00..00..00 | N/A         |
| hd8     | 1   | 1   | 00..00..01..00..00 | N/A         |
| hd1     | 1   | 1   | 00..00..01..00..00 | /home       |
| hd10opt | 1   | 1   | 00..00..01..00..00 | /opt        |
| hd9var  | 1   | 1   | 00..00..01..00..00 | /var        |
| hd3     | 1   | 1   | 00..00..01..00..00 | /tmp        |
| hd5     | 1   | 1   | 01..00..00..00..00 | N/A         |

```
lspv -l hdisk5
```

```
hdisk5:
```

| LV NAME  | LPs | PPs | DISTRIBUTION       | MOUNT POINT |
|----------|-----|-----|--------------------|-------------|
| mydumpLv | 32  | 32  | 00..32..00..00..00 | N/A         |
| hd4      | 1   | 1   | 00..00..01..00..00 | /           |
| hd2      | 4   | 4   | 00..00..04..00..00 | /usr        |
| hd6      | 2   | 2   | 00..02..00..00..00 | N/A         |

```

hd8 1 1 00..00..01..00..00 N/A
hd1 1 1 00..00..01..00..00 /home
hd10opt 1 1 00..00..01..00..00 /opt
hd9var 1 1 00..00..01..00..00 /var
hd3 1 1 00..00..01..00..00 /tmp
hd5 1 1 01..00..00..00..00 N/A

```

```
unmirrorvg rootvg hdisk5
```

0516-1246 rmlvcopy: If hd5 is the boot logical volume, please run 'chpv -c <diskname>' as root user to clear the boot record and avoid a potential boot off an old boot image that may reside on the disk from which this logical volume is moved/removed.

0516-1132 unmirrorvg: Quorum requirement turned on, reboot system for this to take effect for rootvg.

0516-1144 unmirrorvg: rootvg successfully unmirrored, user should performbosboot of system to reinitialize boot records. Then, user must modifybootlist to just include: hdisk1.

```
lspv -l hdisk1
```

```
hdisk1:
```

| LV NAME | LPs | PPs | DISTRIBUTION       | MOUNT POINT |
|---------|-----|-----|--------------------|-------------|
| hd4     | 1   | 1   | 00..00..01..00..00 | /           |
| hd2     | 4   | 4   | 00..00..04..00..00 | /usr        |
| hd6     | 2   | 2   | 00..02..00..00..00 | N/A         |
| hd8     | 1   | 1   | 00..00..01..00..00 | N/A         |
| hd1     | 1   | 1   | 00..00..01..00..00 | /home       |
| hd10opt | 1   | 1   | 00..00..01..00..00 | /opt        |
| hd9var  | 1   | 1   | 00..00..01..00..00 | /var        |
| hd3     | 1   | 1   | 00..00..01..00..00 | /tmp        |
| hd5     | 1   | 1   | 01..00..00..00..00 | N/A         |

```
lspv -l hdisk5
```

```
hdisk5:
```

| LV NAME  | LPs | PPs | DISTRIBUTION       | MOUNT POINT |
|----------|-----|-----|--------------------|-------------|
| mydumplv | 32  | 32  | 00..32..00..00..00 | N/A         |

```
migratepv -l mydumplv hdisk5 hdisk1
```

```
bootlist -om normal
```

```
hdisk1 blv=hd5
```

```
hdisk5
```

```
chpv -c hdisk5
```

```
bootlist -om normal hdisk1
```

```
hdisk1 blv=hd5
```

```
bootlist -om normal
```

```
hdisk1 blv=hd5
```

```
reducevg rootvg hdisk5
```

```
lsvg -p rootvg
```

```

rootvg:
PV_NAME PV STATE TOTAL PPs FREE PPs FREE
DISTRIBUTION
hdisk1 active 273 228 54..21..44..54..55
lspv |grep -i hdisk5
hdisk5 00cb9dbeb3812c53 None
rmdev -dl hdisk5
hdisk5 deleted

```

```

lspv
hdisk0 00cb9dbe18a78c3d sapvg active
hdisk1 00cb9dbe137eba2c rootvg active
hdisk3 00cb9dbef0487445 dell
hdisk4 00cb9dbe22dd1871 vatvg
hdisk6 00cb9dbe136c60a9 obulvg

cfmgr
lspv
hdisk0 00cb9dbe18a78c3d sapvg active
hdisk1 00cb9dbe137eba2c rootvg active
hdisk2 00cb9dbe13a16525 None
hdisk3 00cb9dbef0487445 dell
hdisk4 00cb9dbe22dd1871 vatvg
hdisk5 00cb9dbeb3812c53 None
hdisk6 00cb9dbe136c60a9 obulvg

```

#bootinfo -b -> To verify the booted hard disk.

## Mklvcopy:

By using mklvcopy we can mirror the particular FS or LV,

**Syn:** #mklvcopy -k <lvname><noofcopies><destPV>

EX: #mklvcopy saplv 2 or 3 hdiskX

**Note:** -k automaticly sync the LV or FS after mirrored,

```

lsvg -l rootvg
rootvg:
LV NAME TYPE LPs PPs PVs LV STATE MOUNT POINT
hd5 boot 1 1 1 closed/syncd N/A
hd6 paging 2 2 1 open/syncd N/A
hd8 jfs2log 1 1 1 open/syncd N/A
hd4 jfs2 1 1 1 open/syncd /
hd2 jfs2 4 4 1 open/syncd /usr
hd9var jfs2 1 1 1 open/syncd /var
hd3 jfs2 1 1 1 open/syncd /tmp

```

```

hd1 jfs2 1 1 1 open/syncd /home
hd10opt jfs2 1 1 1 open/syncd /opt
mydump1v dump 32 32 1 open/syncd N/A

```

```
lsvg -p rootvg
```

```
rootvg:
```

| PV_NAME | PV STATE | TOTAL PPs | FREE PPs | FREE DISTRIBUTION |
|---------|----------|-----------|----------|-------------------|
|---------|----------|-----------|----------|-------------------|

|        |        |     |     |                    |
|--------|--------|-----|-----|--------------------|
| hdisk0 | active | 39  | 39  | 08..08..07..08..08 |
| hdisk1 | active | 273 | 228 | 54..21..44..54..55 |

```
lspv -l hdiks0
```

```
0516-306 : Unable to find hdiks0 in the Device Configuration Database.
```

```
lspv -l hdisk0
```

```
lspv -l hdisk1
```

```
hdisk1:
```

| LV NAME  | LPs | PPs | DISTRIBUTION       | MOUNT POINT |
|----------|-----|-----|--------------------|-------------|
| mydump1v | 32  | 32  | 00..32..00..00..00 | N/A         |
| hd4      | 1   | 1   | 00..00..01..00..00 | /           |
| hd2      | 4   | 4   | 00..00..04..00..00 | /usr        |
| hd6      | 2   | 2   | 00..02..00..00..00 | N/A         |
| hd8      | 1   | 1   | 00..00..01..00..00 | N/A         |
| hd1      | 1   | 1   | 00..00..01..00..00 | /home       |
| hd10opt  | 1   | 1   | 00..00..01..00..00 | /opt        |
| hd9var   | 1   | 1   | 00..00..01..00..00 | /var        |
| hd3      | 1   | 1   | 00..00..01..00..00 | /tmp        |
| hd5      | 1   | 1   | 01..00..00..00..00 | N/A         |

```
mklvcopy hd6 2 hdisk0
```

```
lsvg -l rootvg
```

```
rootvg:
```

| LV NAME  | TYPE    | LPs | PPs | PVs | LV STATE     | MOUNT POINT |
|----------|---------|-----|-----|-----|--------------|-------------|
| hd5      | boot    | 1   | 1   | 1   | closed/syncd | N/A         |
| hd6      | paging  | 2   | 4   | 2   | open/stale   | N/A         |
| hd8      | jfs2log | 1   | 1   | 1   | open/syncd   | N/A         |
| hd4      | jfs2    | 1   | 1   | 1   | open/syncd   | /           |
| hd2      | jfs2    | 4   | 4   | 1   | open/syncd   | /usr        |
| hd9var   | jfs2    | 1   | 1   | 1   | open/syncd   | /var        |
| hd3      | jfs2    | 1   | 1   | 1   | open/syncd   | /tmp        |
| hd1      | jfs2    | 1   | 1   | 1   | open/syncd   | /home       |
| hd10opt  | jfs2    | 1   | 1   | 1   | open/syncd   | /opt        |
| mydump1v | dump    | 32  | 32  | 1   | open/syncd   | N/A         |

```
mklvcopy <lvname><noofcopies><desitnationdisk>&^C
```

```
syncvg -l <lvname><vgname> --> To synchronize the particular LV.^C
```

```
syncvg -l hd6 rootvg
```

0516-306 getlvodm: Unable to find rootvg in the Device Configuration Database.

0516-934 /usr/sbin/syncvg: Unable to synchronize logical volume rootvg.

```
which syncvg
/usr/sbin/syncvg
lsvg -l rootvg
```

rootvg:

| LV NAME  | TYPE    | LPs | PPs | PVs | LV STATE     | MOUNT POINT |
|----------|---------|-----|-----|-----|--------------|-------------|
| hd5      | boot    | 1   | 1   | 1   | closed/syncd | N/A         |
| hd6      | paging  | 2   | 4   | 2   | open/syncd   | N/A         |
| hd8      | jfs2log | 1   | 1   | 1   | open/syncd   | N/A         |
| hd4      | jfs2    | 1   | 1   | 1   | open/syncd   | /           |
| hd2      | jfs2    | 4   | 4   | 1   | open/syncd   | /usr        |
| hd9var   | jfs2    | 1   | 1   | 1   | open/syncd   | /var        |
| hd3      | jfs2    | 1   | 1   | 1   | open/syncd   | /tmp        |
| hd1      | jfs2    | 1   | 1   | 1   | open/syncd   | /home       |
| hd10opt  | jfs2    | 1   | 1   | 1   | open/syncd   | /opt        |
| mydumplv | dump    | 32  | 32  | 1   | open/syncd   | N/A         |

```
mklvcopy -k hd2 hdisk0
```

0516-1122 : Invalid number of copies entered, must be 2 or 3.

0516-842 : Unable to make logical partition copies for logical volume.

```
mklvcopy -k hd2 2 hdisk0
```

```
lsvg -l rootvg
```

rootvg:

| LV NAME  | TYPE    | LPs | PPs | PVs | LV STATE     | MOUNT POINT |
|----------|---------|-----|-----|-----|--------------|-------------|
| hd5      | boot    | 1   | 1   | 1   | closed/syncd | N/A         |
| hd6      | paging  | 2   | 4   | 2   | open/syncd   | N/A         |
| hd8      | jfs2log | 1   | 1   | 1   | open/syncd   | N/A         |
| hd4      | jfs2    | 1   | 1   | 1   | open/syncd   | /           |
| hd2      | jfs2    | 4   | 8   | 2   | open/syncd   | /usr        |
| hd9var   | jfs2    | 1   | 1   | 1   | open/syncd   | /var        |
| hd3      | jfs2    | 1   | 1   | 1   | open/syncd   | /tmp        |
| hd1      | jfs2    | 1   | 1   | 1   | open/syncd   | /home       |
| hd10opt  | jfs2    | 1   | 1   | 1   | open/syncd   | /opt        |
| mydumplv | dump    | 32  | 32  | 1   | open/syncd   | N/A         |

# ===== **SYNCHRONIZATION:**

By using synchronization, we will fix stale partitions problems in VGs, PVs, LVs or FSs.

**Syn:** #syncvg -v <vgname> --> To synchronize the VG,  
#syncvg -p <pvname> --> To synchronize the PV  
#syncvg -l <lvname><pv or VG> --> To synchronize the LV or FS.

**Note:** sometimes Manually syncvg command will not fix the stale partitions, this scenario we need set auto sync=yes in VG or PV then reboot the system it may fix,

If auto sync not fixed stale partitions doesn't mean may be hdisk failed (or) disk having problem.

If hdisk is failed we need replace the disk.

```
lsvg rootvg
VOLUME GROUP: rootvg VG IDENTIFIER:
00cb9dbe00004c000000015618487caa
VG STATE: active PP SIZE: 256 megabyte(s)
VG PERMISSION: read/write TOTAL PPs: 312 (79872
megabytes)
MAX LVs: 256 FREE PPs: 229 (58624 megabytes)
LVs: 10 USED PPs: 83 (21248 megabytes)
OPEN LVs: 9 QUORUM: 2
TOTAL PVs: 2 VG DESCRIPTORS: 3
STALE PVs: 1 STALE PPs: 32
ACTIVE PVs: 2 AUTO ON: yes
MAX PPs per VG: 32512
MAX PPs per PV: 1016 MAX PVs: 32
LTG size (Dynamic): 256 kilobyte(s) AUTO SYNC: no
HOT SPARE: no BB POLICY: relocatable
```

```
lsvg -p rootvg
rootvg:
PV_NAME PV STATE TOTAL PPs FREE PPs FREE
DISTRIBUTION
hdisk0 active 39 1 00..00..00..00..01
hdisk1 active 273 228 54..21..44..54..55
lspv hdisk0 |grep -i stale
STALE PARTITIONS: 32 ALLOCATABLE: yes
lspv hdisk1 |grep -i stale
```

```
lsvg rootvg
VOLUME GROUP: rootvg VG IDENTIFIER:
00cb9dbe00004c000000015618487caa
VG STATE: active PP SIZE: 256 megabyte(s)
VG PERMISSION: read/write TOTAL PPs: 312 (79872
megabytes)
MAX LVs: 256 FREE PPs: 229 (58624 megabytes)
LVs: 10 USED PPs: 83 (21248 megabytes)
OPEN LVs: 9 QUORUM: 2
TOTAL PVs: 2 VG DESCRIPTORS: 3
STALE PVs: 0 STALE PPs: 0
```

```

ACTIVE PVs: 2 AUTO ON: yes
MAX PPs per VG: 32512
MAX PPs per PV: 1016 MAX PVs: 32
LTG size (Dynamic): 256 kilobyte(s) AUTO SYNC: no
HOT SPARE: no BB POLICY: relocatable

chvg -sy <vgname>
chvg -sy rootvg
lsvg rootvg
VOLUME GROUP: rootvg VG IDENTIFIER:
00cb9dbe00004c000000015618487caa
VG STATE: active PP SIZE: 256 megabyte(s)
VG PERMISSION: read/write TOTAL PPs: 312 (79872
megabytes)
MAX LVs: 256 FREE PPs: 229 (58624 megabytes)
LVs: 10 USED PPs: 83 (21248 megabytes)
OPEN LVs: 9 QUORUM: 2
TOTAL PVs: 2 VG DESCRIPTORS: 3
STALE PVs: 0 STALE PPs: 0
ACTIVE PVs: 2 AUTO ON: yes
MAX PPs per VG: 32512
MAX PPs per PV: 1016 MAX PVs: 32
LTG size (Dynamic): 256 kilobyte(s) AUTO SYNC: yes
HOT SPARE: no BB POLICY: relocatable

chvg -sn rootvg
lsvg rootvg
VOLUME GROUP: rootvg VG IDENTIFIER:
00cb9dbe00004c000000015618487caa
VG STATE: active PP SIZE: 256 megabyte(s)
VG PERMISSION: read/write TOTAL PPs: 312 (79872
megabytes)
MAX LVs: 256 FREE PPs: 229 (58624 megabytes)
LVs: 10 USED PPs: 83 (21248 megabytes)
OPEN LVs: 9 QUORUM: 2
TOTAL PVs: 2 VG DESCRIPTORS: 3
STALE PVs: 0 STALE PPs: 0
ACTIVE PVs: 2 AUTO ON: yes
MAX PPs per VG: 32512
MAX PPs per PV: 1016 MAX PVs: 32
LTG size (Dynamic): 256 kilobyte(s) AUTO SYNC: no
HOT SPARE: no BB POLICY: relocatable

lspv hdisk0
PHYSICAL VOLUME: hdisk0 VOLUME GROUP: rootvg
PV IDENTIFIER: 00cb9dbe18a78c3d VG IDENTIFIER
00cb9dbe00004c000000015618487caa
PV STATE: active

```

STALE PARTITIONS: 0                      ALLOCATABLE: yes  
PP SIZE: 256 megabyte(s)              LOGICAL VOLUMES: 3  
TOTAL PPs: 39 (9984 megabytes)      VG DESCRIPTORS: 1  
FREE PPs: 1 (256 megabytes)          HOT SPARE: no  
USED PPs: 38 (9728 megabytes)      MAX REQUEST: 256 kilobytes  
FREE DISTRIBUTION: 00..00..00..00..01  
USED DISTRIBUTION: 08..08..07..08..07

### **rmlvcopy:**

How to unmirror particular LV, (or) To remove the mirrored copy in LV or FS

**Syn:** #rmlvcopy <lvname><noofcopies><pvname>

### **Disk availability/un availability**

#chpv -vr <pvname> -> to move PV state removed/unavailability state,  
#chpv -va <pvname> -> To move pv state to availability from removed state.

**Note:** 1) we need umount the FSs in PV before moving to removed state,  
2) Once PV moved to removed state, not possible to use the free space/PPs in PV.

### **VGDA/VGSA/LVCB and Quorum**

#### **VGDA:**

VGDA stands for Volume group descriptor area, VGDA contains the total VG information, such as VG Name, PV names, PPs, LVs and LPs information.

#readvgda <pvname> To display the VGDA information

**Note:** Suppose in VG (sapvg) if you have 3 pvs, 1st pv having 2 VGDA's, next two PVs having 1 VGDA.

Suppose if VG contains =>4 PVs, each PV having only one VGDA,

#### **VGSA:**

Stands for Volume group status area, its contains the PVs and PPs information.

#### **LVCB:**

Stands for Logical Volume Control Block, Its contains the LVs and LPs information.

#getlvcb -AT <lvname> -> To see the LVCB information.

## **ROOTVG / OS VG Mirroring:**

### **To Mirror Rootvg:**

- 1) Take/Add new disk and scan it.  
#cfgmgr
- 2) Add the new disk to rootvg  
#extendvg rootvg <newdiskname>
- 3) Mirror the rootvg with New Disk  
#mirrorvg rootvg <newdisk>
- 4) Verify the mirroring status  
#lsvg -l rootvg

### **Note:**

- 1) In rootvg dumplv (sysdump) will not mirrored, if we need mirror dumplv we can mirror by using #mklvcopy command.

#mklvcopy -k <dumplv> <noof copy> <newdisk>  
-k for to synchronize the stale partitions automaticlly.

- 2) In rootvg mirroring we have run below two addtional steps after mirrored.

- 5) Create BLV (Boot logical volume)/Boot record in new disk.

#bosboot -ad /dev/newdisk

- 6) Set the bootlist order

#bootlist -om normal hdiskX hdiskX

- 7) Verify/display bootlist order

#bootlist -om normal

#To unmirror rootvg

Ex: hdisk4

- 1) Break the mirroring in hdisk

#unmirrorvg rootvg hdisk4

- 2) Verify the mirroring

#lsvg -l rootvg

- 3) Remove the disk from rootvg

#reducevg rootvg hdisk4

- 4) Remove/clear the BLV from the disk

#chpv -c hdisk4

- 5) Remove the unmirrored disk from the bootlist order.

#bootlist -om normal hdisk0

- 6) Display the bootlist order

#bootlist -om normal

hdisk0 blv=hd5

Q1) In rootvg we have two disks (hdisk1, hdisk2) with mirrored, Hdisk1 is failed in rootvg, Now how to replace the failed disk (hdisk1) with new disk (hdisk5) in rootvg.

```
rootvg
hdisk1 (failed)
hdisk2
hdisk5 none
```

- 1) Break the mirror in failed disk
- 2) Clear BLV in failed disk
- 3) Remove the failed disk from the rootvg
- 4) Remove the disk from the OS/server
- 5) Remove the failed disk from the bootlist order
- 6) Add new disk and scan if not available new disk in the system.
- 7) Add the new disk to rootvg
- 8) Mirror rootvg with new disk
- 9) Verify the mirroring
- 10) Create BLV in new disk
- 11) Create bootlist order with new disk
- 12) Verify the bootlist order

**Answer:**

- 1) Break the mirror on failed disk (hdisk1)  
#unmirrorvg <vgname><pvname>
  - 2) Verify the mirroring  
#lsvg -l <vgname>
  - 3) Clear the BLV on failed disk and remove the failed disk from bootlist  
#chpv -c <pvname>  
#bootlist -om normal hdisk2  
#bootlist -om normal  
hdisk2
  - 4) remove the PV from rootvg  
#reducevg <vgname><pvname>
- Note:** if dumpdev presented in failed disk move into existing OS disk (hdisk2)
- 5) remove the disk from OS/ODM  
#rmdev -dl hdisk1
  - 6) remove the disk from server Physically.
  - 7) Add the new disk to the server physically and scan it  
#cfgmgr
  - 8) Add the disk into rootvg  
#extendvg rootvg <newpvname>
  - 9) Mirror the rootvg with new disk (hdisk5)  
#mirrorvg <rootvg><newpvname>

10) Verify the mirroring  
#lsvg -l <vgname>  
11) Create the BLV on new disk  
#bosboot -ad /dev/hdisk5  
12) Create bootlist order  
#bootlist -om normal hdisk2 hdisk5  
13) Verify the bootlist order  
#bootlist -om normal  
hdisk2  
hdisk5

Q2) In rootvg we have one disk (hdisk0) , hdisk0 is failed/performance not good/ I/O errors, Now how to replace the failed disk hdisk0 with new disk(hdisk1), Do we need server downtime for this?

1) Take the new disk and scan it  
#cfgmgr  
2) Add the new disk to the rootvg  
#extendvg rootvg hdisk1  
3) mirror rootvg with new disk  
#mirrorvg rootvg hdisk1  
4) Verify mirroring,  
#lsvg -l rootvg  
5) Create BLV on new disk  
#bosboot -ad /dev/hisk1  
6) Break mirror on failed disk  
#unmirrorvg rootvg hdisk0  
7) Clear the BLV on failed disk  
#chpv -c hdisk0  
8) Remove the failed disk from rootvg  
#reducevg rootvg hdisk0  
9) Remove the disk from server physically  
10) Finally create bootlist order with new disk  
#bootlist -om normal hdisk1

## **QUORUM:**

By using quorum, we can verify the LVM status in VGs, The quorum status/state depends on the VGDA information, if VGDA information => 51% in VG quorum will be enabled, if VGDA information <51% the quorum status will be disabled.

#chvg -Qy <vgname> --> To enable the quorum  
#chvg -Qn <vgname> --> To disable the quorum

## **LTG size:**

Stands for Logical Track Group, By using LTG we can define/see how much data can write/read an I/O operation in VGs/PV.

# lquerypv -M hdisk0 -> To see the LTG size

#chvg -L <LTG size><VGname> -> To change the LTG size of VG,

**Note:** In PVs Max request=LTG size.

### **Rename of LV/FS/VG:**

#### **Rename LV:**

**SYN:**#chlv -n <newname><oldname>

EX: To rename saplv with new name called "oralv"

#chlv -n oralv saplv

#### **Rename of File system:**

#chfs -m <newname><oldname>

EX: To rename /sapfs file system with new name /javafs

#chfs -m /javafs /sapfs

**Note:** To rename mounted file system we have umount/remount to effect on df -gt output.

#### **Rename VG:**

To rename VG we have export VG and import the VG, while importing VG we can rename the Volume Group.

Ex: To rename sapvg VG as javavg follow the below steps:

#### **Export VG:**

##### **Steps:**

1) umount all file systems in the VG

2) varyoffvg

3) exportvg by using below command

**SYN:**#exportvg <VGNAME>

#exportvg sapvg --> To exportvg

**Note:** Once VG exported, what are the file systems are available in the VG, the file systems entries will remove from /etc/file systems configuration file.

## **Importvg VG:**

```
#importvg -y <VGNAME><PVNAME>
#importvg -y javavg hdisk7
```

## **LVM Structure 2:**

1) Take disk & SCAN

```
#cfgmgr
```

2) Create VG

```
#mkvg -y sapvg -s 64 hdiskX hdisk2
```

3) Create FS

```
SYN: #crfs -v <typeofFS> -g <vgname> -a size=<FS size> -A yes -m
</FSNAME>
```

```
EX: #crfs -v jfs2 -g sapvg -a size=1G -A yes -m /sapfs
```

4) Mount FS

```
#mount /sapfs
```

5) Enter FS

```
#cd /sapfs
```

## **Note:**

### **T-Factor:**

T-Factor is one of the technique in LVM, if we need to increase the Max PPs per PV value we can apply the T-Factor formula.

Once applied T-Factor, it will increase the Max PPs per PV and it will reduce the Max PVs for the VG

```
SYN: #chvg -t <T-factor value> <vgname>
```

```
#chvg -t 2 sapvg
```

### **Before:**

Maxs PPs per PV= 1016

Max PVs = 32

### **After applied T-Factor:**

Maxs PPs per PV= 2032

Max PVs = 16

## **File Systems**

File systems is convenient way to store the data in Unix operating systems, The File systems contains the files, directories. We can access the data from PV with help of logical volume and file system.

### **To create File system.**

There are two ways to create the File system

- 1) Single step
- 2) Double step

#### **Single Step FS creation:**

1) `crfs -v <typeoffS> -g <vgname> -a size=<FSsize> -A yes -m </FSname>`

#### **Double step FS Creation:**

2) `mklv -y <lvname> -t <tyoflv><vgname><nooflps><pvname>`

```
#crfs -v <typoffs> -d /dev/lvname -A yes -m </Fsname>
```

### **File system Structure:**

The file system structure is Super block, Allocation group, Inodes, Data Blocks, Fragments and device log.

### **Super Block:**

Super block contains the information about file system, such as file system name, when file system created date & time, used/free Inodes, total size of the file system in 512 byte blocks. list of free data blocks,

**Note:** Super block will be available in 1st block of the file system and same copy is available on 31st block, if 1st super block is corrupted, we can recovery from 31st block by using #dd command.

To recover the super block

**Syn:** #dd count=1 bs=4k skip=31 seek=1 if=/dev/fslvname of=/dev/fslvname

Ex: #dd count=1 bs=4k skip=31 seek=1 if=/dev/lv00 of=/dev/lv00

**Note:** if rootvg file systems super block is corrupted dd command we can execute only on maintenance mode.

### **Allocation group:**

Allocation group contains the inodes and data blocks.

### **Inode:**

Inode block contains the information about file, such as file owner, created date, last access date, modified data and inode number.

```
istat <filename>
```

```
Inode 1339 on device 10/4 File
```

```
Protection: rw-r--r--
```

```
Owner: 0(root) Group: 0(system)
```

```
Link count: 1 Length 0 bytes
```

```
Last updated: Fri Jul 29 11:24:14 CDT 2016
```

```
Last modified: Fri Jul 29 11:24:14 CDT 2016
```

```
Last accessed: Fri Jul 29 11:24:14 CDT 2016
```

```
ls -li aix
```

```
1339 -rw-r--r-- 1 root system 0 Jul 29 11:24 aix
```

#istat <filename> --> To see the inode information of file.

### **Data blocks:**

Data blocks contains actual data of users.

**Fragments:**

If any free blocks left between data blocks, those blocks we can call Fragments.

**Device Log:**

In file system read/write or I/O operations will go via Device log if device log not available, we cannot access the file system.

**Note:** Hard disk space divided as 512 byte blocks in Aix operating system to store the date.

Two 512 byte blocks=1KB

**File systems Types:**

There are 4 types of file systems

- 1) Journal File system (JFS)
- 2) Enhanced Journal File system (JFS2)
- 3) Network File systems (NFS)
- 4) CD-ROM File system (CFS)

Difference Between JFS & JFS2 File systems.

| <b>JFS</b>                                                              | <b>JFS2</b>                                                 |
|-------------------------------------------------------------------------|-------------------------------------------------------------|
| JFS Stands for Journal File system                                      | JFS2 stands for Enhanced Journal File system                |
| it is small file system,                                                | It is large file system                                     |
| We can increase JFS file system but we can not decrease JFS file system | We can Increase as well as we can decrease JFS2 File system |
| Its having Fixed Inodes                                                 | Its having dynamic Inodes                                   |
| JFS max File system size is 1TB                                         | JFS2 max File system size is 4PB                            |
| Max File size is 64GB                                                   | Max File size is 4PB                                        |
| Compression is available                                                | Compression not available                                   |
| Inode size is 128 bytes                                                 | Inode size is 512 bytes                                     |

Difference Between single step and double step file system creation.

| <b>Single step</b>                                                          | <b>Double step</b>                                       |
|-----------------------------------------------------------------------------|----------------------------------------------------------|
| at a time we can create LV and FS                                           | Firstwe need to create LV then we will create FS         |
| LV name decide by the OS, if JFS (lv00, lv01), if JFS2 (fslv00, fslv01 etc) | LV name will decide by US                                |
| File system size we can specify directly in KB, MB, GBs                     | File system size we can define in LPs, while creating LV |

|                                                                                                        |                                              |
|--------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Intra policy not possible while creating FS but after created FS if we need we can enable Intra policy | Intra policy we can define while creating LV |
|--------------------------------------------------------------------------------------------------------|----------------------------------------------|

## **/etc/file systems:**

It is one of the important configuration file for the file systems. Once created file system by using `crfs` command, one stanza will be creating for each file system under `/etc/file systems` configuration file.

EX: `/opt` file system STANZA format in `/etc/filesystems`

`/opt:`

```
dev = /dev/hd10opt
vfs = jfs2
mount = true
log = /dev/hd8
```

`#lsfs ->` To display the total file systems in the system (mounted and unmounted)

`#lsfs </fsname> ->` To display the information about specific FS

`#lsfs -a ->` To display the total file systems in the system (mounted and unmounted) also with extra attributes, (block size, inline log etc)

## **df and mount Differences**

`#df ->` command will display the mounted file systems only also, we can see total size of file system, used, free space and %usage of the File systems.

## **mount:**

`#mount ->` will display the mounted file systems only, also we can see permissions, log device, type of FS and date & time of creation FS etc.

`#mount /fsname ->` To mount the file system

`#mount -a ->` To mount all file systems

## **Umount:**

`#umount /fsname -->` To unmount the file system

**Note:** if any user entered into the file system, the file system will not unmount.

`#fuser -u /fsname -->` To find the users who are accessing the file system.

`#fuser -uk /fsanme -->` To kill the users sessions who are accessing the file systems.

`#lsfs ->` To list the all file systems (mounted & unmounted)

`#lsfs /fsname -->` To list the specific file system

`#lsfs -q -->` To list all file system with extra attributes

#lsfs -q /fsname --> to list specific file system with extra attributes (Ex: inline log size, quota etc).

**FSCK:** By using fsck command we can check the file systems consistency/status/condition. also by using fsck command sometime we can fix the file systems problems (Ex: read only errors, fs corrupted errors etc.)

**Note:** before running fsck command we have to unmount the file systems. if we are not unmounting file system we may lose the file system data.

To run fsck command on /sapfs file system

```
#umount /sapfs
```

```
#fsck -y /sapfs -> To check /sapfs file system consistency.
```

**Note:** if we need to run fsck command on rootvg file systems we have to enter into the maintenance mode and umount the file systems then run fsck command.

```
pwd
/
```

```
cd /testfs
pwd
/testfs
```

```
umount /testfs
umount: 0506-349 Cannot unmount /dev/fslv00: The requested resource is busy.
```

```
fuser -u /testfs
/testfs: 274462c(root)
pwd
/testfs
cd
umount /testfs
```

```
fuser -cuk </fsname> ^C
```

```
mount -a -> to mount the all file systems at a time ^C
```

```
umount -a -> to unmount all file systems. ^C
```

```
lsfs
```

| Name        | Nodename | Mount Pt | VFS             | Size | Options | Auto |
|-------------|----------|----------|-----------------|------|---------|------|
| Accounting  |          |          |                 |      |         |      |
| /dev/hd4    | --       | /        | jfs2 262144 --  | yes  | no      |      |
| /dev/hd1    | --       | /home    | jfs2 262144 --  | yes  | no      |      |
| /dev/hd2    | --       | /usr     | jfs2 2097152 -- | yes  | no      |      |
| /dev/hd9var | --       | /var     | jfs2 262144 --  | yes  | no      |      |
| /dev/hd3    | --       | /tmp     | jfs2 262144 --  | yes  | no      |      |

```

/proc -- /proc procfs -- -- yes no
/dev/hd10opt -- /opt jfs2 262144 -- yes no
/dev/fslv00 -- /testfs jfs2 262144 -- yes no

```

```
df -gt
```

```

Filesystem GB blocks Used Free %Used Mounted on
/dev/hd4 0.12 0.01 0.11 9% /
/dev/hd2 1.00 0.55 0.45 55% /usr
/dev/hd9var 0.12 0.01 0.12 6% /var
/dev/hd3 0.12 0.00 0.12 1% /tmp
/dev/hd1 0.12 0.00 0.12 1% /home
/proc - - - - /proc
/dev/hd10opt 0.12 0.04 0.09 30% /opt

```

```
lsvg -l rootvg |grep -i /testfs
```

```
fslv00 jfs2 1 1 1 closed/syncd /testfs
```

```
mount
```

```

node mounted mounted over vfs date options

/dev/hd4 / jfs2 Jul 30 11:03 rw,log=/dev/hd8
/dev/hd2 /usr jfs2 Jul 30 11:03 rw,log=/dev/hd8
/dev/hd9var /var jfs2 Jul 30 11:04 rw,log=/dev/hd8
/dev/hd3 /tmp jfs2 Jul 30 11:04 rw,log=/dev/hd8
/dev/hd1 /home jfs2 Jul 30 11:04 rw,log=/dev/hd8
/proc /proc procfs Jul 30 11:04 rw
/dev/hd10opt /opt jfs2 Jul 30 11:04 rw,log=/dev/hd8

```

```
lsfs
```

```

Name Nodename Mount Pt VFS Size Options Auto
Accounting
/dev/hd4 -- / jfs2 262144 -- yes no
/dev/hd1 -- /home jfs2 262144 -- yes no
/dev/hd2 -- /usr jfs2 2097152 -- yes no
/dev/hd9var -- /var jfs2 262144 -- yes no
/dev/hd3 -- /tmp jfs2 262144 -- yes no
/proc -- /proc procfs -- -- yes no
/dev/hd10opt -- /opt jfs2 262144 -- yes no
/dev/fslv00 -- /testfs jfs2 262144 -- yes no

```

```
lsfs /home
```

```

Name Nodename Mount Pt VFS Size Options Auto
Accounting
/dev/hd1 -- /home jfs2 262144 -- yes no

```

```
lsfs -a
```

```

Name Nodename Mount Pt VFS Size Options Auto
Accounting

```

|              |    |         |        |         |    |     |    |
|--------------|----|---------|--------|---------|----|-----|----|
| /dev/hd4     | -- | /       | jfs2   | 262144  | -- | yes | no |
| /dev/hd1     | -- | /home   | jfs2   | 262144  | -- | yes | no |
| /dev/hd2     | -- | /usr    | jfs2   | 2097152 | -- | yes | no |
| /dev/hd9var  | -- | /var    | jfs2   | 262144  | -- | yes | no |
| /dev/hd3     | -- | /tmp    | jfs2   | 262144  | -- | yes | no |
| /proc        | -- | /proc   | procfs | --      | -- | yes | no |
| /dev/hd10opt | -- | /opt    | jfs2   | 262144  | -- | yes | no |
| /dev/fslv00  | -- | /testfs | jfs2   | 262144  | -- | yes | no |

# lsfs -a

| Name         | Nodename | Mount Pt | VFS    | Size    | Options | Auto   |
|--------------|----------|----------|--------|---------|---------|--------|
| Accounting   |          |          |        |         |         |        |
| /dev/hd4     | --       | /        | jfs2   | 262144  | --      | yes no |
| /dev/hd1     | --       | /home    | jfs2   | 262144  | --      | yes no |
| /dev/hd2     | --       | /usr     | jfs2   | 2097152 | --      | yes no |
| /dev/hd9var  | --       | /var     | jfs2   | 262144  | --      | yes no |
| /dev/hd3     | --       | /tmp     | jfs2   | 262144  | --      | yes no |
| /proc        | --       | /proc    | procfs | --      | --      | yes no |
| /dev/hd10opt | --       | /opt     | jfs2   | 262144  | --      | yes no |
| /dev/fslv00  | --       | /testfs  | jfs2   | 262144  | --      | yes no |

# lsfs -q

| Name                                                                                                                                                       | Nodename | Mount Pt | VFS    | Size    | Options | Auto   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|--------|---------|---------|--------|
| Accounting                                                                                                                                                 |          |          |        |         |         |        |
| /dev/hd4                                                                                                                                                   | --       | /        | jfs2   | 262144  | --      | yes no |
| (lv size: 262144, fs size: 262144, block size: 4096, sparse files: yes, inline log: no, inline log size: 0, EAformat: v1, Quota: no, DMAPi: no, VIX: no)   |          |          |        |         |         |        |
| /dev/hd1                                                                                                                                                   | --       | /home    | jfs2   | 262144  | --      | yes no |
| (lv size: 262144, fs size: 262144, block size: 4096, sparse files: yes, inline log: no, inline log size: 0, EAformat: v1, Quota: no, DMAPi: no, VIX: no)   |          |          |        |         |         |        |
| /dev/hd2                                                                                                                                                   | --       | /usr     | jfs2   | 2097152 | --      | yes no |
| (lv size: 2097152, fs size: 2097152, block size: 4096, sparse files: yes, inline log: no, inline log size: 0, EAformat: v1, Quota: no, DMAPi: no, VIX: no) |          |          |        |         |         |        |
| /dev/hd9var                                                                                                                                                | --       | /var     | jfs2   | 262144  | --      | yes no |
| (lv size: 262144, fs size: 262144, block size: 4096, sparse files: yes, inline log: no, inline log size: 0, EAformat: v1, Quota: no, DMAPi: no, VIX: no)   |          |          |        |         |         |        |
| /dev/hd3                                                                                                                                                   | --       | /tmp     | jfs2   | 262144  | --      | yes no |
| (lv size: 262144, fs size: 262144, block size: 4096, sparse files: yes, inline log: no, inline log size: 0, EAformat: v1, Quota: no, DMAPi: no, VIX: no)   |          |          |        |         |         |        |
| /proc                                                                                                                                                      | --       | /proc    | procfs | --      | --      | yes no |
| /dev/hd10opt                                                                                                                                               | --       | /opt     | jfs2   | 262144  | --      | yes no |
| (lv size: 262144, fs size: 262144, block size: 4096, sparse files: yes, inline log: no, inline log size: 0, EAformat: v1, Quota: no, DMAPi: no, VIX: no)   |          |          |        |         |         |        |

```
/dev/fslv00 -- /testfs jfs2 262144 -- yes no
(lv size: 262144, fs size: 262144, block size: 4096, sparse files: yes, inline
log: no, inline log size: 0, EAformat: v1, Quota: no, DMAPI: no, VIX: no)
```

```
lsfs -q /testfs
```

```
Name Nodename Mount Pt VFS Size Options Auto
Accounting
/dev/fslv00 -- /testfs jfs2 262144 -- yes no
(lv size: 262144, fs size: 262144, block size: 4096, sparse files: yes, inline
log: no, inline log size: 0, EAformat: v1, Quota: no, DMAPI: no, VIX: no)
```

```
df -m
```

```
Filesystem MB blocks Free %Used Iused %Iused Mounted on
/dev/hd4 128.00 117.35 9% 1486 6% /
/dev/hd2 1024.00 465.48 55% 17739 15% /usr
/dev/hd9var 128.00 121.33 6% 319 2% /var
/dev/hd3 128.00 127.61 1% 11 1% /tmp
/dev/hd1 128.00 127.64 1% 5 1% /home
/proc - - - - - /proc
/dev/hd10opt 128.00 90.51 30% 700 4% /opt
```

```
df -g
```

```
Filesystem GB blocks Free %Used Iused %Iused Mounted on
/dev/hd4 0.12 0.11 9% 1486 6% /
/dev/hd2 1.00 0.45 55% 17739 15% /usr
/dev/hd9var 0.12 0.12 6% 319 2% /var
/dev/hd3 0.12 0.12 1% 11 1% /tmp
/dev/hd1 0.12 0.12 1% 5 1% /home
/proc - - - - - /proc
/dev/hd10opt 0.12 0.09 30% 700 4% /opt
```

```
df -k
```

```
Filesystem 1024-blocks Free %Used Iused %Iused Mounted on
/dev/hd4 131072 120164 9% 1486 6% /
/dev/hd2 1048576 476656 55% 17739 15% /usr
/dev/hd9var 131072 124244 6% 319 2% /var
/dev/hd3 131072 130676 1% 11 1% /tmp
/dev/hd1 131072 130708 1% 5 1% /home
/proc - - - - - /proc
/dev/hd10opt 131072 92680 30% 700 4% /opt
```

```
df
```

```
Filesystem 512-blocks Free %Used Iused %Iused Mounted on
/dev/hd4 262144 240328 9% 1486 6% /
/dev/hd2 2097152 953312 55% 17739 15% /usr
```

```

/dev/hd9var 262144 248488 6% 319 2% /var
/dev/hd3 262144 261352 1% 11 1% /tmp
/dev/hd1 262144 261416 1% 5 1% /home
/proc - - - - - /proc
/dev/hd10opt 262144 185360 30% 700 4% /opt

```

# mount

| node         | mounted | mounted over | vfs          | date            | options |
|--------------|---------|--------------|--------------|-----------------|---------|
| /dev/hd4     | /       | jfs2         | Jul 30 11:03 | rw,log=/dev/hd8 |         |
| /dev/hd2     | /usr    | jfs2         | Jul 30 11:03 | rw,log=/dev/hd8 |         |
| /dev/hd9var  | /var    | jfs2         | Jul 30 11:04 | rw,log=/dev/hd8 |         |
| /dev/hd3     | /tmp    | jfs2         | Jul 30 11:04 | rw,log=/dev/hd8 |         |
| /dev/hd1     | /home   | jfs2         | Jul 30 11:04 | rw,log=/dev/hd8 |         |
| /proc        | /proc   | procfs       | Jul 30 11:04 | rw              |         |
| /dev/hd10opt | /opt    | jfs2         | Jul 30 11:04 | rw,log=/dev/hd8 |         |

## File systems Housekeeping or How to manage the file systems if they have reached threshold limit as per standards.

In office environment we have two mode of FSs

1) OS file systems (/ , /var, /home, /opt, /usr and /tmp) -85%

2) Users File systems (Ex: /sapfs, /javafs, /dbfs, /orafs etc) -90%

if any OS related file systems reached threshold limit we have to take the action plan.

if any user related file systems reached threshold value limit we have to inform to the respected team.

## OS FSs housekeeping:

### /home:

# df -gt

| Filesystem   | GB blocks | Used | Free %Used | Mounted on |
|--------------|-----------|------|------------|------------|
| /dev/hd4     | 0.12      | 0.01 | 0.11 9%    | /          |
| /dev/hd2     | 1.00      | 0.55 | 0.45 55%   | /usr       |
| /dev/hd9var  | 0.12      | 0.01 | 0.12 6%    | /var       |
| /dev/hd3     | 0.12      | 0.00 | 0.12 1%    | /tmp       |
| /dev/hd1     | 0.12      | 0.00 | 0.12 1%    | /home      |
| /proc        | -         | -    | -          | /proc      |
| /dev/hd10opt | 0.12      | 0.04 | 0.09 30%   | /opt       |

# cd /home

# du -sm \* |sort -nr|head -20

1700.68 dougyork

170.93 rduadm

167.38 shawg

86.65 schotten

- 1) enter into the /home
- 2) find the more usage users under /home  
#du -sm \* |sort -nr| head
- 3) Inform to the respected users who are using more space  
**Note:** if users tell delete unwanted files then only we have to delete file/dirs (rm, rm -rf)
- 4) if all files/dirs are important, then final action will be need to increase the file system by using #chfs command.

```
ls
amar lost+found ram sai
guest meerza ravi venkat
du -m ram
0.00 ram
du -m *
0.00 amar
0.00 guest
0.00 lost+found
0.00 meerza
0.00 ram
0.00 ravi
0.00 sai
0.00 venkat
du -m <file/dir> ^C
df <fsame>
```

#### **/usr:**

- 1) enter /usr
- 2) find the more usage of files under /usr  
#du -sm \* |sort -nr|head
- 3) Find the applied file sets/software.
- Note: if applied softwares listed doesn't mean we have two version for the same software, 1 is old version, another one new version**  
if applied file sets listed, we can commit them if they are installed by before one month  
#installp -c all -> to commit the softwares. it will be remove the old version.
- 4) if there is no applied softwares, everything is important in /usr file system. Finally, we can increase the /usr file system.

#### **/dev/hd3 - /tmp:**

- 1) enter into /tmp
- 2) find the more usage file/dirs in /tmp
- 3) if data not important we can delete them.
- Note:** mostly under /tmp temporary data will available, we can remove them.
- 4) if data important, finally we can increase FS.

### **/dev/hd4 - / (root):**

1) enter into /

2) find the more usage file/dirs in /

**Note:** 1) once listed more usage dirs, verify in system, do we have any file system with same dir name in the system, if file system available we should not touch the directory.

2) mostly /etc/ directory configuration files may using more space,

Ex /etc/security/failedlogin, /etc/security/lastlog etc.

Ex: if you see failed login file is using more space.

1) take the back up into /tmp

#cp /etc/security/failedlogin /tmp/failedlogin\_07302016

2) remove the content of failed login file

#>/etc/security/failedlogin

3) gzip the backup file /tmp/failedlogin\_07302016

#gzip /tmp/failedlogin\_07302016

once zipped file will be, /tmp/failedlogin\_07302016.gz

4) Move to zipped file to original location

#mv /tmp/failedlogin\_07302016.gz /etc/security.

5) If everything is important in / file system, finally we can go the increase file system.

#gunzip <filename> -> to unzip the file

### **/dev/hd9var - /var:**

1) enter /var

2) find the more usage files/dir /var.

#du -sm \* |sort -nr|head

**Note:** mostly in /var system log files may using more space, we have to verify the which log file using more space by using du cmd.

Ex: errorlog, sulog, cronlog, nimlog etc.

3) if you see any log file using more space, take the below action.

a) take backup of log file any one of the FS (ex:/tmp)

b) remove the original log file content (>logfile)

c) gzip the backup logfile

d) move the zipped log file to original location.

4) if everything is important, finally we can increase the file system.

### **/dev/hd10opt - /opt:**

1) enter /opt

2) find more usage file/dir /opt

```
#du -sm * |sort -nr|head,
```

**Note:** /opt is optional file system, users can use this file system depends on their requirement. Ex: install software, keep temporary data, install applications etc.

4) if we see applicaion/database data we can inform to them,  
if we see unused/temporary data we can remove it.

5) if total data important we can increase the /opt file system.

### **Users Related FS house keeping:**

Ex: /sapfs, /javafs, /dbfs, /oraclefs, /apachefs etc.

In case /sapfs is 90% usage, need to take below action plan

1) enter into /sapfs #cd /sapfs

2) find the more usage file/dirs under /sapfs

```
#du -sm * |sort -nr|head
```

3) Find the owner of more usage files/dirs.

```
#ls -l <file/dir>
```

4) inform to the respected user to take action plan on /sapfs FS.

**Note:** 1) until confirmed by the users, we should not delete file/dirs on user related file system.

2) until confirmed by the respected users, we should not increase the user related FSs.

### **FSCK:** File system checking.

1) By using FSCK we can check/verify the file system consistency/status.

2)#fsck command will fix the file system problems sometimes.

**Note:** We need to unmount the file system before run fsck command.

if we run fsck command on mounted file systems, we may lost/corrupt the existing data in file system.

### **Problems:**

file/dirs will not create in FS ever we had 777 permissions.

sometimes file systems will go read-only mode.

sometimes file systems will not be accessible.

```
df -gt /testfs
```

| Filesystem  | GB blocks | Used | Free %Used | Mounted on |
|-------------|-----------|------|------------|------------|
| /dev/fslv00 | 0.12      | 0.00 | 0.12 1%    | /testfs    |

```
pwd
```

```
/testfs
```

```
cd
```

```
umount /testfs
```

```
fsck -y /testfs
```

The current volume is: /dev/fslv00

Primary superblock is valid.

\*\*\* Phase 1 - Initial inode scan  
\*\*\* Phase 2 - Process remaining directories  
\*\*\* Phase 3 - Process remaining files  
\*\*\* Phase 4 - Check and repair inode allocation map  
\*\*\* Phase 5 - Check and repair block allocation map

File system is clean.

**Note:** in rootvg file system if we need to run fsck command, we need to enter into the maintenance mode to run fsck command. take the shell before mount file systems in maintenance mode.

To remove the File system

- 1) umount the file system, if mounted
- 2) rmfs </fsname> remove the file system

**Note:** rmfs will not delete the file system's directory.

- 3) rmdev </fsdirname>

### **Device Log:**

#### **To Increase/Decrease the File system size:**

- 1) Find the VG of file system
- 2) Verify the free space of volume group, if free space available you can increase by using #chfs command. if free space not available follow the below steps.
  - a) Take the new/empty disk
  - b) Add to the VG
  - c) Increase the file system by using #chfs command.

#chfs -a size=+<sizeofFS></Fsname> --To Increase the File system

#chfs -a size=+1G /sapfs --> To increase the /sapfs file system with 1G.

#chfs -a size=-1G /sapfs --> To reduce the 1GB space from /sapfs file system.

Q) There is one file system (EX: /var), total size is 10G, Actually in the file system all files/dirs usages is 2G but when you run the #df -gt command the file system usage listing 100% what is problem how to fix it.

### **Device Log/Log Device:**

In file system read/write or I/O operations will go via Device log if device log not available, we cannot access the file systems data. one more important use of log device while doing I/O operations if system crashed/rebooted the mid data will stored on log device once server is up the data will move into the hard disk.

- 1) In VG if we create jfs file system one default log device (jfslog) will create.

- 2) In VG if we create jfs2 file system one default log device (jfs2log) device will create.
- 3) On each VG default log devices will be available
- 4) Log devices will create with one LP
- 5) In rootvg default log device is /dev/hd8
- 6) All jfs file systems will access jfslog device only, if dedicated log devices not created to file systems
- 7) All jfs2 file systems will access jfs2log device only, if dedicated log devices not created to file systems.
- 8) Log device nothing it is a logical volume

### **Dedicated Log Device:**

If we need to increase file system I/O operations performance we can create dedicated log device and assign to the file system which is required to high speed performance.

### **To create Dedicated log device & assigned to file system:**

```
#mklv -y <loglvname> -t <typeofloglv> <vgname> <nooflps> <pvname>
#mklv -y oralogdevicelv -t jfs2log sapvg 2 hdisk0 -> To create jfs2
dedicated log device lv (oralogdevicelv) with two lps
#mklv -y javalogdevicelv -t jfslog sapvg 2 hdisk0 -> To create jfs dedicated
log device lv (javalogdevicelv) with two lps
#logform /dev/loglvname --> To formate the log device.
```

**Note:** Once created dedicated log device we should formate the log device before assigning to the file system

EX: #logform /dev/oralogdevicelv --> To formate ora log device lv.

```
#chfs -a log=<loglvname> </fsname>
#chfs -a log=/dev/oralogdevicelv /orafs2 --> To assign the dedicated log
devices (oralogdevicelv) to /orafs2 file system.
```

Q) Create the /dev/saslv and assigned to /sasfs file system with dedicated log device (/dev/saslogdevlv) on sasvg

```
#mklv -y saslv -t jfs2 sasvg 10 hdisk3
#mklv -y saslogdevlv -t jfs2log sasvg 1 hdisk3
#logform /dev/saslogdevlv
#crfs -v jfs2 -d /dev/saslv -a logname=/dev/saslogdevlv -A yes -m /sasfs
```

Q) Create the /apachefs with dedicated log device (/dev/apacheloglv) on apachevg

```
#mklv -y apacheloglv -t jfs2 apachevg 1 hdisk5
#logform /dev/apacheloglv
```

```
#crfs -v jfs2 -g apachevg -a size=10G -a logname=/dev/apacheloglv -A yes
-m /apachefs
#mount /apachefs
```

### **INLINE LOG:**

INLINE log is one of the log device, INLINE log device will create on same LV, which logical volume is we are using to create file system.

Q) create /appfs with INLINE log device on appvg

```
#crfs -v jfs2 -g appvg -a size=5G -a logname=INLINE -A yes -m /appfs
/dev/fslv00 /appfs
```

To add INLINE log device to existed file system (/sapfs)

```
#chfs -a log=INLINE /sapfs
```

### **To convert jfs file system to jfs2 file system:**

```
#chlv -t <typeofFS><lvname>
```

## **Boot Process**

In this chapter we will discuss, how Aix server is booting once power on the system.

Power on

POST (power on self test) --> Here it will configure the base devices (mouse, keyboard, audio devices, memory etc).

FIND THE AIX BOOT IMAGE

LOCATE THE AIX BOOT IMAGE

LOAD THE AIX BOOT IMAGE

INITIAL PROGRAM LOAD (IPL) START NORMAL MODE

FINALLY SYSTEM WILL BOOT FROM DEFAULT RUN LEVEL 2

Actually in the boot process 3 important phases will run.

### **1st Boot Phase.**

1) The init process start from the RAMFS or Temporary RAM.

- 2) Then rc.boot script will run first time.
- 3) Then "restbase" command will call the ODM partial image from BLV and it will copy into the RAMFS.
- 4) cfgmgr -f command will read "config\_rules" file,  
This file contains base devices information, which are required to access the rootvg.
- 5) bootinfo -b will run and it will identify the last booted device.  
1st phase completed then start the second phase

## **II Boot Phase:**

- 1) rc.boot script execute the 2nd time on this phase
- 2) ipl\_varyonvg will activate the rootvg
- 3) "/" root file system will checked by fsck -y command last shutdown time root file system successfully unmounted or not.
- 4) then "/" root file system will verify by the fsck -y command and it will temporarily mounted on RAMFS as /mnt.
- 5) Then "/usr" file system will verify by the fsck -y command and it will temporarily mounted on RAMFS.
- 6) Then "/var" file system will verify by the fsck -y command and it will temporarily mounted.
- 7) "copycore" command will verify the dump is created or not, if dump is created it will copy into /var/adm/ras location.
- 8) Default Paging Space /dev/hd6 will be activated.
- 9) "mergedev" process will read the all devices files under /dev directory and it will scan them.
- 10) At the end of second phase all rootvg file systems (/home,/usr,/,/var,/opt) will mounted permanently.

## **III Boot Phase:**

- 1) rc.boot script file will execute 3rd time.
- 2) /etc/init process will be started and it will read the /etc/inittab configuration file.
- 3) /tmp file system will activated or mounted
- 4) syncvg command will synchronize the rootvg.
- 5) Then errdaemon and syncd daemon will start
- 6) Finally system will boot from the default run level 2.

## **Runlevels,**

- 0,1 --> These run levels will not use by the users, This runlevels are reserved may be future purpose.
- 2 --> It default run level, and this run level multiuser run level.
- 3-9 --> These run levels users related run levels, users can use depends on their requirement.
- a,b,c,h --> These are not original run levels.

M,m --> Maintenance mode run levels (OS corrupted, OS file systems issues, ODM corrupted)  
S,s --> Service mode run levels.

#cat /etc/.init.state --> To display current running run level.  
#who -r --> To display current running run level

#lsinitd init --> To display the default run level.

#telinit <runlevel> --> To change the run level.  
Ex: telinit 4

#telinit -q --> To refresh the /etc/inittab file.

**Note:** Every 60 seconds /etc/inittab file will refresh by init process.

### **/etc/inittab file**

It is one of the important configuration file. This file will be execute the scheduled jobs/entries at the time of system booting. If you want run any scripts or commands continuously in the system we can add those entries in the /etc/inittab file. this file will be refresh/update every 60 seconds by init process.

### **inittab file Stanza:**

#### **Identifier:RunLevel:Action:Command/Script**

**Identifier:** This is id name of the entry

**RunLevel:** Which run level we need run command/script depends on user requirement.

**Action:** Which action users required,

#### **Actions:**

**once:** This is one the action field in inittab entry if we need to run any job/cmd only once after server booted we can specify the action plan as "once"

**Note:** if action field is once the job will run only one time, if job time completed or killed/stopped by users it will not run again until restart the system.

#### **respawn:**

if action field is "respawn" the job will run continuously in the system, if anyone killed/stopped the job, init process will start the job automatically from /etc/inittab file.

**/etc/inittab** file will access by using below commands

#mkitab -> To add the entry in inittab file

Ex: mkitab "javasrpt:2:respawn:/tmp/javascript"

#lsitab <idname> -> To list the entries in the inittab

#chitab -> To change/modify the entries in the inittab file

#rmitab <idname> -> To remove entries in the inittab file

#telinit q -> to refresh/update /etc/inittab file

#telinit <runlevel> --> to change the run level

#cat /etc/.init.state -> To list the current runlevel

or #who -r

#lsitab init -> To list the default runlevel

How to put commentthe entry?

By adding colon (:) in front of the job

mkitab "test:5:respawn:/tmp/test>> /tmp/todayop"

0c2

once or respawn

## **Configuration**

a) SMIT (System Management Interface Tool)

b) ODM

c) Devices

d) Paging space

e) Error Log

f) SRC/ Demons/Daemons

- g) FTP/SCP or CoreFTP/WinSCP
- h) PMR/Perf PMR/Snap
- i) System dump
- j) Network Configuration
- k) NFS

## **SMIT**

### **System Management Interface Tool:**

It is alternate way to run the commands, smitty mode is graphical mode interface, this has been designed like applications. just we need to fill the fields which are required.

#### **Note:**

- \* --> We should fill the field
- + --> We have listing options
- / --> We have to specify the path of location/directory

#### **Function Keys:**

Ex: F1 or esc+1 --> both are same  
F2 or esc+2 --> both are same

- F1 --> Help
- F2 --> Refresh the current window
- F3 --> Cancel the current window
- F4 --> List the available options
- F5 --> Reset the filled fields
- F6 --> To list the command
- F7 --> Select the multiple items/resources
- F9 --> To take the new shell
- F10 --> Exit from the smitty
- Enter --> if everything ok, then click on enter key to continue.

### **SMIT (System Management Interface Tool).**

smitty is alternate way to run the commands, In aix any activity we can run two ways one is command mode, another one is smitty mode.

Ex: To create VG

#### **Command mode:**

#mkvg -y <vgname>

#### **Smitty mode:**

#smitty mkvg

In smitty everything we will be defined all fields like application form, just we have to fill the required fields.

- 1) if we see left side star (\*) symbol in smitty mode we must fill them.
- 2) if we see right side plus (+) we can list the available options by using F4 key.
- 3) smitty we have below function keys

F1 (or) Esc + 1 = Help

F2 (or) Esc + 2 = Refresh the current window

F3 (or) Esc + 3 = To cancel current window

F4 (or) Esc + 4 = To list the available options (we can see right side plus (+) symbol.

F5 (or) Esc + 5 = Reset the filled fields

F6 (or) Esc + 6 = To provide the command, if we need to run in command line.

F7 (or) Esc + 7 = To select the required multiple items.

F8 (or) Esc + 8 = Image

F9 (or) Esc + 9 = Shell, to take the new shell without exit from the current window

F10 (or) Esc + 0 = To exit from entire smitty window.

Enter=do = To execute the final window after filled fields

### **Object Data Manager (ODM)**

**ODM:** it is a repository, total system data managed by the ODM or ODM contains total system data information Such as:

- 1) Device configuration information
- 2) SMIT Information
- 3) Vital product data information for installation and update procedures
- 4) TCP/IP Communication configuration information
- 6) System resource controller (SRC) data information
- 7)Error log and dump information
- 8)Network Installation Manager (NIM) information

ODM data stored in binary format, we cannot modify/edit the ODM files by using cat, vi commands, if we need to access the ODM files we have to use ODM special commands Such as

#odmadd

#odmshow

#odmcreate

#odmget

#odmchange

```
#odmdrop
#odmdelete
```

### **Actually ODM data will stored on three important directories:**

Such as

- 1) /etc/objrepos
- 2) /usr/lib/objrepos
- 3) /usr/share/lib/objrepos

/etc/objrepos -> This directory contains the customized data information

/usr/lib/objrepos -> This directory contains the predefined data information

/usr/share/lib/objrepos -> This directory contains the Software Vital Product data information (SWVPD).

## **DEVICES**

**Devices:** This chapter we will discuss how to manage the system devices (hdisks, adapters, cdrom, Processors, Memory, tape drive, mother board etc).

### **Commands:**

cfgmgr, lsdev, lscfg, lsattr, chdev and rmdev, rmdev -Rdl, diag, parent and child devices

#### **cfgmgr:**

This command we will use to scan the new devices, which are added to the system.

Ex: #cfgmgr

#cfgmgr -l <devicename> --> To scan the particular device

#cfgmgr -v --> To scan new configure devices in verbose mode

#### **rmdev:**

By using rmdev command we can remove the devices from the ODM/OS/system.

Ex: rmdev -dl <devicename>

**Note:** The device should empty before delete

#rmdev -l <devicename> --> To move the device into defined state.

#cfgmgr -l <devicename> --> To move the device from defined to available state

## **LSDEV:**

By using lsdev command we can display the system devices.

#lsdev --> To display the total system devices

it will display the 4 fields

- 1) device name
- 2) state of device
- 3) location code of device
- 4) description of the device

### **Device Class names:**

harddisks Class name is "disk" (hdisk0, hdisk1)

adapters class name is "adapter" (ent0, ent1, fcs0 fcs1, scsi0 )

cd-roms class name is "cdrom" (cd0, cd1, cd2)

processors class name is "processor" (proc0, proc1, proc2)

mother board - sys (sys0)

tape drive class name is "tape" (rmt0, rmt1, rmt2)

Memory --> mem (mem0, mem1)

**Predefined devices:** What are model devices system will be support we can see in the predefined list

**Customized devices:** What are devices currently installed or attached to the system.

#lsdev -P --> To display the total pre-defined devices models in the system.

#lsdev -C --> To display the total added devices in the system.

#lsdev -Pc <class name>

#lsdev -Cc <class name>

#lsdev -Pc disk --> To display system support disk models

#lsdev -Cc disk --> To display the present added disks to the system

#lsdev -Pc cdrom -> To display the supported cdrom models.

#lsdev -Cc cdrom -> To display the present added cdroms.

#lsdev -Cc sys -> to display the mother board

#lsdev -Cc processor --> To display the processors

#lsdev -Cc tape --> To display the tape drives.

## **lscfg:**

By using lscfg command we can list the vital product data (VPD) information for the devices such as Device model number, serial number, port number, FRU number and location code of the devices etc.

#lscfg -vpl <devicesname> --> To display the VPD information for the device.

ex:# lscfg -vpl cd0

cd0 U7879.001.DQDTTFV-P4-D1 IDE DVD-ROM Drive

Manufacturer.....IBM

Machine Type and Model.....DROM0020541

ROS Level and ID.....P633

Device Specific.(Z0).....058002028F000030

# lscfg -l hdisk0

hdisk0 U9117.570.65B9DBE-V4-C4-T1-L870000000000 Virtual SCSI  
Disk Drive

### lsattr:

By using lsattr command we can display the devices attributes.

**Syn:** #lsattr -El <devicename>

# lsattr -El hdisk0

|                 |                                  |                       |       |
|-----------------|----------------------------------|-----------------------|-------|
| PCM             | PCM/friend/vscsi                 | Path Control Module   | False |
| algorithm       | fail_over                        | Algorithm             | True  |
| hcheck_cmd      | test_unit_rdy                    | Health Check Command  | True  |
| hcheck_interval | 0                                | Health Check Interval | True  |
| hcheck_mode     | nonactive                        | Health Check Mode     | True  |
| max_transfer    | 0x40000                          | Maximum TRANSFER Size | True  |
| pvid            | 00cb9dbe18a78c3d0000000000000000 | Physical volume       |       |
| identifier      | False                            |                       |       |
| queue_depth     | 3                                | Queue DEPTH           | False |
| reserve_policy  | no_reserve                       | Reserve Policy        | True  |

### chdev:

By using chdev command we change the devices attributes values or settings.

**Syn:** #chdev -l <devicename> -a <attribute=value>

**Note:** while changing attributes if we got device busy error, we need to run below command to change the attributes.

#chdev -l <devicename> -a <attribute=value> -P --> This command will update on only ODM. after reboot the system it will reflect on the device.

## **Parent & Child Devices:**

`#lsdev -l <devicename> -F parent ->` To display parent device.

Ex: `# lsdev -l ent0 -F parent`  
`pci3`

above example pci3 is parent device of ent0 device.

`#lsdev -p <devicename> -->` To display the child devices of the parent device.

## **#rmdev:**

By using rmdev command we can delete the devices or move the devices into defined state.

`#rmdev -dl <devicename> -->` To remove the device from the OS/ODM

`#rmdev -l <devicename> -->` To move the device into defined state.

`#rmdev -Rdl <parentdevice> -->` To delete parent device and child devices at a time.

## **DIAG:**

By using "diag" command we can replace the devices remotly from the Data Centers.

Ex:

On one of the Aix server (aix1) located on USA, we need to replace the failed device (hdiskX, entX, cdrom etc). from India on aix1 server, how to perform this activity.

- 1) First we have to collect the device loction code, model number, serial number, port number and provide the details to DC Technician.
- 2) We need to ensure the device should be empty not used by any resources.
- 3) We can replace the device by using diag command by using below commands.

a) `#diag`

b) Click on enter key to continue

c) Select the " Task Selection " option

d) Select the "Hot Plug Task" option

e) Select the "SCSI device" option for the hard disks

F) Replace/Remove a Device Attached to an SCSI Hot Swap Enclosure Device

G) Select failed "disk or device" which is need to be replace.

h) Finally click on enter button, the diag command will remove the device by using rmdev command from ODM.  
and yellow color LED light will blinking on top of the disk.

on "h" step we should not exit from the diag command until confirmed by the DC Technician. once DC technician confirmed after replace the failed disk with new one. we can exit from diag by select the F10 option.

While existing from the diag command, automatically diag command will scan the new disk.

**Note:** if device hot swapable, Server downtime not required to replace the devices.  
if device non-hot swapable, Server downtime required to replace the devices.

### **SMT (simultaneous multithreading)**

SMT feature is available from P5 servers, if we enable SMT to any server/LPAR single physical processor will become two logical processors.

```
#smtctl -m <mode> -w <when>
#smtctl -m on -w now --> To enable smt now
#smtctl -m on -w boot --> To enable smt after reboot the server
#bindprocessor -q (or) #sar --> To check the total logical Processors in system
#smtctl -m off -w now --> To disable SMT now
#smtctl -m off -w boot --> To disable SMT after reboot system.
#lsattr -El proc0 -> To check the SMT enabled or not and smt threads we can verify.
```

### **Paging Space**

**Paging space:** it is the alternate memory in Aix operating system, Paging also called like swap space, secondary memory, alternate memory and virtual space.

The main use of Paging space when ever Real Memory full the programs/applications/processes are use Paging space as alternate memory.

#### **Note:**

- 1) The paging space we can create from the Hard disks.
  - 2) We can create Max Paging space size double size of the existing Real memory.
- Ex: if real memory is 5GB, Paging Space we can create upto 10GB.

3) Default Paging space is /dev/hd6 Logical Volume.

**Page Out:** When ever Real Memory become full the Applications/Programs/Process are moved from the real memory to Paging space is called Page out.

(Or)

If process are moving from Real Memory to Paging space

**Page In:** If Applications/Programs/Process are moving from Paging Space to Real Memory we can call Page IN.

**Page Replacement:** If any Application/Process completed Page out & Page In, this is called Page Replacement.

#mkps -s <noofLPs><vgname><pvname> --> To create Paging Space.

#mkps -a -s <noofLPs><vgname><pvname> --> To create Paging Space with Auto on Yes.

(or)

#mklv -y <lvname> -t <paging><vgname><nooflps><pvname>

#mklv -y sappaginlv -t paging sapvg 10 hdiskX

#lsps -a -> To list total Paging Spaces

#lsps <PSname> --> To list particular Paging Space

#lsps -s -> To display total paging space size and usage of total paging space.

#swapon /dev/PSname --> To activate the paging space.

#swapoff /dev/PSname --> To deactivate the paging space.

#chps -s <nooflps><PSname> --> To increase the Paging Space size.

#chps -d <nooflps><PSname> --> To Decrease Paging Space size.

#rmpps <PSname> --> To remove the paging space

#rmlv <PSname> --> To remove the paging space

**Note:** PS should be deactivate before remove from the system.

#chps -ay <PSname> --> To set Auto On=yes for the PS.

#chps -an <PSname> --> To set Auto on=no for the PS.

**/etc/swapspaces** --> It is paging space configuration file

**Note:** if Paging Space is Auto=yes, PS entry will add into the /etc/swapspaces.

#chlv -n <NewPSname><OldPSname> To rename the paging space.

**Note:** Deactivate the paging space before rename.

```
lsps -a
```

| Page Space Type | Physical Volume | Volume Group | Size   | %Used | Active | Auto |
|-----------------|-----------------|--------------|--------|-------|--------|------|
| sappaginglv     | hdisk5          | sapvg        | 512MB  | 1     | yes    | no   |
| paging02        | hdisk4          | sapvg        | 512MB  | 0     | no     | yes  |
| paging01        | hdisk5          | sapvg        | 512MB  | 0     | no     | no   |
| paging00        | hdisk4          | sapvg        | 1024MB | 0     | no     | no   |
| hd6             | hdisk3          | rootvg       | 512MB  | 1     | yes    | yes  |

```
rmpps sappaginglv
```

0517-062 rmpps: Paging space sappaginglv is active.

0517-061 rmpps: Cannot remove paging space sappaginglv.

```
lsps sappaginglv
```

| Page Space Type | Physical Volume | Volume Group | Size  | %Used | Active | Auto |
|-----------------|-----------------|--------------|-------|-------|--------|------|
| sappaginglv     | hdisk5          | sapvg        | 512MB | 1     | yes    | no   |

```
swapoff /dev/sappaginglv
```

```
rmpps sappaginglv
```

rmlv: Logical volume sappaginglv is removed.

```
rmlv paging00
```

Warning, all data contained on logical volume paging00 will be destroyed.

rmlv: Do you wish to continue? y(es) n(o)? y

rmlv: Logical volume paging00 is removed.

```
lsps -a
```

| Page Space Type | Physical Volume | Volume Group | Size  | %Used | Active | Auto |
|-----------------|-----------------|--------------|-------|-------|--------|------|
| paging02        | hdisk4          | sapvg        | 512MB | 0     | no     | yes  |
| paging01        | hdisk5          | sapvg        | 512MB | 0     | no     | no   |
| hd6             | hdisk3          | rootvg       | 512MB | 1     | yes    | yes  |

```
chps -ay paging01
```

```
lsps -a
```

| Page Space Type | Physical Volume | Volume Group | Size  | %Used | Active | Auto |
|-----------------|-----------------|--------------|-------|-------|--------|------|
| paging02        | hdisk4          | sapvg        | 512MB | 0     | no     | yes  |
| paging01        | hdisk5          | sapvg        | 512MB | 0     | no     | yes  |
| hd6             | hdisk3          | rootvg       | 512MB | 1     | yes    | yes  |

```
ls -l /etc/swapspaces
```

-rw-r--r-- 1 root system 473 Aug 09 11:13 /etc/swapspaces

# cat /etc/swapspaces

\* /etc/swapspaces

\* This file lists all the paging spaces that are automatically put into  
\* service on each system restart (the 'swapon -a' command executed from  
\* /etc/rc swaps on every device listed here).

\* **WARNING:** Only paging space devices should be listed here.

\* This file is modified by the chps, mkps and rmeps commands and  
referenced

\* by the lmps and swapon commands.

**hd6:**

dev = /dev/hd6

**paging02:**

dev = /dev/paging02

**paging01:**

dev = /dev/paging01

# set -o vi

# mkps -s 4 sapvg hdisk4

paging00

# cat /etc/swapspaces

\* /etc/swapspaces

\* This file lists all the paging spaces that are automatically put into  
\* service on each system restart (the 'swapon -a' command executed from  
\* /etc/rc swaps on every device listed here).

\* **WARNING:** Only paging space devices should be listed here.

\* This file is modified by the chps, mkps and rmeps commands and  
referenced

\* by the lmps and swapon commands.

**hd6:**

dev = /dev/hd6

**paging02:**

dev = /dev/paging02

**paging01:**

dev = /dev/paging01

# mkps -an -s 4 sapvg hdisk4

paging03

# cat /etc/swapspaces

\* /etc/swapspaces

\* This file lists all the paging spaces that are automatically put into  
\* service on each system restart (the 'swapon -a' command executed from  
\* /etc/rc swaps on every device listed here).

\* WARNING: Only paging space devices should be listed here.

\* This file is modified by the chps, mkps and rmpps commands and  
referenced  
\* by the lspas and swapon commands.

### hd6:

dev = /dev/hd6

### paging02:

dev = /dev/paging02

### paging01:

dev = /dev/paging01

### paging03:

dev = /dev/paging03

# lspas -a

| Page Space | Physical Volume | Volume Group | Size  | %Used | Active | Auto |
|------------|-----------------|--------------|-------|-------|--------|------|
| paging03   | hdisk4          | sapvg        | 256MB | 1     | yes    | yes  |
| paging02   | hdisk4          | sapvg        | 512MB | 0     | no     | yes  |
| paging01   | hdisk5          | sapvg        | 512MB | 0     | no     | yes  |
| paging00   | hdisk4          | sapvg        | 256MB | 0     | no     | no   |
| hd6        | hdisk3          | rootvg       | 512MB | 1     | yes    | yes  |

# chlv -n myps paging03

0516-708 chlv: Cannot rename open logical volume. If LV  
contains a filesystem, unmount to close logical volume paging03.

# chlv -n myps paging02

# lspas -a

| Page Space | Physical Volume | Volume Group | Size  | %Used | Active | Auto |
|------------|-----------------|--------------|-------|-------|--------|------|
| paging03   | hdisk4          | sapvg        | 256MB | 1     | yes    | yes  |
| mysps      | hdisk4          | sapvg        | 512MB | 0     | no     | no   |
| paging01   | hdisk5          | sapvg        | 512MB | 0     | no     | yes  |
| paging00   | hdisk4          | sapvg        | 256MB | 0     | no     | no   |
| hd6        | hdisk3          | rootvg       | 512MB | 1     | yes    | yes  |

# lsconf |grep -i memory

Memory Size: 3072 MB

Good Memory Size: 3072 MB

0516-010 : Volume group must be varied on; use varyonvg command.

```

+ mem0
prtconf -m
Memory Size: 3072 MB

lsvg
rootvg
sapvg
aixvg

lsvg sapvg |grep -i free
MAX LVs: 256 FREE PPs: 2506 (160384 megabytes)
mkps -s <noofLPs><vgname><pvname>
lsvg sapvg |grep -i ppsize
lsvg sapvg |grep -i pp
VG STATE: active PP SIZE: 64 megabyte(s)
VG PERMISSION: read/write TOTAL PPs: 2506 (160384
megabytes)
MAX LVs: 256 FREE PPs: 2506 (160384 megabytes)
LVs: 0 USED PPs: 0 (0 megabytes)
STALE PVs: 0 STALE PPs: 0
MAX PPs per VG: 30480
MAX PPs per PV: 3048 MAX PVs: 10

lsvg -p sapvg
sapvg:
PV_NAME PV STATE TOTAL PPs FREE PPs FREE
DISTRIBUTION
hdisk4 active 319 319 64..64..63..64..64
hdisk5 active 2187 2187 438..437..437..437..438

mkps -s 16 sapvg hdisk4
paging00

mkps -s 8 sapvg hdisk5
paging01

mkps -ay -s 8 sapvg hdisk4
mkps: Not a recognized flag: y
0517-050 Usage: mkps [-a] [-n] [-t lv] -s NumLPs Vgname Pvname
 Makes a paging space using logical volume.
mkps [-a] [-n] -t nfs hostname pathname
 Makes a paging space using an NFS server
mkps -an -s 8 sapvg hdisk4
paging02

lsps -a

```

| Page Space Type | Physical Volume | Volume Group | Size   | %Used | Active | Auto |
|-----------------|-----------------|--------------|--------|-------|--------|------|
| paging02        | hdisk4          | sapvg        | 512MB  | 1     | yes    | yes  |
| paging01        | hdisk5          | sapvg        | 512MB  | 0     | no     | no   |
| paging00        | hdisk4          | sapvg        | 1024MB | 0     | no     | no   |
| hd6             | hdisk3          | rootvg       | 512MB  | 1     | yes    | yes  |

```
mklv -y sappaginglv -t paging sapvg 4 hdisk5
sappaginglv
```

```
lsps -a
```

| Page Space Type | Physical Volume | Volume Group | Size   | %Used | Active | Auto |
|-----------------|-----------------|--------------|--------|-------|--------|------|
| sappaginglv     | hdisk5          | sapvg        | 256MB  | 0     | no     | no   |
| paging02        | hdisk4          | sapvg        | 512MB  | 1     | yes    | yes  |
| paging01        | hdisk5          | sapvg        | 512MB  | 0     | no     | no   |
| paging00        | hdisk4          | sapvg        | 1024MB | 0     | no     | no   |
| hd6             | hdisk3          | rootvg       | 512MB  | 1     | yes    | yes  |

```
lsps -s
```

| Total Paging Space | Percent Used |
|--------------------|--------------|
| 1024MB             | 1%           |

```
swapon /dev/sappaginglv
```

```
lsps -a
```

| Page Space Type | Physical Volume | Volume Group | Size   | %Used | Active | Auto |
|-----------------|-----------------|--------------|--------|-------|--------|------|
| sappaginglv     | hdisk5          | sapvg        | 256MB  | 1     | yes    | no   |
| paging02        | hdisk4          | sapvg        | 512MB  | 1     | yes    | yes  |
| paging01        | hdisk5          | sapvg        | 512MB  | 0     | no     | no   |
| paging00        | hdisk4          | sapvg        | 1024MB | 0     | no     | no   |
| hd6             | hdisk3          | rootvg       | 512MB  | 1     | yes    | yes  |

```
lsps -s
```

| Total Paging Space | Percent Used |
|--------------------|--------------|
| 1280MB             | 1%           |

```
swapoff /dev/paging02
```

```
lsps -a
```

| Page Space Type | Physical Volume | Volume Group | Size   | %Used | Active | Auto |
|-----------------|-----------------|--------------|--------|-------|--------|------|
| sappaginglv     | hdisk5          | sapvg        | 256MB  | 1     | yes    | no   |
| paging02        | hdisk4          | sapvg        | 512MB  | 0     | no     | yes  |
| paging01        | hdisk5          | sapvg        | 512MB  | 0     | no     | no   |
| paging00        | hdisk4          | sapvg        | 1024MB | 0     | no     | no   |
| hd6             | hdisk3          | rootvg       | 512MB  | 1     | yes    | yes  |

```
chps -s 4 sappaginglv
```

```
lsps sappaginglv
Page Space Physical Volume Volume Group Size %Used Active Auto
Type
sappaginglv hdisk5 sapvg 512MB 1 yes no lv
```

### **Paging Space Problems:**

```
"INIT: Paging space is low"
"ksh: cannot fork no swap space"
"Not enough memory"
"Fork function failed"
"fork () system call failed"
"Unable to fork, too many processes"
"Fork failure - not enough memory available"
"Fork function not allowed. Not enough memory available."
"Cannot fork: Not enough space"
```

## **ERROR LOG**

**Error log:** By Using error log we can identify the system errors/problems, it is one of the advanced facility in Aix to identify the system software or hardware errors.

### **Hardware Errors:**

```
Harddisk Failed
Adapters Failed
Adapters Link problems
Sysplanner Failed
VRM (Voltage Regulator Module) Failed
DIMM/Memory Failed
CD-ROM Errors
Tape Drive Errors
```

### **Software Errors:**

```
Low paging space
File systems problems
Logical volumes Problems
Super block corrupted
Application Problems
Deamons Problems etc.,
```

### **Location:**

**/var/adm/ras/errorlog**

**#errpt -->** To display the total system errors with short description.

```
errpt |more
IDENTIFIER TIMESTAMP T C RESOURCE_NAME DESCRIPTION
BFE4C025 0809102216 P H sysplanar0 UNDETERMINED ERROR
BFE4C025 0809092916 P H sysplanar0
```

**IDENTIFIER:** ID name of the error

**TIMESTAMP:** The date & time of error generated.

**T/type:** Type of error (**TEMP/PERM/INFO/UNKN**).

**C/Class:** Class of error (**Software/Hardware/OPERATOR**)

**RESOURCE\_NAME:** What is main resource to generated error (Ex: hdisk, sysplanner, FS, LV and Daemon).

**DESCRIPTION:** Error short Description:

```
#errpt -j <IDname> --> To list the particular ID related errors,
#errpt -a --> To list the total system errors with full details/description.
#errpt -aj <ID name> To display the particular error with full
details/description.
#errpt -d S --> To list the software errors with short details.
#errpt -a -d S --> To list the software errors with full details.
#errpt -d H --> To list the hardware errors with short detail.
#errpt -a -d H --> To list the hardware errors with full details.
#errpt -d O --> To list the other errors
```

### **Error daemon: /usr/lib/errdemon**

```
#ps -ef |grep -i err --> To check the errdemon status.
```

```
ps -ef |grep -i err
```

```
root 200866 1 0 09:19:32 - 0:00 /usr/lib/errdemon
```

```
#!/usr/lib/errstop --> To stop the error demon.
```

```
#!/usr/lib/errdemon --> To start the error demon.
```

```
/usr/lib/errdemon -l --> To display the error log location and size of log.
```

Error Log Attributes

```

Log File /var/adm/ras/errlog
Log Size 1048576 bytes
Memory Buffer Size 32768 bytes
Duplicate Removal true
Duplicate Interval 10000 milliseconds
Duplicate Error Maximum 1000
```

```
#!/usr/lib/errdemon -s <sizeinbytes> --> To Increase the errorlog.
```

### **Clear the errors from the Log:**

```
#errclear -d S 0 --> To clear Software errors
```

#errclear -d H 0 --> To clear Hardware errors  
#errclear -d O 0 --> To clear Other errors  
#errclear 0 --> To clear total system errors (hardware/software/Other etc.)

**Note:** Before clear the errors take the backup of errorlog safe side to check old errors in future.

#errlogger "my test error in aix1 server" -> To generate the error into error log manually.

**Note:** if errlog file not available or removed by some one. follow the below steps to create log file.

**/usr/lib/errstop**  
**/usr/lib/errdemon**

Once stop/start the errdemon the errlog file will create automatically.

#errpt -s <startdate> -e <enddate> To display the errors on particular dates.

Date Formate: 0809123016 (MMDDhhmmYY)

=====  
**System Resource Controller (SRC)**  
=====

**SRC:** By Using SRC we can manage the demons, SRC commands are:

#startsrc ---> To start the Deamons/Services  
#stopsrc ---> To stop the Deamon/Services  
#lssrc ---> To display the Deamon/Services status.

**Demons/Deamons or Services:**

Demon is a process or protocol, it is running on the background of the server, the Demons or managed by the SRC commands such as (startsrc, stopsrc and lssrc). We have several deamons on Aix/Unix.

Ex:

Network Deamons  
Installation Deamon  
Error demon  
Telenet/SSH demons  
NFS Demons

LVM Deamons etc.

### **lssrc:**

By using lssrc command we can list demons status.

#lssrc -a --> To list the all demons in the system with their status.

#lssrc -g <groupname> --> To display the group of deamons with their status.

#lssrc -g nfs --> To display all NFS group deamons status.

#lssrc -s <Demon> --> To display the single deamon status, or particular demon status.

#lssrc -s biod,

### **stopsrc:**

By using stopsrc we can stop the deamons or services.

#stopsrc -g <groupname> --> To stop the group of deamons at a time.

#stopsrc -s <Demon> --> To stop the single demon.

### **startsrc:**

By startsrc we can start the Deamons.

#startsrc -g <groupname> --> To start the group of deamons at a time.

#startsrc -s <daemon> --> To start the single demon.

### **Restart the Deamons:**

if we need any daemon restart, we have to stop the daemon and start it.

#refresh -s <demon> --> To refresh the demon

### **Note:**

1) All Deamons are presented on /etc/services file or /etc/inetd.conf configuration files.

2) If any demons are commented (#) on above configuration files, the demons will not work.

3) Except errdemon, all demons are managed by SRC commands.

## **Errorlog**

cp /var/adm/ras/errlog /tmp/errlogbkp\_08092016

# errclear -d S 0

# errpt -d S

# errclear -d H 0

# errpt -d H

```
/usr/lib/errrdemon -l ----> listing errorlog attributes
errpt -d O
IDENTIFIER TIMESTAMP T C RESOURCE_NAME DESCRIPTION
9DBCfDEE 0809121516 T O errrdemon ERROR LOGGING TURNED
ON
192AC071 0809121516 T O errrdemon ERROR LOGGING TURNED
OFF
9DBCfDEE 0809120816 T O errrdemon ERROR LOGGING TURNED
ON
192AC071 0809120816 T O errrdemon ERROR LOGGING TURNED
OFF
A6DF45AA 0809091916 I O RMCdaemon The daemon is started.
9DBCfDEE 0809091916 T O errrdemon ERROR LOGGING TURNED
ON
F89FB899 0808150016 P O dumpcheck The copy directory is too small.
E87EF1BE 0808150016 P O dumpcheck The largest dump device is too
small.
A6DF45AA 0808143216 I O RMCdaemon The daemon is started.
9DBCfDEE 0808143116 T O errrdemon ERROR LOGGING TURNED
ON
192AC071 0808122316 T O errrdemon ERROR LOGGING TURNED
OFF
A6DF45AA 0808112316 I O RMCdaemon The daemon is started.
9DBCfDEE 0808112316 T O errrdemon ERROR LOGGING TURNED
ON
A6DF45AA 0808101616 I O RMCdaemon The daemon is started.
9DBCfDEE 0808101516 T O errrdemon ERROR LOGGING TURNED
ON
A6DF45AA 0807164816 I O RMCdaemon The daemon is started.
9DBCfDEE 0807164816 T O errrdemon ERROR LOGGING TURNED
ON
192AC071 0806192516 T O errrdemon ERROR LOGGING TURNED
OFF
```

```
errclear -d O 0
errpt -d O
errpt
errclear -d 0
```

### **errclear:**

0315-181 Bad value supplied for error class: 0  
Rerun the command with a valid error class.

### **Usage:**

```
errclear -J err_label_list -K err_label_list -N resource_name_list
-R resource_type_list -S resource_class_list -T err_type_list
-d err_class_list -i filename -m machine_id -n node_id
```

```

errclear 0
errclear 0
errpt

set -o vi
cp /var/adm/ras/errlog /tmp/errlogbkp_08092016
cp /tmp/errlogbkp_08092016 /var/adm/ras/errlog
errpt
IDENTIFIER TIMESTAMP T C RESOURCE_NAME DESCRIPTION
9DBCfDEE 0809121516 T O errdemon ERROR LOGGING TURNED
ON
192AC071 0809121516 T O errdemon ERROR LOGGING TURNED
OFF
9DBCfDEE 0809120816 T O errdemon ERROR LOGGING TURNED
ON
192AC071 0809120816 T O errdemon ERROR LOGGING TURNED
OFF
BFE4C025 0809102216 P H sysplanar0 UNDETERMINED ERROR
BFE4C025 0809092916 P H sysplanar0 UNDETERMINED ERROR
BFE4C025 0809092816 P H sysplanar0 UNDETERMINED ERROR
BFE4C025 0809092716 P H sysplanar0 UNDETERMINED ERROR
BFE4C025 0809092616 P H sysplanar0 UNDETERMINED ERROR
BFE4C025 0809092116 P H sysplanar0 UNDETERMINED ERROR
A6DF45AA 0809091916 I O RMCdaemon The daemon is started.
E142C6D4 0809091916 T H sysplanar0 EEH temporary error for
adapter
E142C6D4 0809091916 T H sysplanar0 EEH temporary error for
adapter
9DBCfDEE 0809091916 T O errdemon ERROR LOGGING TURNED
ON
BFE4C025 0808162616 P H sysplanar0 UNDETERMINED ERROR
F89FB899 0808150016 P O dumpcheck The copy directory is too small.
E87EF1BE 0808150016 P O dumpcheck The largest dump device is too
small.
BFE4C025 0808145116 P H sysplanar0 UNDETERMINED ERROR
BFE4C025 0808144616 P H sysplanar0 UNDETERMINED ERROR
BFE4C025 0808144416 P H sysplanar0 UNDETERMINED ERROR
BFE4C025 0808144416 P H sysplanar0 UNDETERMINED ERROR
BFE4C025 0808143516 P H sysplanar0 UNDETERMINED ERROR
A6DF45AA 0808143216 I O RMCdaemon The daemon is started.
E142C6D4 0808143216 T H sysplanar0 EEH temporary error for
adapter
E142C6D4 0808143116 T H sysplanar0 EEH temporary error for
adapter
2BFA76F6 0808122416 T S SYSPROC SYSTEM SHUTDOWN BY USER

```

## FTP

### FTP/SCP or CoreFTP/WinSCP:

By Using FTP/SCP or CoreFTP/WinSCP we can transfer/copy the files & Directories/software (data) from One server to another server and we can transfer the files/directories/software from Windows (laptops, desktops) servers to Unix servers (Aix, Linux etc)

FTP/SCP ---> Command mode tools  
CoreFTP/WinSCP --> GUI Mode tools.

**Note:** Network is should be available both servers to transfer the data.

### File Transfer Protocol (FTP):

How to transfer the data from Server1 (192.168.1.10) to Server3 (192.168.1.20) via FTP.

**Source:** Server1: Location /sapfs/fico/aix1, aix2, India (dir)

**Destination:** Server3: Location /var/sapdata/

#### Steps:

- 1) Login to the source server server1
- 2) Enter into the Source location  
`#cd /sapfs/fico/`
- 3) Connect the Destination Server via FTP  
`#ftp <hostname/IP Address>`  
`#ftp 192.168.1.20`  
User name: ravi  
Passwd: \*\*\*  
ftp>
- 4) Enter into the destination Location  
ftp>`cd /var/sapdata/`  
ftp>`bin` --> To accept the binary format files  
ftp>`hash` --> hash (#) symbol will be display until copy the data.  
ftp>`prompt` --> To disable the prompt.
- 5) Copy the required files/dirs (data)  
ftp> `put filename/dir` --> To copy single file/dir  
or  
ftp> `mput filename1 file2 file3` --> To copy the multiple files/Directories  
or  
ftp> `mput *` --> To copy the total data (files/dirs) from the source location to destination.  
ftp> `bye` --> To exit from the FTP

## **To Transfer the data from Server3 (/var/sapdata) to Server1 (/desktop/data/)**

ftp> get <file/dir> To receive the single file/dir (data) from destination to source.

ftp> mget <file1 file2 dir1> --> To receive the multiple files/dirs

ftp>mget \* --> To receive the total files/dirs from destination to source location.

ftp> help --> FTP help commands will display

**Note:** FTP By default will be available in the server once installed OS.

FTP port number is 21

## **SCP (Secure Copy):**

By using scp we can copy/transfer the data from one server to another server, but scp having more security to copy the data, it is not default software in Aix OS, we need download from the internet and install on the Unix servers (Aix, Linux) if required.

**Source:** Server1: Location /sapfs/fico/aix1, aix2, India (dir)

**Destination:** Server2: Location /var/sapdata/

1) Login to the source server

2) Copy the files/dirs by using scp

#scp <file/dir> User@destServer:<DestLocation>

#scp aix1 root@192.168.1.20:/var/sapdata/

## **Passwd of root:**

#scp file1 file2 root@192.168.1.20:/var/sapdata/ --> To copy multiple files/dirs

#scp \* root@192.168.1.20:/var/sapdata/ --> to copy total files/dirs from source to destination.

## **CoreFTP/WinSCP:**

By using CoreFTP/WinSCP we can transfer the data from one server to another server on GUI mode. CoreFTP/WinSCP tools we can download from the internet and install on the windows laptops/desktops

/etc/ftpusers --> If we add any users in this configuration file, for that user ftp access can be disabled.

```
vi /etc/ftpusers
sachin
:wq
```

## **PMR/Snap/Perf PMR**

### **Problem Management Request (PMR):**

When we are facing any software/hardware minor or major problems in the Aix servers, if Aix admin not able to fix the problems then we can contact to the IBM to open the PMR.

**IBM Contact Number:** 18004267378

**Option1:** Hardware Problems

**Option2:** Software problems

IBM will ask below details before Help us:

Hardware/Software problem?

Your Name

Your Company Name

Customer ID Ex: 21281138 (Disney Project)

Server Model `#prtconf |more`

Server Serial number

What is Aix OS Version installed

What is your problem

Your contact details (Phone, Mail IDs)

Priority of the PMR

Take your PMR number (63721 082 000)

63721 PMR number

082 Branch code

Then IBM will ask us to upload the server snap file into IBM site.

### **SNAP File:**

The snap file contains/collect the total information about server.

#### **To create SNAP file:**

`#snap -r -->` To remove the old SNAP file, before create new snap file.

`#snap -gc or ac -->` To create snap file

`#cd /tmp/ibmsupt ->` **To enter snap file location**

`#ls -l snap*`

rename the snap file with PMR number

`#mv snap.pax.Z 63721.082.000.snap.pax.Z`

**Once snap is created/ready, Now we can upload the snap file into the IBM Site via FTP:**

**Software Site:** testcase.software.ibm.com

**Hardware Site:** testcase.boulder.ibm.com

**Suppose if Hardware case:**

```
#ftp testcase.boulder.ibm.com
User name: anonymous
Passwd: surendra.s@hcl.com
#cd /toibm/hw --> Hardware snap file upload location
or /toibm/aix (For software snap file upload location)
#bin
#hash
#prompt
#put 63721.082.000.snap.pax.Z
```

**Note:** If we dont know how to upload the snap file, ask IBM technician He will provide total details/commands step by step to upload the snap file.

## **System Dump**

**System dump:** it contains the information about system booting errors. (or) if system having any booting related problems, we can create/generate the System Dump file (vmcore file) to analyze the system booting related problems by IBM.

In Aix we have two dump devices

- 1) primary dump device
- 2) secondary dump device

### **1) Primary dump device:**

If real memory below 4GB (<4GB) primary dump device is /dev/hd6

If real memory =>4GB primary dump device is /dev/lg\_dumplv

### **2) Secondary dump device:**

Default secondary dump device is /dev/sysdumpnull

**Note:**

- 1) If we need additional dump devices, we can create by using #mklv command.
- 2) If primary dump device is /dev/hd6 or /dev/lg\_dumplv, once generated dump file the vmcore file will create and copy into the /var/adm/ras location.

3) If primary dump device is /dev/sysdumpnull, once dump generated, vmcore file will create but vmcore core will not copy into the /var/adm/ras location, because the vmcore file delete by the sysdumpnull.

so if we need to create dump file (vmcore) we should not keep primary dump device as /dev/sysdumpnull.

### **Pre-Checks before create dump file:**

- 1) Estimate the dump size
- 2) Verify the /var file system size, /var file system free space should => estimated dump size.

```
#sysdumpdev -l --> To list the primary & secondary dump devices
information.
#mklv -y <lvname> -t sysdump <vgname><nooflps><pvname>
#sysdumpdev -p /dev/dumplv --> To change the primary device temporarily.
#sysdumpdev -Pp /dev/dumplv --> To change the primary dump device
permanently
#sysdumpdev -s /dev/dumplv --> To change the secondary dump device
temporarily.
#sysdumpdev -P-s /dev/dumplv --> To change the secondary dump device
permanently.
#sysdumpdev -e --> To estimate the dump file size before creating. the size
it will list on bytes. we can convert the bytes size into MBs to estimate
required space on dump logical volumes (/dev/lg_dumplv etc.)
#sysdumpstart -p --> To create/generate system dump file (vmcore) from
the primary dump device.
#sysdumpstart -s --> To generate system dump file (vmcore) from the
secondary dump device.
#savecore -d <location> --> To generate the dump file (vmcore) manually
after create.
#savecore -d /var/adm/ras
#sysdumpdev -L --> To list the last time or previously created dump device
information.
```

### **LED code of Dump Device: 0c2**

#### **Recent Problems:**

**Problem:** Dump file not generated

**Reason:** No enough space on /var file system to copy the dump file (vmcore).

(or) Primary dump device if /dev/sysdumpnull.

(or) No enough space Primary or Secondary dump device.

#### **Issue Fixed:**

Increased /var file system.

Increased Primary/Secondary dump device space

## NETWORK

**Network:** By using network we can able to communicate the servers remotely from one server to another servers.

### Network Types:

LAN

WAN

MAN

- 0) Ethernet adapter or NIC (Network Interface Card)
- 1) Switch/Router
- 2) IP Configuration/Deconfiguration
- 3) IP Aliasing Configuration/Deconfiguration
- 4) Media Speed and MTU Size
- 5) Etherchannel/Link Aggregation/Teaming
- 6) Network troubleshooting Part
- 7) Routes Adding/Removing

**Ethernet Adapter:** By using ethernet adapter we can configure the network IP address of the servers. Each ethernet adapter having two interfaces

Ex: ent0 - Ethernet adapter

en0 & et0 - Two interfaces for ent0

Note: en0 Interface we will use to configure the IP address.

### Ethernet adapters we having two types

- 1) Physical Ethernet adapter
- 2) Virtual Ethernet adapter

any adapter we can use to configure network.

### Aix Servers Network configuration:

#### Pre-requirements:

At least we need one Ethernet Adapter (Ex: ent0) to configure the IP address.

**Note:**

- 1) the adapter (ent0) should be free before configure ip address means any IP should not configured to ent0 adapter
- 2) If we are configuring first IP address for the server we have to login via console to the server.

**To Configure the IP Address:**

- 1) `#smitty mktcpip` --> To configure the IP address
- 2) Select the Ethernet Interface (Ex: enX)
  - \* Hostname [server1] --> The name of the server
  - \* Internet Address [192.168.1.60] -> Server IP address
  - \* Internet Interface [en4] -> EAI
  - Network Mask [255.255.255.0]
  - Default Gateway [10.1.3.5]
  - Start Now [yes] --> To start or up IP address.

**Note:** if we have netmask & gateway we have to fill or else not required to fill them. Mostly netmask & gateway address Network team provide us.

`#ifconfig -a` --> To list configured IP addresses in Server.  
`#ifconfig <Interface>` --> To check the particular Interface or IP status.  
`#ifconfig en4`  
`#hostname` --> To display the server hostname.  
`#lsdev -Cc adapter |grep ent*` --> To list the ethernet adapters and their status.  
`#lsdev -Cc if` --> To list the Network interfaces and their status.  
`#entstat -d <EA> |grep -i link` --> To check the link status of the ethernet adapter between switch and server.  
EX:  
`#entstat -d ent0 |grep -i link`  
Link Status: up --> Good  
Link Status: down, unknown --> The server having cable connectivity problem.  
`#hostname <newhostname>` --> To change the server hostname temporarily.  
`#chdev -l inet0 -a hostname=newname` --> To change the hostname permanently  
`#smitty mktcpip` --> To configure/change the ip address/hostname permanently.

**Note:** once configured IP address by using `smitty mktcpip` command. Server IP and hostname will update under `/etc/hosts` file automatically.

`#entstat -d <EA> |grep -i hardware` --> To display the EA hardware address or Media Access Control (MAC) address.

### **To remove the IP address:**

```
#ifconfig <Interface><down>
#ifconfig <Interface><delete>
#ifconfig <Interface><detach>
#ifconfig enX down
#ifconfig enX delete
#ifconfig enX detach
#ifconfig <Interface><state>
#ifconfig en3 down --> To bring down the IP Address.
#ifconfig en3 up --> To bring up the IP Address
```

### **Deconfigure or Remove the IP Address:**

1) Bring down the IP Address

```
#ifconfig <Interface><down>
#ifconfig en0 down
```

2) Delete the IP Address from the ODM or Interface card.

```
#ifconfig <interface><delete>
#ifconfig en0 delete
```

3) Detach the IP address from Interface card

```
#ifconfig <interface><detach>
#ifconfig en0 detach
```

Q) In aix1 server ent1 ethernet adapter failed and its having 192.168.1.200 ip addresss. How to replace the ent1 adapter with new one.

### **ANS:**

- 1) Remove the ipaddress for ent1
- 2) Take ent1 details (location, model, serial numbers etc.)
- 3) Remove the adapter by using diag command
- 4) Once added new adapter exit from the diag command.
- 5) Configure the 192.168.1.200 ip for the new adapter (ent2)

### **2) IP Aliasing Configuration/Deconfiguration:**

**Aliasing:** If we need to configure more than one IP address for single NIC card, we will configure the first IP address as original IP Address, remaining IP addresses we can configure as Alias IPs.

**Note:** Alias IPs we can configure Permanent or Temporarily in the server.

If we configure alias IP as Permanent the IP will be available in the server, after rebooted system.

If we configure alias IP as Temporarily the IP will not be available after reboot the server.

### **To Configure the Alias IP Temporarily:**

```
#ifconfig <interface><Alias IP><alias>
```

```
#ifconfig en0 192.168.1.11 alias --> To configure the alias IP (192.168.1.11) for en0 Interface.
```

```
#ifconfig <interface><Alias IP><Netmask><gateway><alias>
```

```
#ifconfig en0 192.168.1.12 255.255.255.254 10.3.1.66 alias --> To configure the alias IP (192.168.1.12) with netmask value and gateway value as alias.
```

```
#ifconfig <interface><Alias IP><delete> --> To delete the temporarily alias IP.
```

```
#ifconfig en0 192.168.1.11 delete
```

### **To Configure the alias IP Permanently:**

```
#chdev -l <Interface> -a alias4 or 6=<alias IP>
```

```
#chdev -l en0 -a alias4=192.168.1.12 --> To add the alias IP (192.168.1.12) as permanently.
```

```
#chdev -l en0 -a alias4=192.168.1.13 -a netmask 255.255.255.0 -a gateway 10.3.6.60 --> To add the IP alias (192.168.1.12) with netmask,gateway IPs as permanently.
```

```
#chdev -l <Interface> -a delalias4=<Alias IP>
```

```
#chdev -l en0 -a delalias4=192.168.1.13 --> To delete the permanent alias IP.
```

## **3) Media Speed and MTU (Maximum Transfer Unit) Speed:**

**Media Speed:** By using media speed attribute we can verify/Increase/Decrease the IP address speed.

### **Media Speed Values:**

10\_Full\_Duplex

10\_Half\_Duplex

100\_Full\_Duplex

100\_Half\_Duplex

1000\_Full\_Duplex

1000\_Half\_Duplex

Auto\_Negotiation

```
#chdev -l <EA> -a media_speed=<value> --> To change the media speed to the Ethernet adapter or NIC card.
```

```
#chdev -l ent0 -a media_speed=1000_Full_Duplex --> We can change ent0 adapter media speed as 1000_Full_Duplex.
```

**Note:** When we are changing the media speed to the ethernet adapter (Ex: ent0,en0), The adapter (en0) should not have the any IP address or if IP configured should not up, If IP is up the media speed will not change.

Q) How to change the IP address for the EA adapter,the adapter (ent0) having IP (192.168.1.10) and IP is up status. without bring down IP how to change media speed.

A)

```
#chdev -l ent0 -a media_speed=1000_Full_Duplex -P
```

**Auto\_Negotiation:** This is one of the value in media\_speed attribute, if we set media\_speed=Auto\_Negotiation for any EA, the adapter media\_speed automatically will change depends on situation.

```
#lsattr -El <EA> |grep -i media_speed --> To verify the media_speed of EA.
#entstat -d <EA> |grep -i media --> To verify the current running media speed.
#entstat -d <EA> |grep -i link --> To verify the link status
#entstat -d <EA> |grep -i hardware --> To verify the hardware address/MAC address.
```

#### 4) Maximum Transfer Unit (MTU):

By using MTU size we can define how much data (packet) need to transfer for each operation through the Network Interface.

The MTU range: Min 60 bytes -- Max 65535 Bytes.  
Default value is -- 1500 bytes.

```
#lsattr -El <Interface> |grep -i mtu --> To display the MTU size.
#ifconfig <Interface> mtu <mtusize>
#ifconfig en0 mtu 3000 --> To change the MTU size as 3000 bytes for en0 Interface.
```

#### 5) Etherchannel/Link Aggregation:

Etherchannel is one of the network mechanism in AIX Operating System, By using Etherchannel we can achieve the network or IP address high availability and we can Increase network speed also if we are configure as Link Aggregation mode.

**Etherchannel:** To configure etherchannel we need minimum two ethernet adapters, one for primary adapter and another one for backup adapter. In

Etherchannel configuration only one adapter active at a time, if active adapter fail then backup adapter will provide the network service to IP address. This configuration High Network Availability available but network load balance will not be share by the both adapters at a time. Only active adapter will balance the network load. Backup adapter always idle until active adapter fail/down.

**Link Aggregation:** To configure Link Aggregation we need minimum two ethernet adapters but here both adapters we can configure as primary adapters, by using Link Aggregation configuration we can Increase the Network speed, and load balance will be share by the both adapters at a time. If one adapter fail, second adapter will provide the network service.

### **To create Etherchannel or Link aggregation adapters:**

To create Etherchannel

- 1) #smitty etherchannel
- 2) Select "Add An EtherChannel / Link Aggregation" option.
- 3) Select the "Primary adapter"
- 4) Select the "Backup adapter"

### **To create Link Aggregation Etherchannel:**

- 1) #smitty etherchannel
- 2) Select "Add An EtherChannel / Link Aggregation" option.
- 3) Select the both adapters as "Primary adapters"

**Note:** Which are adapters we are using to create etherchannel/Link Aggregation, the adapters should be empty, we should not configure any IP address.

We will configure the IP address for etherchannel/Link aggregation adapter after created.

### **Testing Etherchannel:**

#entstat -d <Etherchanneladapter> |grep -i active --> To display the active adapter.

#!/usr/lib/methods/ethchan\_config -f <EEA> --> To manually failover to from primary adapter to backup adapter or Backup adapter to primary adapter.

#entstat -d <EEA> ---> To verify network statistics of primary and backup adapters (Transfer, Receive packets, media speed, links status and Hardware address).

#entstat -d <EEA> |grep -i link --> To verify the status of primary and backup adapter links.

#entstat -d <EEA> |grep -i hardware --> To list the hardware address or mac address.

Q) In etherchannel/Link aggregation have (two adapters) which adapter mac address will add to the EEA.

A) Primary adapter mac address will assign to the EEA.

### **How to remove Etherchannel:**

- 1) Remove the IP, if configured.
- 2) Remove the etherchannel (smitty etherchannel)

Q) On one of the server we have Etherchannel/Link Aggregation, Primary adapter is (ent3), Backup adapter is (ent4). Now primary ent3 adapter having problems, IBM suggested us replace the primary adapter with new adapter (ent6), Do we need to bring down IP or network on the server for this activity?

A) IP or Network down time not required, we can replace the problem adapter with new adapter online by following below steps.

- 1) #smitty etherchannel
- 2) Select " Change / Show Characteristics of an EtherChannel / Link Aggregation" Option
- 3) Select the EEA (ent7)
- 3) First delete the Main adapter (ent3)
- 4) Add the main new adapter (ent6)

### **To replace the backup adapter:**

- 1) #smitty etherchannel
- 2) Select " Change / Show Characteristics of an EtherChannel / Link Aggregation" Option
- 3) Select the EEA (ent7)
- 3) First delete the Backup adapter (ent4)
- 4) Add the Backup new adapter (ent6)

### **Recent Activity or Troubleshooting:**

I have replace the EEA (primary adapter), due to the adapter is failed.

### **5) Network troubleshooting Part:**

Q) We are unable to login to the server by using network, How to identify the problem and how to fix the problem.

### **Action Steps:**

- 1) We have to login server by using Console to identify the problem.

- 2) Verify the error logs
- 2) `# ifconfig -a` -> To verify the IP status.
- 3) `# lsdev -Cc adapter |grep -i ent` -> Check the adapter status.
- 5) `# lsdev -Cc if --` -> To verify interface status.
- 6) `# entstat -d <EA> |grep -i link, --` -> To verify the link status, if link is down, we have to inform the data center technician to put cable connection properly.
- `# lspcfg -l entX` --> to take the location code of adapter.
- 7) Verify the telnet/SSH demons. should be active.

### **Recent Activity or Issue fixed:**

**Problem:** We are not able to login to the server via network or Putty.

#### **Solution1:**

- 1) logged in server via console.
- 2) Verified in error logs, there is link problem for the adapter
- 3) verified link status, link is down.
- 4) informed DC technician check the cable connections and put them properly.

#### **Solution2:**

Routes are missed I have added.

### **6) Routes Adding/Removing**

**Routing:** When we have many servers/ipaddress and less networks we can make the separate routes to reduce network traffic between servers.

#### **The routes we can add two ways**

- 1) Permanent routes adding (by using `#smitty mkroute`) command
- 2) Temporarily routes adding (By using `#route add` ) command

`#netstat -nr` -> To display the current routes information in the system.

#### **To add the route Temporarily:**

EX: `192.168.1./24 192.168.1.226 en3`

`#route add <IP Address><Netmask><gateway><Interface>`

`#192.168.1.0 255.255.255.0 192.168.1.226 en3`

**Note:** /24 = 255.255.255.0 netmask values

Ex2: `192.168.1.0 192.168.1.200 en0`

`#route add 192.168.1.0 192.168.1.200 en0`

#### **To Add the routes permanently**

```
#smitty mkroute
Destination Address [192.168.1.0]
Gateway [192.168.1.226]
Netmask [255.255.255.0]
Network Interface [en3]
```

### **To Delete route Temporarily:**

```
#route delete <IP Address><netmaskvalue><gateway><Interface>
```

### **To Delete Route Permanently:**

```
#smitty rmroute
*Destination address []
Gateway []
Interface []
Netmask []
```

### **/etc/hosts -->**

It is one of the network related configuration file. By using this file we can resolve the server IPs/Hostnames between the servers to communicate with each other.

Ex:

```
192.168.1.200 server1
192.168.1.250 server2
```

To make network communication between servers (server1,server2) we need to update the both servers IP & hostname on /etc/hosts file on both sides.

**Note:** /etc/hosts normal ASCII format file we can modify by using vi command.

### **Login server1**

```
#vi /etc/hosts
192.168.1.200 server1
192.168.1.250 server2
:wq
```

### **Login server2**

```
#vi /etc/hosts
192.168.1.200 server1
```

192.168.1.250 server2  
:wq

### **DNS:**

Stands for domain name server, by using DNS all servers will be able to communicate with each other. To communicate all servers with each other we need to add the all server's IPs & hostnames in DNS server. DNS server will configure mostly Big organizations.

On ALL Servers we have to update/add the DNS server IP & Hostname in /etc/resolv.conf configuration file.

**/etc/resolv.conf** -> it is one of the configuration file it contains the DNS server IP and hostname.

### **/etc/services:**

This file contains the all daemons/services information with respected port numbers

### **Note:**

- 1) If you want change the daemon port numbers we can change/modifying under /etc/services file.
- 2) if you want stop the daemons we can put the hash (#) symbol in front of the daemon.

### **Port numbers:**

telnet - 23  
ssh- 22  
ftp - 21  
rlogin - 513  
DNS - 443  
NFS - 2049  
Nim - 1058  
Nim register - 1059  
apache 8080

**/etc/inetd.conf** --> It is one of the Demons configuration file, this file contains the daemons information (Ex: telnet, ssh, ftp, sftp etc demons).

**Note:** if we modify /etc/inetd.conf configuration file we need to restart the inetd demon to reflect the modifications on system.

### **Network Time Protocol (NTP):**

By using NTP we can maintain the same date & time on all servers at a time.

**NTP client servers:**

/etc/ntp.conf -> NTP configuration file  
xntpd -Daemon

## NFS

**NFS:** stands for Network File System, By using NFS we can share the file systems data between the servers through Network without copying the data on remote servers. The remote servers users can access file system's data as a local users once implemented NFS successfully.

(Or)

If we add any file system in Network sharing the file system we can call NFS file system.

Q) We have two servers (Server1 and Server2), On server1 /sapfs file system presented, now we need to share the /sapfs file system to Server2 via Network (NFS).

Server1 - 192.168.1.100 (NFS Server)

Server2 - 192.168.1.30 (NFS Client)

**Pre-checks:**

0) Configure the IPs and Hostnames if not configured on servers By using #smitty mktcpip

1) Add the IPs and Hostnames on /etc/hosts file on both servers.

2) Test the network communication between both servers by using ping command.

a) Login server1

#ping IP or hostname

#ping 192.168.1.30

TTL = 255 (Ping Success)

b) Login to the Server2

#ping 192.168.1.100

TTL = 255 (Ping Success)

**NFS Implementation:**

Once ping is success on both servers, go ahead to implement NFS

**Server1 Side Steps:**

1) Login to server1

2) `exportfs -i <NFSName> -->` to export the file system in network, or add FS in Network sharing temporarily.

Ex: `#exportfs -i /sapfs`

3) Verify file system exported or not.

`#showmount -e`

**O/P:** /sapfs

### **Server2 side steps or Client side:**

1) Login to Server2

2) list the exported file systems from NFS server.

`#showmount -e <NFS Server IP or Hostname>`

`#showmount -e 192.168.1.100`

**O/P:** /sapfs

**Note:** above output should be list /sapfs file system, if it is not listed not possible to access the NFS.

3) Mount the NFS file system on server2 in any local directory.

`#mkdir /localdir`

`#mount <NFServer:NFSFS> </localdir>`

Ex:

`#mkdir /sap`

`#mount 192.168.1.100:/sapfs /sap`

`#cd /sap`

`#ls`

aix1

aix2

aix3

### **Actually in NFS**

We have two types of exports and three types of mounts.

### **Exports Types:**

1) Temporary Export

2) Permanent Export

#### **1) Temporary Export:**

`#exportfs -i <FS name>`

`#showmount -e -->` List the exported file systems.

#### **2) Permanent Export:**

a) `vi /etc/exports`

/sapfs

:wq

b) exportfs -a --> Refresh the /etc/exports file.

c) showmount -e -> To list the exported file systems

/sapfs

**Note:** If File system exported as temporarily mode the NFS sharing will be lose after reboot the server.

b) Permanent export always available on every reboot of the server.

c) Mostly ALL office environments we will use the permanent exports only.

### **Mounts are Three types:**

1) Explicit mount or Temporary mount

2) Predefined mount or Permanent mount

3) Auto FS mount

#### **1) Explicit mount:**

It is temporarily mount of the NFS file system on NFS clients.

##### **Steps:**

a) showmount -e NFSServerIP --> To check Server's NFS exported FSs

b) #mkdir /local --> To create Local Directory

c) #mount <NFS Server:/NFS FS></Localdir> --> To mount the NFS file system into local directory.

**Note:** Temporary mount will remove from the system after reboot the system.

#### **2) Predefined mount or Permanent mount:**

#crfs -v nfs -d </NFSFS> -a nodename=<NFSServer> -a options=<requiredoptions> -A yes -m /LocalFS

EX: crfs -v nfs -d /sapfs -a nodename=192.168.1.100 options=bg,soft,nointr,rw -A yes -m /localsapfs

b) mount /localsapfs

c) cd /localsapfs

d) ls

**Note:** a) when we are mount the NFS FS by using crfs command doesn't mean we are creating FS in NFS clients to access the data, crfs command will update the NFS share (/sapfs) file system as local file system (/localsapfs) on /etc/filesystems configuration file.

b) options=fg,hard,intr,rw --> for Hard mount

(or)

options=bg,soft,nointr,rw --> for Soft mount

**Hard mount:** If we mount any NFS file system as hard mount on NFS clients, in case NFS server is down or not accessible, this situation if try to access the NFS File system from clients, client session will go hung state. it will not release session until NFS server is up. Even if reboot client server issue will not fix, only solution NFS server should be up.

**SoftMount:** If we mount any NFS file system as soft mount on NFS clients, in case NFS server is down or not accessible, this situation if try to access the NFS share from the clients, client session will go hung state for few seconds, then will release the session.

Always soft mount will be good if we are mounting the NFS file system permanently.

#exportfs -u </fsname> --> To un-export the file system temporarily.

(or)

vi /etc/exports --> Permanent un-export  
remove FS the entry

:wq

#exportfs -a

#exportfs -u /fsname

### **NFS Deamons:**

NFS deamons are

biod

nfsd

rpc.lockd

rpc.statd

rpc.mountd

**NFS Server:** Which is providing the file systems service access that is called server.

or

Which is providing the service we can call server.

**Client:** Which is taking the service we can call client.

**Note:** some times a server can act as server also as client.

**biod daemon:** The client server will take the Network file system access from the server via biod daemon.

**nfsd:** The server will provide the Network file system access to clients via nfsd daemon

**rpc.mountd:** It is main daemon on both servers, to mount the Network file system from server to client.

### **Server Side Deamons should be run:**

nfsd, rpc.mountd, rpc.statd, rpc.lockd

**Client Side Deamons should be run:**

biod, rpc.mountd, rpc.statd, rpc.lockd.

**NFS not working properly how to troubleshoot it.****(OR) Recently I fixed NFS issue by following below steps.**

1) We have verify the IP status and ping test, if ok then.

2) Verify the NFS and Port map deamons. if not running restart the deamons by using below commands.

#stopsrc -g nfs --> To stop nfs deamons

#stopsrc -g portmap --> To stop the portmap daemon

#startsrc -g portmap --> To start portmap daemon.

#startsrc -g nfs --> To start the NFS deamons.

3) Verify the exported list

To check the NFS version

#nfsstat -v |grep -i version

#topas

**To check the rpc state information.**

#rpcinfo -s

## INSTALLATION

In this chapter we will discuss, how to install the Aix Operating System, TL Levels Upgrades, Service Pack (SP) Level Upgrades and user required softwares/optional softwares.

AIX OS Installation (5.3, 6.1, 7.1 and 7.2)

Q) Create Logical Server (Sapserver) and install Aix 5.3 OS on Logical Server.

**Pre-requirements:**

1) OS CD

2) CD Drive

3) Hard Disk

4) Minimum CPU/Memory.

5) Network Adapter (entX). if IP address/network required.

**INSTALLATION STEPS:**

Once above resources are added to the system we can follow the below steps to Install OS via HMC.

1) Start the system and enter into the SMS (System Management Service) menu to select 1st boot device as CD-ROM/DVD-ROM to install Aix OS via CD.

To enter the SMS menu.

- a) Select server --> Operations --> Activate --> Profile.
- b) Then profile window will open,
- c) Select Advanced button --> Select boot mode as "SMS menu" option.
- d) Select the check box, if we need console.
- e) Finally click ok button.

To select 1st boot device as CD-ROM.

Once enter into SMS menu it will display the 5 options. Here we have to select the option 5 "select boot options".

- a) Select boot options (option 5)
  - b) Configure boot device order (option 2)
  - c) Select 1st boot device (Option 1)
  - d) Select CD-ROM (Option 3)
  - e) Select the CD-ROM Model, Ex: IDE (Option 4), if we dont know CD-ROM model, we can select list all devices (Option 7).
  - f) Select cd-rom device (Option 1).
  - g) Select boot sequence: Configure 1st boot device. (option 2).
  - h) Once 1st boot device configured, we can exit from SMS menu (Option x).
- x= Exit from SMS menu.

Installation options need setup one by one.

1) Please define the system console

Here select 1 and enter.

2) Select the Language (Option 1 English).

3) Then it will display the below 3 options

**a) Start install now with default settings**

**b) Change/Show Installation settings and Install**

**c) Start Maintenance mode for system recovery**

**a) Start install now with default setting:**

If we need to install OS with old settings we can select the 1 option

**b) Change/Show Installation settings and Install:**

If we need to change installation method and disk we can select the 2 option.

3) Select "0" to proceed with selected settings.

4) Select "1" to continue the installation.

Then installation will be started, it will take 45 minutes to 1 hour to install Aix OS.

**Note:** Once 93% OS installation completed. it will ask the Insert the Volume 2 CD (Graphical AIX CD).

if need graphical mode we can insert 2 nd CD, if not required select the "q" quit from the option.

## **INSTALLATION TYPES/METHODS:**

In Aix there are three types of installation methods such as:

### **1) New and completed overwrite Installation Method. (NCOIM)**

### **2) Migration Installation Method (MIM)**

### **3) Preservation Installation Method (PIM)**

#### **1) New and completed overwrite Installation Method:**

It is one of the installation method, When we are installing Aix OS on new servers first time we will select this method OR if we need overwrite completely existing OS, means if we wish remove the old OS we can choose this method these scenario existing rootvg data/file systems all will remove and recreate newlly.

#### **2) Migration Installation Method:**

It is one of the Installation method, When we are upgrade the OS from lower version (Ex: 5.3) to higher version (Ex:6.1,7.1) we can select migration installation method, Migration installation method will not remove the existing rootvg data/file systems, total rootvg data will be available after OS upgrade except /tmp file system, only /tmp file system will remove and recreate on this method, if we have any data on /tmp file system we will be lost after Migration Installation.

#### **3) Preservation Installation Method:**

It is one of the installation method, when we need to install the same existing OS version on servers we will select the Preservation installation method.

Ex: If server having 6.1 OS, Due to OS problem, if any OS packages are removed/corrupted etc.

This situation we are wish to install again 6.1 version on the server, We will select the preservation installation method. This installation method will remove and recreate the rootvg file systems (/ , /var, /usr, /tmp and /opt) except /home file system. /home file system data will not be lost this method, if we need save/preserve the remaining file systems (/ , /var, /usr, /tmp and /opt) data we have add the file systems entries on /etc/preserve.list file like below.

```
vi /etc/preserve.list
.^/var
.^/usr
:wq:
```

above example says while doing preservation installation, /var and /usr file systems data will not be lose.

**Note:**

- 1) NCOIM Option will be available while doing all installations,
- 2) MIM Option will be available only when we are upgrading OS version from lower to higher
- 3) PIM Option will be available if are installing same OS version on server.
- 4) By default 3 configuration files will be available in preserve.list file such as: /etc/filesystems, /etc/resolve.conf file, /etc/swapspaces.

**INSTALLATION OF OPTIONAL SOFTWARES** (Java, Apache, SSH, SSL, sudo, NTP (network time protocol). etc:

**Fileset:** File set nothing but it is smallest installable software, a file set contains the set of files.

**Package:** A package contains the set of file sets

**LPP (Licenced Program Products):** LPP Contains the set of packages

**BUNDLES:** A Bundle contains the set of file sets, packages and LPPs.

We can install any file sets or softwares by using command mode or smitty mode. if we are installing one or two softwares we can install command mode, if we are installing bulk softwares we can install by using smitty mode.

**Software can be installed in two modes/ways:**

- 1) Commit mode.
- 2) Applied mode.

**1) Commit mode Installation:**

a) When we are installing the software first time in the server we can install in commit mode.

b) when we are upgrade the software from lower version to higher version, Ex: java 5.0 to 6.0, if we install latest version (6.0) in commit mode, the old vesion (5.0) will be remove from the system.

**Note:** when we install software firsttime in the system, if we install software in applied mode, it will install only in commit mode.

## 2) Applied Mode:

a) When we are upgrade the software from lower version to higher version  
Ex: Java 5.0 to 6.0, if java software installed in applied mode, both versions are available in the system.

if we are facing any problem with new version we can remove it by reject the new version or if there is no problem with latest version we can remove the old version after monitor the server one or two weeks.

**Note:** All applied mode file sets stored under /usr/lpp/filesets.

Q) After upgrade software in applied mode, how to remove the latest version.

```
#installp -r java.rte
```

Q) After upgrade software in applied mode, how to remove the old version.

```
#installp -c software -> To commit the new software with old one.
```

```
#installp -c java.rte -->
```

Q) After upgrade software in applied mode, how to remove the old and new versions.

```
#installp -ug <software>
```

To list the applied file sets

```
#installp -s (or)
```

```
#lslpp -l |grep -i applied
```

To list the commit mode file sets

```
#lslpp -l |grep -i commit
```

To list the applied & commit mode file sets

```
#lslpp -l
```

### Command mode Installation:

```
javar.te.5.0.bff
```

The softwares will be available in file systems or cdrom.

Ex: /softwarefs/java.rte (5.0) or /dev/cd0/java.rte

**#installp:** By using installp command we can installp the software.

```
#installp -cgXY -d <Dir/CD Location><software>
```

```
#installp -cgXY -d /softwarefs java.rte --> To install java.rte file set from /softwarefs file system.
```

```
#installp -cgXY -d /dev/cd0 java.rte --> To install the java.rte software from cd.
```

#installp -cgXYp -d /dev/cd0 java.rte --> To install the java.rte software from cd.

c - commit mode  
g - group of softwares  
X - Expand the rootvg file systems if required  
Y - Accept the License  
p - Preview  
d - Location of software (FS/DIR/CD)

### **To Install software in smitty mode:**

smitty install  
Install and Update Software  
Install software  
\*INPUT Device /Directory for software [FS/DIR/CD]  
ex: cd [/dev/cd0]

### **Final Window Options:**

\*INPUT Device/Directory [/dev/cd0] --> Location sw  
\*Software install [ java.rte] -> Select the required sw to install  
Preview Only [ yes/No]  
Committed Software [Yes/No] if applied mode (no)  
Save Replaced Files [yes]  
Expand File systems if space is required [yes]  
Detailed Output [yes]  
Accept New License Agreement [ yes ]  
Preview New License Agreement [Yes/No]

#installp -agXY <UpdateSW> -d <location of SW>  
#installp -agXY java.rte.6.0 -d /software/java/ -> To install the java software from /software/java/ location in applied mode.  
#installp -agXY fileset1 fileset2 -d /software/java/ --> To install the two softwares/multiple softwares at a time.  
#installp -agXY -d /softwares/ all --> To install all softwares from the /softwares file system.

### **a - Applied mode**

#installp -r java.rte.6.0 --> To reject/remove the latest version 6.0 of java.  
#lslpp -l --> To list the total installed softwares in the system.  
  
#lslpp -l <software> or #lslpp -l |grep -i <software> --> To list the particular installed software.  
#lslpp -l |grep -i java --> To check java software installed or not.

#lspp -h <softwarename> -> To check the history of the particular software means when installed date& time, state, description etc.

#lspp -w <cmd location> --> To list the respected software of specified command.

Ex: #lspp -w /usr/sbin/lspv

#lspp -f <softwarename>

#lspp -f bos.rte.lvm --> To list the respected files of the software.

### **To uninstall the software:**

#installp -ug <softwarename> To install the software.

### **u - uninstall**

#smitty remove

\*select software [java.rte] +

Preview [yes/No]

Remove depend software [no]

Detailed output [Yes]

**Recently Softwares Installed:** Java (6.0/7.0), open SSH (6.0), open ssl (1.0), TL upgrades,

RPM [Redhat Package Manager]

sudo.1.9.rpm

#rpm -ivh <rpmsw>

i - Initially

v -verbose

h - software name

Ex: #rpm -ivh suo.1.9.rpm --> To install RPM packages.

#rpm -uvh <RPM SW> --> To update the rpm packages.

u - update/upgrade

#rpm -qa --> To list the all installed rpm packages

#rpm -q <rpmSW> or #rpm -q|grep -i <RPM SW>

#rpm -q sudo-->To list the particular installed rpm packages.

#rpm -e <rpmSW> --> To uninstall the rpm software.

**RPM SWs:** sudo, tripware etc.,

### **FIX Management:**

#instfix -k <FixName> --> To install the flx

#instfix -ik <FixName> --> To list the installed Fix or To verify the fix is installed or not.

#instfix -iak <Fixname> --> To list the installed fix full details.

### **E-FixManagement:**

E- Emergency

#emgr -e <E-fixname> --> To install e-fix

#emgr -l --> To list the installed e-fixs

#emgr -e <E-fixname> -p -> To check the preview of the e-fix installation.

#emgr -r <E-fix/fix> --> To remove the E-fix.

#installp -L -d /dev/cd0 -> To list the software of cdrom.

#instfix -T -d /dev/cd0 -> To list the fixs from the cdrom.

### **To mount cdrom into local directory.**

#mkdir /localdir -> Create local directory

#mount -v cdrfs -o rw /dev/cd0 /localdir

### **Difference between PTF and APAR Fixes:**

### **Alternate Disk Installation / Rootvg Cloning:**

When we are performing the major Installations/Activities (Ex: OS Installation, Migrations, TL/ML/SP upgrades and Firmware upgrades).

We will create alternate/clone rootvg safe side, in case Installation/performing activity failed or rootvg data corrupted we can boot the server from the alternate rootvg with single reboot. So to recovery system it will take only 5 to 10 minutes if we are having alternate/clone rootvg.

if we have alternate rootvg we can rollback/recovery the system in short time (ex: below 5 to 10 minutes) we can reduce the more down time to recovery the system.

**Note:** If rootvg is corrupted, we can recovery the rootvg by using mksysb or alternate rootvg.

if we are recovery the system by using mksysb it will take time 1 to 2 hours because, need to configure mksysb, spot, client server and need assign the resources (mksysb & spot) to client server. then go to client side, ping test, configure 1st boot device as network adapter then exit from SMS menu then follow the installation steps.

if we are recovery the system by using alt rootvg, it will take less time (5 to 10 minutes). Just boot the server with old version rootvg by single reboot.

## **Configure/Create Alt rootvg:**

### **Pre-requirements:**

Take the one empty hard disk. The disk size should be equal or morethan rootvg existing disk size.

By using below commands we can create alternate rootvg.

```
alt_disk_install
alt_disk_copy
alt_disk_clone
```

In my environment we will configure alternate rootvg by using below command.

```
alt_disk_install -C -B <new/emptydisk>.
```

```
#lspv
hdisk0 - rootvg - size 100GB
hdisk1 - none - size 100GB
```

Ex:  
#alt\_disk\_install -C -B hdisk1

**Note:** once executed above command, It will create alt\_lvs (Ex:alt\_hd1, alt\_hd2) then it will create the alternate file systems

(Ex:/alt\_inst/home, /alt\_inst/usr etc) then it will take data backup from old rootvg and restore into the alt\_rootvg.

Once created alternate rootvg (altinst\_rootvg), We can perform new version installation any one of rootvg.

### **To Install latest version installations on old rootvg:**

#bootinfo -b hdisk0 --> System is booted from hdisk0 so now hdisk0 is active disk, if we perform any installation it will install only in old rootvg.

EX: #installp -agXY -d /dev/cd0 <softwarename>

#installp -agXY -d /dev/cd0 bos.sysmgt.nim.master -> It will install on hdisk0 disk because it is active disk.

### **To Install latest version installations on Alternate rootvg (hdisk1).**

```
#alt_disk_install -C -P2 -b update_all -l agXY -l
<softwarelocation><altdiskname>
```

Ex: #alt\_disk\_install -C -P2 -b update\_all -l agXY -l  
/dev/cd0/bos.sysmgt.nim.master hdisk1

Q) Create alternate rootvg and upgrade the TL level from 6.1-08 to 6.1-10 on alternate rootvg.

**TL Location:** /software/aix/61/TL10/

A) Follow the below three alternate disk installation steps/phases.

1)alt\_disk\_install -C -P1 hdisk1 --> Create the alternate rootvg on new disk (hdisk1).

2)alt\_disk\_install -C -P2 -b update\_all -l agXY -l /software/aix/61/TL10/ -d hdisk1 --> To install the TL 61-10 on alternate rootvg (hdisk1)

**Note:** TL upgrade will take 30 to 45 minutes time. once TL upgrade completed run below command to boot the system with hdisk1.

3) alt\_disk\_install -r -C -P3 hdisk1 --> To reboot the system and boot with hdisk1 disk.

**Note:** Once server is up altinst\_rootvg name channel as rootvg. original rootvg name changed as old\_rootvg.

### **Once Installation/upgrade done and rebooted system with alternate\_rootvg.**

as per standard we will monitor the system 15 to 30 days. if everything is fine with alternate\_rootvg will remove the old rootvg by using below command.

```
#alt_disk_install -X old_rootvg --> To remove the old rootvg.
```

if we are facing problems with alternate rootvg after installations/upgrade we will remove the alternate rootvg by using below steps.

1) set the bootlist order as old\_rootvg disk (hdisk0).

2) Boot the system old\_rootvg

3) remove the alternate rootvg by using below command.

```
#alt_disk_install -X altinst_rootvg
```

**Note:** if we need to remove old\_rootvg, alt VG should be active.  
if we need to remove alisnt\_rootvg, old rootvg should be active.  
#bootinfo -b --> to find the active VG disk.

Q1) In rootvg we have two disks (hdisk0, hdisk1) with mirrored. We don't have empty hard disk to take alternate rootvg. Now how to take alternate/clone rootvg?

Q2) In rootvg we have only one disk (hdisk0) and we don't have empty hard disk to take alternate/clone rootvg. Now how to take alternate/clone rootvg.  
/home, /usr, /tmp, /var, / (root), /opt

Ans) Yes, we can take, but not entire rootvg cloning only we can create alternate file systems by using #multibos command.

**Note:** In multibos we can upgrade/install only TL/ML/SP levels only. OS migration not possible.

### **MULTIBOS:**

### **TL/ML/SP Upgrade process**

**TL/ML/SP Packages:** TL packages are nothing but these are OS updated/upgrade packages. These packages we can download from the IBM site  
[www.ibm.com/support/fixcentral/](http://www.ibm.com/support/fixcentral/)

### **Pre-works:**

Once get the downtime from the clients/server owners we will do the below.

As per server owners we got maintenance window Sat 12 AM to 6AM for two servers.

### **pre-checks**

1) We will take the mksysb backup & place it in nim server.

2) We will configure the rootvg cloning/alternate rootvg

3) We will take the important commands output,  
oslevel -s, netstat -nr, instfix -i |grep ML, ifconfig -a, df -gt, mount, lsvg, lsvg -o, lsfs -a

4) We will verify the applied packages

#installp -s or #lspp -l |grep -i applied

**Note:** if any applied packages are available we will commit them by using  
#installp -c all

5) We will verify the system filesets consistency

#lppchk -v

**Note:** this command will list if any in-consistency/broken/failed/practically installed packages are available we have to fix the problem by installing

software or if Broken packages are not important we will remove from the server by using `#installp -C -->` To clean `lppchk -v` output

6) We will verify the Latest TL Packages on which server available.

a) Nim Server (EX: `/software/TLpackages/5308`)

or

b) Local servers, if already downloaded from IBM site under(`/tmp/5308`) location.

c) If packages are not available on NIM server or Local server then we will download from IBM site and copy into the nim server or local server.

**Note:** Pre-checks can be done anytime/anyday it will not effect anything on server no need to wait for customer maintenance window.

### **Implementation:**

**Note:** Implementation steps we have to perform only on maintenance window as per the customer.

Once stopped applications/database then we will install the TL packages,

1) Mount the TL file system from the nim server on aix1 (client)

`#mkdir /mnt ->` Create local directory

`# mount nimserver:/software /mnt`

`# cd /mnt/TLpackages/5308`

2) `inutoc . ->` To create the inutoc. file (.toc)

3) `smitty update_all`

\*INPUT Device/Directory for software [`/mnt/TLpackages/5308`] -> location of TL packages

Fill below fields in Final window:

Preview only [yes/no]

Commit software update [no]

SAVE replaced files [yes]

Expand file system if space need [yes]

Detailed output [yes]

Accept License Agreement [yes]

Preview License Agreement [yes/no]

(or )

`#installp -agXY -d /mnt/TLpackages/5308 all`

**Note:** TL Upgrade will take 30 to 45 minutes

4) `oslevel -s ->` Verify the latest TL level

5) Reboot the server.

### **Once Server UP, Post validation:**

1) Verify the latest/installed version

#oslevel -s

O/p: 5300-08-00

2) lppchk -v -> Verify the filesets consistency

3) instfix -i |grep -i ML or TL --> To verify all TL level status

4) Verify the important commands outputs

ifconfig -a, netstat -nr, df -gt, oslevel -s, lsvg, lsvg -o,lsfs -a

5) once post validation everything is fixed ask customer/client to start applications.

**Recent troubleshooting:**

**Recent critical activity:**

**Recent major activity:**

**Question:**

After Upgrade TL from low level to higher level (Ex: 06 to 08 TL), In the server still we can see the low TL level 06 only, what is the problem and how to fix it?

**Answer:**

After TL upgrade may be few filesets/software are missing from the TL installation. we have to find missing softwares.

By using below command we can find the missing file sets,

#oslevel -rl <Latest TL Level>

#oslevel -rl 5300-08

**Note:** if below 10 missing packages are listed we will install only which are missing file sets or if more than 20/50/100 file sets are missing we will re-install the total latest TL level again.

**TL Packages:** Few updated OS packages, IBM will release as TL packages,

**SP Packages:** SP packages also OS packages only, SP packages updated from TL packages.

**Difference between TL & ML:**

TL (Technology Level) & ML (Maintenance Level) both are same only, Actually in 5.3 version from 5.3-01 to 5.3-03, IBM use to call TL levels, after 03 TL level IBM name changed as ML.

5.3 TL levels, 01 - 12

6.1 - 01 - 11

7.1 01 - 08

**Maintenance Mode:**

If we are facing any OS related problems we recovery the few problems from the maintenance mode.

### **The Problems are:**

- 1) If ODM files are corrupted
- 2) If rootvg file systems corrupted/not accessible.
- 3) if /etc/inittab corrupted
- 4) if root password recovery
- 5) if BLV is corrupted/missing

To enter into maintenance mode we need OS CD/lpp\_source/mksysb/spot/

### **Maintenance Mode:**

In Maintenance mode we can fix the below problems such as.

- 1) Root Password Recovery
- 2) BLV (Boot logical volume) Recovery
- 3) Rootvg file systems problems
  - a) To run `#fsck` command, for read only file systems.
  - b) To recovery super block (`#dd`)
- 4) Corrupted ODM files recovery
- 5) Corrupted /etc/inittab file recovery

### **To Enter into the maintenance mode:**

To enter into the maintenance mode we have to assign the OS packages to the server. CD OS packages or NIM OS packages (lpp\_source or mksysb) with spot.

- 1) On assigned OS packages to the server via nim
- 2) Go to SMS menu, do ping test, select first boot device then exit from sms menu.
- 3) select define console
- 4) select language
- 5) then next window below three options will be list.
  - a) Start install now with Default settings
  - b) change/show installation settings & install
  - c) Start Maintenance mode for system recovery
- 6) on above options we have to select start maintenance mode (Option 3).
- 7) Next it will list below five options
  - a) Access a volume group
  - b) Copy a system dump
  - c) Access advanced maintenance
  - d) Erase disks
  - c) Install from a system backup
- 8) on above options we have to select Options "a"

- 9) Select "0" to continue
- 10) Select "1" rootvg and VGID will be list
- 11) Next window it will list rootvg logical volumes and below two important options
  - a) Access the volume group and start a shell
  - b) Access the volume group and start a shell before mounting file systems.

**Note:**

1) if we select Option "a" we can recovery below problems.

- a) Root Password Recovery
- b) BLV (Boot logical volume) Recovery
- c) Corrupted ODM files recovery
- d) Corrupted /etc/inittab file recovery

2) if we select option "b" we can recovery below problems

Rootvg file systems problems

- a) To run `#fsck` command, for read only file system
- b) To recovery super block (`#dd`)

3) If we select option "a" first rootvg all file systems will be mounted and provide the shell

if we select option "b" all rootvg file systems without mounting it will provide the shell, here we can run `#fsck` command on rootvg file systems.

**To Recovery root password:**

- 1) Select option "a"
- 2) `#passwd root`  
     enter rootvg new password  
     enter rootvg new password again
- 3) reboot the system

**To recovery BLV:**

- 1) Select option "a"
- 2) Run `bos boot` command on rootvg disk  
`#bosboot -ad /dev/rootvgdisk`
- 3) Reboot the system

**To recovery /etc/inittab file**

- 0) Select option "a"
- 1) Take backup of old /etc/inittab file

```
#cp /etc/inittab /etc/inittab_todaydate
#rm /etc/inittab
#vi /etc/inittab
init:2:initdefault
:wq
#reboot system
```

### **To recover rootvg file systems problems:**

- 1) Select option "b"
- 2) run fsck command on all rootvg file systems or only run on specific file system which is having problem,  
#fsck -y /fsname
- 3) Reboot the system

### **LED Codes:**

After started system/LPAR we can see the right side reference codes (LED) on HMC. once all LED codes completed successfully server will up then we can login to the server.

if server having any problems it will not boot also server will stop with problem LED code. By seeing LED codes we find easily find the system problem to fix it.

### **Mostly Important LED codes are:**

201 - Damaged boot image  
223-229 - Invalid boot list (OS disk not in order)  
551, 555, and 557 - Corrupted file system, corrupted JFS log, and so on  
552, 554, and 556 - Super block corrupted and corrupted customized ODM database  
553 - Corrupted /etc/inittab file

## **BACKUP and RESTORE**

In This chapter we will discuss how to take the system OS backup (rootvg) and Users data backup (Ex: sapvg, javavg etc).

Q) If we lose/removed data in system how to restore from backup.

## **RootVG or OS Backup:**

By using mksysb command we can take the rootvg backup. the rootvg backup we can take into file systems/tape drive or Nim Servers file systems.

**Note:** My environment we will take all servers rootvg backups in NIM server under /export/nim/mksysb file system.

EX:

```
/export/nim/mksysb
/export/mksysb
/export/nim/image
/export/image
```

To take rootvg backup into NIM server's file system.

### **Two ways we can take OS VG backup.**

- 1) Manually run the mksysb and copy(FTP) the file into nim server under (/export/nim/mksysb)
- 2) Mount the NIM mksysb file system by using NFS and run the mksysb manually.

```
client1 (192.168.1.10)
nimserver (192.168.1.200)
```

#### **1) Manually run the mksysb and copy(FTP) the file into nim server under (/export/nim/mksysb)**

```
a)SYN: #mksysb -ieX </FSName/bkpfilenamae>
EX: #mksysb -iex /backup/aix1_rvgbkp_08172016_53_04_02 -> To take the
aix1 server rootvg backup in /backup file system.
b) #cd /backup
c) #ftp 192.168.1.200
d) #cd /export/nim/mksysb
e) #bin, hash, prompt
f) #put aix1_rvgbkp_08172016_53_04_02
```

#### **2) Mount the NIM /export/nim/mksysb file system by using NFS and run the mksysb manually.**

```
a) login client server
b) #showmount -e 192.168.1.200
O/P: /export/nim/mksysb
```

c) mkdir /mnt  
d) mount 192.168.1.200:/export/nim/mksysb /mnt  
e) mksysb -ieX /mnt/aix1\_rvgbkp\_08172016\_53\_04\_02

-i - To create the image.data file  
-e - Exclude the file systems from backup if required.  
-X - Expand the rootvg file systems space if required.

**image.data file:** This file presented under "/" root file system. it contains total rootvg information such as PP size, PV names, PPs, LV Names, LPs, total copies (1,2 or 3) etc.

**/etc/exclude.rootvg:** if we dont want take any file systems backup while taking mksysb backup add the file systems entries in /etc/exclude.rootvg.

Ex: #vi /etc/exclude.rootvg  
^./tmp  
:wq

### **We are excluding/removing /tmp file system from the mksysb backup**

#mkszfile --> To create the image.data file.  
#lsmksysb -l -f <BKPfilelocation> --> To display the mksysb information such as oslevel, TL level, date & time of backup and which are file systems taken in backup.

#lsmksysb -L -f <BKP filename> --> To display to package/software in backup.  
#lsmksysb -f <BKP filename> --> To list the all file & Dirs, with full paths on rootvg filesystems.  
#lsmksysb -r -f <BKP Filename><./filename> To restore the particular file from the mksysb backup.

### **To take rootvg backup into tape drive**

#lsdev -Cc tape -> To list/verify the tape drive  
#mksysb -ieX /dev/rmtX

### **To take the users created vgs backup:**

Ex: sapvg backup

All user created vgs we can take by using #savevg command.

#savevg -ief <BKP file location><vgname>

Ex: #savevg -ief /backup/sapvg\_bkp\_170820116 sapvg --> To take the sapvg backup into /backup file system.

-i - Create image.data file for sapvg

-e - exclude the file systems from sapvg backup  
-X - Expand the rootvg file systems space, if required.

## **Restoration:**

**Rootvg Restore:** if rootvg data lose/corrupted, if we have mksysb backup we can restore through nim server.

## **User VGs:**

```
#restvg -f <backupfilelocation><pvname><pvname>
#restvg -f /backup/sapvg_bkP_08172016 hdiskx hdiskX
#mksysb -ieX <tapedrive>
#mksysb -ieX /dev/rmt0 --> To take the rootvg backup into tape drive.
#savevg -ief /dev/rmtX sapvg --> To take the sapvg backup into tape drive.
```

## **Note:**

Actually in office Aix Admins will take only rootvg backup.

User created VGs backup will taken by the backup team.

Ex: Legato backup team, TSM Backup team, Net Backup etc.

In my office environment we are using legato backup  
Legato Backup Team

```
#ps -ef |grep -i nsr -> To verify the legato process running or not.
#/etc/rc.nsr -> To start the NSR services
#/bin/shutdown -> To stop NSR services
```

client server: 192.168.1.10  
backup server: 192.168.1.20

**SYN:** #mminfo -s <bkp server> -c <client server> -t <time>

EX: #mminfo -s 192.168.1.20 -c 192.168.1.10 -t one week ago --> To list the file systems on 192.168.1.10 server, which are taken backup by the backup server (192.168.1.20).

#recovery -> To recovery the data from legato backup.

## **Backup Types:**

- 1) Full backup
- 2) Incremental backup or Daily backup
- 3) Differencial backup

## **Full backup:**

To take the total file system backup Ex: weekly once or 15 days once or monthly once.

### **Incremental backup:**

This also called daily backup, this backup will not take the total file systems backup, It will take the backup of newly added data/modified data only.

### **3) Differencial backup:** its date to date backup.

Ex: from Dec 05th to 15th to take backup

```
#ps -ef |grep -i nsr --> To check the legato process.
/etc/rc.nsr start -> To start the legato /services
/etc/rc.nsr stop -> To stop the legato services
```

### **Tape Drive Layout:**

In tape drive layout 4 blocks will be available, each block is 512MB by default.

### **TAPE Drive:**

We will take data backup into tape drive (rootvg, User VGs)  
#mksysb -iex /dev/tapedrive (ex: rmt0, rmt1)

### **Tape drive Layout:**

Tape drive layout contains the 4 blocks.

### **Block 1) Boot image:**

Boot image is available on 1st block of the tape drive, its contains the kernel and device drivers information. this information will help to boot from the tape drive. it will create the bos boot command.

### **2) Mkinsttape image.**

second block contains the mkinsttape image. this image contains the three important files.

- a) tapeblksz
- b) image.data
- c) boinst.data and commands

**tapeblksz** --> its contains tape block size information. Ex:512, 1024

**image.data** -> This file contains the, total vg information, such as LV, LPs, PVs, PPs and PP size etc.

#mkszfile -> to create image.data file.

bosinst.data -> This file we can customize how we can restore the data from tape drive.

Location: /var/adm/ras/bosinst.data

INSTALL\_METHOD = overwrite or migration or preservation

PROMPT = yes or no

ACCEPT\_LICENSES = yes

**target\_disk\_data:**

PVID = 00cb9dbe3372131d

HDISKNAME = hdisk3

### **3) Dummy.toc**

dummy.toc is available on third block of the tape drive. its contains structure of backup data information.

### **4) rootvg data.**

4th block contains the total rootvg data information (mounted file systems information).

## PERFORMANCE MONITORING/ANALYSIS AND TUNING

As a system administrator we should have general knowledge about system performance. This knowledge can help to find/analyze the system behaviour and fix the system performance related issues.

Generally, system performance depends on 5 key resources such as

- 1) Processor/CPU
- 2) Memory
- 3) I/O Devices (Disks/Adapters/Cd-rom/tape)
- 4) Network
- 5) File systems

If server performance is low we have to find the issue and need to fix the issue. To find performance related issues we have to analyze above 5 resources.

To analyze 5 key resources follow the below procedures/process one by one.

### 1) Processor/CPU (or) #sar:

CPU we can analyze by using #sar command,

**SYN:** #sar <seconds><lines>

Ex: #sar 2 10 -> To list CPU performance status every 2 seconds 10 lines.

In sar command it will list 4 fields such as %usr, %sys, %wio %idle

**%usr:** if we see more CPU usage under %usr, nothing but some users are using more CPU in the system we have to find the which user using more CPU by using below command.

#ps aux -> To list the users which are using more CPU/Memory

**Note:** if applications (apache/java/tomcat/sas/pega) or Databases (oracle/DB2) guys are using more CPU. We have to inform to them, if root/admin users are using more CPU we have to fix the issue.

**%sys:** if we see more CPU usage under %sys nothing but kernel may be using more CPU means in the system may be Zombie/Defunct/Dead processes are running in the system. if any zombie processes are running in the system we can kill them.

**Zombie Process:** This process also we can call defunct or dead process means after process work completed still it is listed in processing table and it may be using more CPU/Memory in system. Because of zombie processes sometimes system performance will become low.

`#ps -ef |grep -i defunct -->` To list zombie/defunct processes in processing table.

**Note:** if we see any zombie process in server first we can try to kill them PID, if PID not killing we can PPID (Parent PID) if still zombie process not killing finally we can reboot the system.

**%wio:** if we see more CPU usage under %wio means some I/O devices may be using more CPU so we have to find the which are I/O devices are using more CPU by using `#iostat` command and take required action.

**%idle:** In the idle field, we can see percentage CPU is free in the system. under idle field if we see above 30% we dont have any CPU related issues in the system if we see below 30% the server having CPU bound related issues.

## Memory Analysis:

We can perform memory analysis by using below commands

```
#vmstat
#svmon
#lsps
```

In the system we have two types or memories one is real memory another one is paging space.

Both memories we can analyze by using `vmstat` command.

EX: `#vmstat 2 10 -->` To list memory statistics every 2 seconds one line upto 10 lines.

in `vmstat` command 5 columns will be listed such as

- 1) kthr (Kernel Thread)
- 2) Memory
- 3) Page
- 4) faults
- 5) CPU

### 1) kthr:

under Kthr we can see two fields (r, b) under "r" field we can value of processes how many are currently running in the system.

Under "b" field we can see how many value of processes how many are waiting for the memory.

**Note:** if we see any value under "b" field doesn't mean we have memory related issues in the system. if "b" having value is zero we don't have any memory related issues in the system.

### 2) Memory:

under memory we see below fields

- a) avm
- b) fre

**avm:** this field contains the value of pages, means how many pages are free in paging space we can list.

**fre:** this field contains the value of frames, means how many frame are free in real memory we can list.

### 3) Page:

Under page column we can below 3 important fields

- a) pi - Page In
- b) po - Page out
- c) re - Replacement

The above 3 fields having any value doesn't mean we have memory related issues in the system, if 3 values are zero value we don't have any memory related issues in the system.

replacement means if any process is moved from real memory to paging space and when real memory become free the same process is moving paging space to real memory is called replacement.

### 4) Faults:

under faults column we can analyze one important field such as "in" in means interruption, if we see any value under interruption doesn't mean while doing read/write operations on I/O devices server is facing performance related issues in this situation we have to analyze the I/O devices again by using #iostat command.

### 5) CPU:

This column contains same attributes which are listed by #sar command. we have to analyze them as discussed in #sar command.

### I/O Devices:

I/O Devices we can analyze by using `#iostat` command. `#iostat` command will list the below fields

- 1) `tm_act`
- 2) `Kbps` (Kilo bits per second)
- 3) `tps` (Transfer per second)
- 4) `Kb_read` (Kilo bit read operations)
- 5) `Kb_write` (Kilo bit write operations)

#### 1) **tm\_act:**

under this field we can see how usage of I/O devices if we see any I/O device having more percentage usage we have analyze whats going in the I/O device Ex: `hdisk10` is using more usage then we have to analyze in `hdisk10` performance by using below commands.

`#lsnv -l hdisk10 ->` To find the content of `hdisk10`, if we Application/Database related file systems we have to inform to them on your file systems more I/O operation going please check them.

`#lsdev -Cc disk |grep -i hdisk10 ->` To find disk available status

`#errpt |grep -i hdisk10 ->` To find `hdisk10` errors if anything listed in `errorlog`.

**2) Kbps:** This field will list the value of (read & write) operations per second.

**3) tps:** This field will list the value of only transfer I/O operations per second

**4) Kb\_read:** This field will list the value of only read operations.

**5) Kb\_write:** This field will list the value of only write operations.

#### 4) **Network Analysis:**

By using `#netstate` command we can analyze the network statistics.  
`#netstate -i >` This command will list the network data transactions information such as

- 1) `Ipkts`
- 2) `Ierrors`
- 3) `Opkts`
- 4) `Oerrors`

**1) Ipkts:** Under Input packets we can see value of receiving packets means how much data server is receiving we can see.

**2)Ierrors:** Stands for Input errors, while receiving data if any packets (data) lose. we can missing packets value under `Ierrors`.

**3)Opkts:** Stands for Output packets. under this field we can see value of how many packets (data) the server sending in network.

**4)Oerrors:** Stands for Output errors. while sending data if any packets (data) lose. we can see missing packets value under Oerrors.

**Note:** Ierrors/Oerrors value if zero we dont have network related issues in the system.

if anyone/both are having any value, may be system having network related issues then we have to find the network issues.

**Topas/NMON:** By using topas and nmon commands we can analyze the total system resources statistics at a time.

**Note:** topas by default available in system but NMOM will not be available if we need to use nmon on system we have to download NMON file from the google and copy the file into server under below location then only nmon will work.

**NMON file copy location:**

5) File system analysis/File system House keeping

## NETWORK INSTALLATION MANAGEMENT

By Using NIM Configuration/NIM Server, we can perform the OS Installations, TL Upgrades, and any optional softwares (java, sap etc) installation through network, without cd-rom interruption.

The main advantage of the nim server we can perform any installations at a time on bulk of servers.

In NIM configuration, Network is the main role to perform the installations.

**NIM Server Configuration/To Build NIM Server:**

a) Create NIM Server/Lpar

b) Install Aix OS (5.3,6.1 or 7.1) on NIM Server.

c) Install the NIM supported file sets /softwares on the NIM Server from the same OS CD.

**Note:** If server is running with 6.1 version, NIM supported file sets we have to install from 6.1 CD only.

```
#installp -agXY -d /dev/cd0 bos.sysmgt.nim.master bos.sysmgt.nim.spot
(or)
#smitty install -> Install and update software -> Install software -> INPUT
device/ director for software [/dev/cd0]
```

|                                         |                         |   |
|-----------------------------------------|-------------------------|---|
| * INPUT device / directory for software | /dev/cd0                |   |
| * SOFTWARE to install                   | [Select Nim software] + |   |
| COMMIT software updates?                | No                      | + |
| SAVE replaced files?                    | yes                     | + |
| EXTEND file systems if space needed?    | yes                     | + |
| DETAILED output?                        | yes                     | + |
| ACCEPT new license agreements?          | yes                     | + |

```
#lsipp -l |grep -i nim --> To list installed nim file sets.
bos.sysmgt.nim.client
bos.sysmgt.nim.client
bos.sysmgt.nim.master
bos.sysmgt.nim.spot
```

**Note:** bos.sysmgt.nim.client file sets will be install, when we are installing operating system. if we need to configure any aix server as NIM server we have to install master and spot file sets.

d) Configure the NIM Network.

Configure the IP address for available ethernet adapter then proceed with below steps

```
smitty nim
Configure the nim environment
Advance Configuration
Initialize the NIM Master only
```

```
* Network Name [nim_network]
* Primary Network Install Interface [en0]
```

```
lsnim -c networks
nim_network networks ent
```

**Note:** Once configure the nim network, two nim demons will be started such as: nimesis, nimd

3) Configure the lpp\_source

**lpp\_source Defination:** lpp\_source is a resouce in NIM server, in lpp\_source file system we will copy the OS images from the CD so it is equal

to OS cd only. by using lpp\_source we can install OS on multiple nim clients at a time.

**Note:** before configure lpp\_source we have to create the one lpp\_source file system to store/copy the OS images.  
(Ex: /export/lpp\_source or /export/nim/lpp\_source)

Ex:

```
#crfs -v jfs2 -g nimvg -a size=10G -A yes -m /export/lpp_source
#mount /export/lpp_source
```

**Note:**

- 1) lpp\_source file system minimum 5G size required to copy one version OS.
- 2) lpp\_source file system we can create any one of the VG, which VG we are having free space.

Once create lpp\_source file system follow below steps to copy OS images from CD to FS (/export/lpp\_source).

```
smitty nim
Perform nim administration tasks
Manage Resources
Define a resource
Select the resource "lpp_source"
Fill the below final window
*Resource Name [lpp5304]
*Resource Type [lpp_source]
*Server of Resource [master]
*Location of Resource [/export/lpp_source]
source of installation images [cd0]
```

**Note:** once click enter cd0 images will copy into /export/lpp\_source file system.

lpp\_source configuration will take 30 to 45 minutes, if we have more free memory and CPU, it will take 10 to 20 minutes time.

4) Configure the SPOT

**SPOT Definition:** SPOT stands for Shared Product Object Tree, SPOT we can configure from the lpp\_source or mkysyb resources, if we need to install OS (lpp\_source) or rootvg mkysyb restoration we should configure the respected spot for them. without spot we can't install OS (lpp\_source) or we can not restore rootvg mkysyb backup.

SPOT is equal to /usr file system, it is contain the booting related commands, the nim clients initially boot from the SPOT only.

Before configure the SPOT we need to create separate file system to store SPOT related data. Minimum 512MB space is required to configure one OS version spot

EX: /export/spot or /export/nim/spot

```
#crfs -v jfs2 -g nimvg -a size=5G -A yes -m /export/spot
#mount /export/spot
```

**Note:** minimum spot file system space is required 512MB to store one version OS or one server mkysb.

Once spot file system created & mounted follow the below steps to configure the spot from lpp\_source.

```
smitty nim
Perform nim Administration tasks
Manage Resources
Define a Resource
Select a resource "spot"
fill the below final window step
*Resource Name [spot5304]
*Resource type [spot]
*Server of Resource [master]
*Source of Install images [lpp5304]
*Location of resource [/export/spot]
```

Once filled fields hit enter to configure the spot from lpp\_source.

**Note:**spot configuration will take 30 to 45 minutes, if we have more free memory and CPU, it will take 10 to 20 minutes time.

## 5) Configure the NIM Client:

Before configure the nim client we have to add client server IP and hostname in /etc/hosts file on NIM Server. once added client entries on /etc/hosts file follow the below steps to configure the nim client

```
#smitty nim
Perform NIM Administration tasks
Manage Machines
Define a machine
*Host name of the machine [nimclient1]
hit two times enter, client will be configure.
```

#lsnim -c machines --> To list the configured client servers.

### **To remove the nim configuration:**

- 1) #smitty nim --> perform nim admin tasks --> unconfigure nim
- 2) remove the nim file systems
- 3) uninstall nim softwares
  
- 6) Add/Allocate the resources (lpp\_source & spot) to nim clients, on which are the servers we have to install OS.
  - a) #smitty nim\_bosinst
  - b) Select the client server (ex: sapserver),
  - c) Select Installation Environment Ex: rte for lpp\_source installation.
  - d) Select the lpp\_source which version going to install [ex: 5.3) "lpp5304"
  - e) Select respected spot Ex: "spot5304"

On Final window fill the below steps.

```
* Installation Target [aix3]
*Installation Type [rte]
*Spot [spot5304]
*lpp_source[lpp5304]
Accept License Agreement [yes]
Force PUSH Installation [no]
Initiate reboot and Install now [no]
Accept new license Agreement [yes]
```

**Note:** Before install OS, we need add the minimum below hardware configuration.

- a) one hard disk (minimum 5 to 10G)
- b) One Ethernet Adapter (Network)
- c) Minimum CPU and Memory.

7) Go to client side to install the OS by using below steps.

Once added required hardware, Start/Activate the server and enter SMS menu.

Select Server --> Operations --> Activate --> Profile --> Advanced --> Select SMS menu --> click OK

**Once we have entered into SMS menu we have to perform below two steps to start installation.**

- a) ping test between nim server & client server
- b) Select first boot device as Ethernet adapter because installation will done through the network.

**a) PING TEST STEPS:**

- 1) Enter SMS menu
- 2) Select "Setup Remote IPL (initial program load)
- 3) Select the Ethernet adapter
- 4) Select the "IP Parameters" option to add Server & Client IPs
- 5) Once added IPs, return back previous window (esc)
- 6) Select the "Ping test" option.
- 7) Execute the ping test by select number "1"

**b) SELECT FIRST BOOT DEVICE AS NETWORK ADAPTER FOLLOW THE BELOW STEPS:**

- 1) Come to the Main menu in SMS (m)
- 2) Select boot options (5)
- 3) Configure boot device order (2)
- 4) Select 1st boot device (1)
- 5) Select Network (6)
- 6) Select the Ethernet adapter as 1st boot device
- 7) Select Set boot sequence: configure 1st boot device (2).
- 8) Select number "1" Ethernet adapter device
- 9) Exit from SMS menu.

**Next:** Once exit from SMS menu Installation will start then follow/select the installations options as usual like cd installation.

- 1) Define Console window
- 2) Select Language English
- 3) Installation setting window, here three options
  - a) Default settings
  - b) Change/Show installation settings
  - c) Maintenance mode

Once setting completed select "1" and "0" to proceed with installation.

**To remove network:**

Smitty nim  
Perform NIM Administration Tasks  
Manage Networks  
Remove a Network  
Select the Network, which is need to remove.  
click enter button two times.  
To Install OS on servers by using NIM server.

**Follow the below steps on NIM Server:**

- 1) Configure the NIM Server
- 2) Configure the NIM Network
- 3) Configure the lpp\_source

- 4) Configure the spot
- 5) Configure the NIM Client
- 6) Add the resources (lpp\_source and SPOT ) to the nim clients.
- 7) Go to client side install the OS by using below steps.

### **Troubleshootings:**

#### **Problem: Ping test fails**

##### **Find the issue:**

- 1) Verify adapters state on NIM server and Clients.
- 2) Verify Interfaces state both sides
- 3) Verify the LAN cables link status.
- 4) IP Parameters may be wrong.

#### **Problem2:**

Once assigned resources to client. Client not booting to install OS via NIM.

##### **Reasons:**

- 1) Verify the ping test. if success then
- 2) May be bootd / tftpd demons not running on NIM server  
#bootps -d -d -d -d -d
- 3) May be /etc/niminfo file missing on NIM server
- 4) may be lpp\_source or SPOT not good /not configured properly.

### **NIM Activities:**

#### 0) Configure NIM Server

- 1) How to install OS on new servers through NIM Server. (or) How to install OS by using lpp\_source on NIM clients.
- 2) How to perform Migration Installation through NIM Server on nim clients.
- 3) How to take the mkysyb backup of nim clients on NIM Servers.
- 4) How to restore mkysyb backup on nim clients by using NIM Server.
- 5) How to perform TL Upgrade on NIM Clients via NIM server.

(II) How to perform Migration Installation through NIM Server on nim clients.

We can perform migration installation two ways on nim clients by using nim server.

- 1) Manually we can assign the latest version resources to nim clients perform the pull method.
- 2) By using "nimadm" command we can perform.

## Migration Installation via NIMADM command:

nimadm is one of the advanced feature in NIM installations, when we are Migrating OS from lower version to higher version we can use "#nimadm" command to perform the migration installation. Once executed nimadm command on nim server, the command will run the 12 Phases/Steps to complete Migrations Installation on client side.

Ex: sapserver is nim client and running with 5.3 version. Now how to migrate sapserver with 6.1 version through nimadm.

## Follow the below steps to Migrate OS.

### Pre-Works:

- 1) Take one empty hard disk on nim client (sapserver) Ex: hdisk1.
- 2) Make sure network communication between nim server and client by adding both servers IPs on /etc/hosts both sides.
- 3) Configure the 6.1 version lpp\_source and spot.
- 4) Configure the sapserver client server
- 5) /etc/niminfo file should be presented on nim server and nim client side also (aix1).
- 6) Enable the NIM operation permissions to nim server on nim clients.
- 7) Nim client side and NIM server alt\_rootvg supported file sets should be installed.  
bos.alt\_disk\_install.rte  
bos.alt\_disk\_install.boot\_images

**Resources:** lpp6108 and spot6108

**NIM client:** sapserver, Empty disk: hdisk1

### NIMADM COMMAND:

**Note:** nimadm command we will run on the NIM Server.

```
#nimadm -j <vgname> -l <lpp_source> -s <spot> -c <client server> -d
<client emptydisk> -Y
```

```
EX:#nimadm -j nimvg -l lpp6108 -s spot6108 -c sapserver -d hdisk1 -Y
```

j - vgname means on Which VG lpp\_source & Spot file systems are presented.

-l -> lpp source name, which OS version going to Migrate/Install

-s -> respected SPOT of lpp\_source

-c -> Client server name, which is going to Migrate

-d->Client side empty Hard disk to create Clone/alt rootvg

-Y -> Accept the licence agreement.

Once executed nimadm command it will run the below 12 phases to completed migration installation.

### **nimadm 12 Phases:**

- 1) It will create the alternate rootvg on nim client.
- 2) Nim server will export the all alternate file systems (with read/write permissions).
- 3) On clients side old rootvg data will take backup and it will restore on the altinst\_rootvg.
- 4) Pre-migration script will be execute.
- 5) Client server configuration files will be saved.
- 6) All old version OS packages will be migrated with new version packages
- 7) Once OS packages migrated, Post-Migration script will be executed.
- 8) #bosboot command will create boot image on alternate\_rootvg disk
- 9) all alt filesystem will be removed from NIM server.
- 10) alt filesystems are mounted on nim client.
- 11) again alt\_disk\_install command will run to verify final adjustments in altinst\_rootvg status.
- 12) bootlist order will created as (hdisk1) and reboot the system.

### **MKSYSB BACKUP AND RESTORE THROUGH NIM SERVER:**

#### **MKSYSB BACKUP via NIM:**

**mksysb backup we can take two ways in nim server.**

- 1) Manually mksysb backup
- 2) Automatically mksysb backup through NIM Server

Nim Server: 192.168.1.200

Nim client: 192.168.1.30

**Note:** All servers mksysb backups we will store under /export/nim/mksysb (or) /export/nim/image (or) /export/mksysb, /export/image file systems

#### **1) Manual nim clients mksysb backup:**

Create mksysb image on nim client and copy to the NIM server by using FTP or NFS.

- a) Login client server
  - b) create mksysb image into any file system
- Ex: #mksysb -ieX /tmp/mksysbfilename

c) copy mksysb image on nim server under /export/nim/mksysb file system by using NFS or FTP

## **2) Automatic mksysb backup on nim server:**

- a) Login to the NIM Server and NIM client server.
- b) Verify the network communication between both servers
- c) Ensure the /etc/niminfo file should be present on both servers.  
#nimconfig -r -> To create /etc/niminfo file on NIM server.

To create /etc/niminfo file on nim client server.

#nimit (or) smitty nim -> Configure Network Installation Management Client Fileset -> fill below options

```
* Machine Name [sapserver]
* Primary Network Install Interface [en0] +
* Host Name of Network Install Master [nimserver]
```

- d) Enable the NIM Server permissions on nim client side (ex:aix100)  
#smitty nim -> Manage Network Install permissions -> Select enable option  
-> enter
- e) Configure the NIM client on server. if already configured ignore this step  
(#lsnim -c machines)
- f) Take the mksysb backup by using below steps.

smitty nim

Perform NIM Administrations Tasks

Manage Resources

Define a Resource

Select mksysb resources "mksysb"

Fill the below fields in finally window.

```
*Resource Name [aix100_mksysb_08232016]
*Resource Type [mksysb]
*Server of Resource [master]
*Location of Resource [/export/mksysb/sapserver/sapserver_mksyb_5308]
--> aix100_mksyb_5308 is rootvg image file name.
*Create system backup images [yes]
*NIM Client to Backup [aix100] --> on which client mksysb backup taking.
```

## **Restore MKSYSB Backup image via NIM Server:**

### **Manually backup restoration:**

- a) Create mksysb resource/image on NIM Server by using backup image file (mksysb\_sapser\_5304\_122716).
- b) Create SPOT with respected mksysb
- c) Configure the NIM client

- d) Assign the resources (mksysb & spot) to nim client
- e) Go to the client side SMS menu, do ping test and first boot device setting.
- f) Exit from SMS menu and follow the installation steps.

**Note:** Change show settings we can select installation type New and completed overwrite and same rootvg disk.

- a) Create mksysb image on NIM Server by using backup image file.

```
smitty nim
Perform NIM Administration tasks
Manage resource
Define a resource
Select "mksysb" resource
Fill the below steps in final window
*Resource Name [sapserver_mksysb_5304] -> Restoration image name.
*Resource Type [mksysb]
*Server of resource [master]
*Location of resource [/export/mksysb/sapserver/sapserver_mksyb_5304]
```

- b) Create SPOT with respected mksysb:

```
smitty nim
Perform NIM Administration Tasks
Manage Resource
Define a Resource
Select "spot" resource
Fill the below final window steps:
*Resource Name [sapserver_spot_5304]
*Resource Type [spot]
*Server of Resource [master]
*Source of Install image [sapserver_mksysb_5304] --> Select correct
client server image file which is need to be restore.
*Location of Resource [/export/spot/]
```

- c) Configure the NIM client, if already configured ignore this step (#lsnim -c machines).

- d) Assign the resources (mksysb & spot) to nim client:

```
smitty nim_bosinst
Select client server "sapserver" to restore mksysb
Select Installation environment "mksysb - Install from a mksysb image"
Select correct mksysb image "sapserver_mksysb_5304"
Select correct spot "sapserver_spot_5304"
Fill the below fields on Final window.
* Installation Target [sapserver]
```

\*Installation Type [mksysb]  
\*spot [sapserver\_spot\_5304]  
\*mksysb [sapserver\_mksysb\_5304]  
Accept new license Agreement [yes]  
Force PUSH Installations [no]  
Initiate reboot and Installation Now [no]  
Accept new license Agreement [yes]

- e) Go to the client side SMS menu, do ping test and first boot device setting.
- f) Exit from SMS menu and follow the installation steps.

### **Automatically mksysb backup restoration:**

If we have mksysb image backup by using automatically mksysb backup concept no need to configure NIM mksysb because automatically backup method will configure NIM mksysb image. #lsnim -t mksysb

- a) ls nim -t mksysb  
mksysb\_sapser\_5304
- b) Create SPOT
- c) Configure NIM client (aix100)
- d) Assign the nim resources (mksysb and spot) to client server.
- e) Go to NIM client side in SMS menu, ping test and first boot device select network and exit from SMS menu.
- f) Exit from SMS menu and follow the installation steps.

**Note:** Change show settings we can select installation type New and completed overwrite and same rootvg disk.

### **What is recent activity**

### **What is recent troubleshooting**

### **What is recent critical activity done**

**ANS)** One of the production server rootvg corrupted after TL upgrade installations/OS Migration then we have restored mksysb image from the NIM Server by using below steps.

- a) We have all servers rootvg images on /export/mksysb/ file system in NIM server.
- b) login to the nim server.
- c) Verified network communication between both servers
- d) Configured NIM mksysb by using rootvg image backup file.
- e) Configured spot
- f) Configured NIM client (sapserver)
- g) assigned resources to nim client
- f) enter into SMS menu in client, ping test and network selected first boot device, exit from sms menu.
- h) Follow the installation steps.

**Note:** The restoration will take time 45 minutes to 1 hour.

### **TL Upgrade by Using NIM Server:**

TL Upgrade we can do two ways by using NIM server.

- 1) By using NFS/FTP we can install TL packages
- 2) By using lpp\_source we can install TL Packages

**Note:** To install TL Packages SPOT is not required because client server already running with OS.

- 1) To Install TL packages from 61-06 to 08 Level By using NFS/FTP.

NIM Server: 192.168.1.200

Client Server: 192.168.1.30

TL Packages location : /export/software/TL08

- a) Take all pre-works on nim client before install TL packages
- b) Go to NIM Server export the /export/software file system.

```
#vi /etc/exports
/export/software
:wq
#exportfs -a
```

- c) Go to client side mount the NFS file system

```
#showmount -e 192.168.1.200
Op: /export/software
#mkdir /local
#mount 192.168.1.200:/export/software /local
#cd /local/TL08
#smitty update_all -> To update TL packages
```

- 2) To Install TL Packages by using lpp\_source

- 1) Configure lpp\_source
- 2) Configure client
- 3) Go to client side and Install TL Packages

```
#smitty nim_task_install
```

Resource name [lpp6108TL]

Location of resources [/export/nim/lpp\_source]

Source of install [/export/software/TL08]

**Note:** To install TL Packages SPOT is not required because client server already running with OS.

### **NIM Commands:**

lsnim --> To list all nim resources, machines, networks etc.

lsnim -l --> To list all nim resources, machines, networks with their attributes.

lsnim -l <resourcenam> --> To list the resource attributes.

lsnim -c networks --> To list nim networks.

lsnim -c resources --> To list nim resources (lpp\_source, mkysyb and spots)

lsnim -c machines --> To list NIM client servers.

lsnim -t lpp\_source --> To list only lpp sources

lsnim -t spot --> To list only spots

lsnim -t mkysyb --> To list only mkysybs

**To create lpp\_source/mkysyb/spot by using command mode.**

```
#nim -o define -t lpp_source -a location=<lpp_sourcefilesystem> -a
server=<server of resource> -a source=<source of
images><lpp_source_name> --> To create lpp_source.
```

```
Ex:#nim -o define -t lpp_source -a loaction=/export/lpp_source/ -a
server=master -a source=/dev/cd0 lpp5304
```

### **To create NIM mkysyb**

```
#nim -o define -t mkysyb -a location=<mkysyb filesystem> -a
server=<server of resource> -a source=<source of
images><mkysyb_resource_name> -->
```

```
Ex:#nim -o define -t mkysyb -a location=/export/mkysyb -a server=master -a
source=/export/mkysyb/aix100_mkysyb_53 sapserver_mkysyb_5304
```

```
#nim -o define -t spot -a server=<server of resource> -a
source=<lpp_source/mkysyb> -a location=<spot file system><spot
resource name> --> To create spot
```

```
Ex:#nim -o define -t spot -a server=master -a
source=/export/nim/lpp_source/lpp5403 -a location=/export/spot/ spot5403
--> To create spot for aixlpp53 installation
```

```
#nim -o define -t spot -a server=master -a source=/sapserver_mkysyb_5304
-a location=/export/spot aixmkysyb_spot_53 --> To create spot for aix100
server mkysyb restoration.
```

```
#nim -o reset -a force=yes client/master --> To reset the client server or
master server.
```

```
EX:#nim -o reset -a force=yes aix100
```

### **by using smitty**

smitty nim  
Perform nim Administration tasks  
Manage machines  
Perform operations on Machines  
Select server  
Select reset option and enter

### **How to deallocate/remove the resources from the nim client.**

smitty nim  
Perform nim Administration tasks  
Manage machines  
Perform operations on Machines  
Select server  
Select "reset"  
Fill fields final window  
Target name [sapserver]  
Deallocate all resources [yes]  
Force [yes]  
and enter.  
#nim -Fo deallocate -a subclass=all clientname -> To deallocate the resources.  
#nim -Fo check <lpp/mksysb/spot resource> --> To check the resource condition/status/health checking etc.,

### **To create /etc/niminfo file on nim server side and nim client side.**

#### **NIM Server:**

smitty nim  
Perform nim administration Tasks  
Rebuild the niminfo file on master enter  
(or)  
#nimconfig -r --> To create nim info file

#### **NIM Client side**

smitty nim  
Configure Network Installation Management client files  
\*Machine Name [aix100]  
\* Primary Network Install Interface [en0]  
\* Hostname of Network Install Master [nimserver]  
(or)  
#niminit -a pif\_name=en0 -a name=aix100 -a master=nimserver

### **To remove the machines/networks/lpp\_sources/spots/mksysb.**

EX:smitty nim

Perform NIM Administration Tasks

Manage Networks --> To remove network

Manage resources --> To remove resources (spot/mksysb/lpp\_sources)

Manage machines --> To remove clients.

### **NIM Demons:**

Nim network demons:

1) nimesis

2) nimd

above demons will create after configured network

### **Client side booting supported demons are"**

3) bootpd

4) bootps

5) tftpd

NFS demons also main role in NIM

### **NIM Configuration Files:**

/etc/niminfo

/etc/bootptab

/etc/hosts

/etc/exports

/ftfpboot --> It is directory.

### **NIM Ports:**

1058 --> NIM server port

1059 --> NIM Registration port

### **/etc/niminfo:**

It is important nim configuration file should be presented on both sides to perform nim operations. this file contains the nim server and client server hostnames, IPs and port numbers etc.

### **/etc/bootptab:**

This is one of the nim configuration file, once assigned the resources to nim client from NIM server, one entry will create on /etc/bootptab.

**Note:** Once installation completed entry will remove from /tftpboot directory

### **Entry Format:**

nimclientname: client IP: nimserver IP: netmaskvalue

Ex: aix100: ip=192.168.1.10:192.168.1.200:255.255.255.0

### **/etc/hosts:**

This is network configuration file both sides should be presented with both servers IPs and Hostname

### **/etc/exports:**

It is NFS configuration file in nim NFS is main role to perform the installations on nim clients. NIM server export the nim file systems and import into client side in the time of lpp\_source installations or mkysyb restoration etc.

### **/tftpboot:**

It is very important directory in NIM, once assigned the resources to client server 3 files/entries will create under /tftpboot, after installation completed the entries will be remove from /tftpboot directory.

### **3 entries format:**

a) file1

sapserver\_mkysybspot\_53.chrp.mp.ent

sapserver - client server

mkysybspot\_53 -> Spot name

chrp.mp.ent --> mp means 32 bit kernel

chrp.64.ent --> 64 kernel

b) file2

sapserver -> /tftpboot/aix53spot.chrp.mp.ent

One soft link file will create with client name and spot file.

c) file3

sapserver.info

this client information file, by using this file we can see installation completed percentage/information in the time of installation.

#lsnim -a info (or) lsnim -a aix100 --> To see the client information/ Installation Percentage status.

## **HMC (Hardware Management Console)**

HMC stands for Hardware Management Console by using HMC we can manage the IBM Power Servers (P5, P6 and P7 servers).

HMC OS: HMC OS is combination of Linux and Java Versions.

HMC Version: 7.7 and 8.0

HMC Models: CR-7045 and CR7046

HMC Login: We can login HMC two ways

1) By using Web Browser - GUI mode

2) By using Putty - Command mode.

### **1) Web Browser Login:**

1) Open Internet Explorer/ Chrome/Fire Fox

2) Give the HMC IP or hostname in title/site bar

- 3) Click on launch HMC button
- 4) Type HMC username (hscroot) and password (abc123)

## **2) Putty/CMD HMC login:**

- a) Open putty
- b) give HMC IP/Hostname in Putty & Select SSH
- c) Click on open button

## **HMC Activities:**

HMC activities are two types

- 1) Managed Systems activities
- 2) LPAR Activities

## **HMC:**

Default Username & Password:  
hscroot & abc123

## **HMC First Window:**

Welcome  
System Management (VVV IMP)  
System plans  
HMC Management (IMP)  
Updates (Firmware Upgrade)

## **System Management:**

This view having two options:

- a) Servers --> To list the Connected Managed Servers to HMC
- b) Custom Groups --> To list total lpars of all Managed Systems,

## **Managed Servers Activities via HMC:**

### **Properties**

#### **General:**

Managed System name, serial number, model number and Max lpars can create options available.

#### **Processors:**

Total Installed Processors, Available/Free Processors, Configured Processors Deconfigured Processors we can see.

#### **Memory:**

Total Installed Memory, Available Memory, Configured memory and Deconfigured Memory, Region Size (LMB) we can see

### **I/O:**

I/O Devices are:

- 1) Ethernet adapters ( Network)
- 2) FCS/HBA or SCSI Adapters (For Hard disks)
- 3) RAID Controller/Mass Storage Controller (CR-ROM)

a) Select Manage System (MS) --> Click on properties

### **Operations:**

Power ON / Power OFF MS -> To activate and deactivate the Managed Systems.

**Note:** Before power off Managed System we should shutdown the all LPARs (if we have VIO server first power off the all LPARs then power off the VIO server).

**LED Status** --> To activate/deactivate and test the LED symbol  
ASM or ASMI (Advance System Management Interface)

### **ASMI Default username & password:**

admin & admin

In ASMI Setting we can do MS power off/ Power on and reboot, also we can MS IPs, LMB size we can verify, if we want change LMB we can change it for memory allocation to partitions EX: 16MB, 32MB, 64, 128, 256, 512M, 1024M

### **HMC Points:**

- 1) Single HMC can connect to 48 Managed Systems, if Power 7 servers single HMC can connect only 32 Managed Systems.
- 2) One Managed System can connect to Max two HMCs
- 3) Single HMC can connect difference types of hardware servers (p5, p6 and p7).
- 4) Suppose, HMC having 30 MSs and 400 Lpars all are up and running. if HMC is down/shutdown, for all Lpars and MSs nothing will happen all are running normally but only disadvantage is we are not to perform HMC activities on them.

### **3) CONFIGURATION:**

### **a) Create Logical Server:**

To create Logical server

Connect HMC --> System Management --> Servers --> Select Managed System -> Configuration --> Create Logical Partition --> Select OS

**Environment:** Aix & Linux or VIOS -> fill the Partition ID and Name --> Profile Name (normal/default) --> Processors (Shared or Dedicated) --> Fill the Minimum/Desired/Maximum Processors -> Minimum/Desired/Maximum Memory -> Select required I/O devices (Add as Required/Desired) -> Create Virtual Adapters if required --> Next --> Next --> Final window we can see Creating Lpar/Server

configuration summary information -> Click the Finish Button.

### **Lpar ID and Lpar Name:**

Each lpar having one unique ID and Name, the lpar ID start from number 1, Max IDs numbers range is depend on Installed processors.

Ex: If Frame having 2 processors. Partion ID should be between (1 - 20). if Frame having 3 Processors range should be between (1 -30).

Once create Lpar, ID can not change but lpar name can be changed.

### **Profile or Profile Name:**

Profile contains the total Lpar configuration information such total memory, number of CPUs assigned, I/O devices etc to the lpar.

We can create multiple profiles for single lpar, we can create/edit/copy/activate/delete the profiles.

### **Processors:**

We can assign the processors to lpar in two modes.

- 1) Dedicated mode
- 2) Shared mode

#### **1) Dedicated Mode:**

While adding processors to lpar, if we add the processor as dedicated mode, the entire processor can be used by the assigned lpar. so, if we have 4 processors in Frame, Max we can create 4 Lpars, if we select dedicated mode, To create Lpar minimum 1 processor required

#### **2) Shared Mode or Micro Partitioning:**

By Shared mode, We can share the processors between the Lpars, In shared mode processors will be devided into units. 1 processor equal to 10 units. To create lpar, minimum 0.1 unit required. so by using 1 processor we can

create Max 10 lpar. if we have 2 processors we can create 20 Lpars in Frame. Shared mode processors also we can call Micro partitioning.

### **Difference Between Minimum/Desired/Maximum Values:**

#### **Minimum:**

While activating the LPAR minimum value should required, if minimum value not available in Frame The lpar will not activate.

Minimum value always  $\leq$  Desired and Max values.

#### **Desired:**

While activating the LPAR if desired value available in FRAME, the lpar will take if not available, LPAR will satisfy with minimum value and activate it.

Desired Value always  $\Rightarrow$  Minimum value,  $\leq$  to Maximum Value.

#### **Maximum:**

We can Increase/change the desired value upto Maximum value by using DLPAR operations without bring down the server.

The Max value  $\Rightarrow$  To Minimum & Desired Values.

### **Shared Processors:**

In shared processors, we can add processors/CPUs in Units and we have below advanced features.

Processing Units

Shared Processor Pool (SPP)

Virtual Processors

Capped/Uncapped

Weight Value

### **To Create SPP:**

Select Manage System --> Configuration --> Shared Processor Pool Management --> Click on the spool --> Give the SPP name --> Add the Maximum Processing units---> Click on ok button to create pool.

by using SPP we can share the processing units between the LPARs. SPP feature available only on shared mode processors.

in P5 only one shared processor pool available, P6 & P7 servers we can create Max 64 Shared Processor Pools. In SPP we can add capped & uncapped both type of lpar. SPP IDs range between 0 to 63.

### **Capped Mode/Capped Partition:**

if we create any LPAR in capped mode, after activated LPAR, it will not release the free units to the SPP and It will not take the free units from the SPP.

### **Uncapped Mode:**

If we create any LPAR in uncapped mode, after activated LPAR. it will release free/un-used units to the SPP also if required more units it will take from SPP if available.

**Weight Value:** Weight Value available only for Un-capped partitions/LPARs. By using weight value we can define the priority of the LPAR to take the processing units from the pool. weight value range is 1 to 255. zero is low priority, 255 is high priority. Default weight value is 128.

Ex1: Server1 weight value is 127. Server9 weight value is 180, here SPP will consider server9 as higher priority server.

Ex2: Server1 (ID:5) and server9 (ID:13) both are having same weight value 150, then SPP will give priority to first created LPAR. now SPP will consider Server1 is higher priority server.

### **Virtual Processors:**

Physical processors we can call as a virtual processor. in Virtual processors we define the LPAR how many physical processors have to use.

### **Memory:**

Memory we can add as minimum/desired/maximum. The memory size we Increase/Decrease by using LMB.

### **I/O Devices:**

The I/O devices we can assign to LPARs in two modes.

- 1) Required Mode
- 2) Desired Mode

**1) Required Mode:** If add any I/O devices to LPAR as required mode, if added I/O device available it will take, if not available the LPAR will not be activate.

**2) Desired Mode:** If we add any I/O devices to LPAR as desired mode, If I/O device available it will take, if not available without I/O device it will activate.

### **Managed System's LPARs or Partitions Backup:**

1) Select MS --> Configuration -> Manage Partition Data -> Backup --> Fill the backup file name (Ex: bkp\_08262016)

backup file Location: /var/hsc/profiles/MSSerialnumber/bkp\_08262016

## **SYSTEM PROFILE:**

In system profile we can add multiple LPAR's profiles. By using system profile we can manage few activities on group of servers at a time such as activate/shutdown etc.,

### **To Create System Profile:**

1) Select MS -> Configuration --> System Profile -> Actions -> New -> Type System Profile name -> Select LPAR's profiles, which are need be grouped in system profile -> Click add button --> finally click ok button.

### **Connections:**

Select MS --> Connections -->

In Connections we have below 4 options

- a) Service Processor status --> To check the FSB cable status.
- b) Reset/remove Connection --> Reset means reconnect MS. remove means to remove the MS from HMC.
- c) Disconnect to another HMC --> if we have two HMC to the MS. if we want disconnect one HMC, we can use it.
- d) Add the Managed System --> We can add another/new managed system by using this option.

### **To add MS to HMC**

Select MS -> Connections --> Add the Managed System --> Give the MS IP and passwd --> click ok button.

(or)

Login HMC -> System Management -> select Servers -> Connections -> click on add Managed System -> give the MS IP and Passwd.

### **Updates:**

By using updates we can upgrade/update firmware level from low level to Higher level on Managed System.

Select MS -> Updates -> Change Licence Internal Code for the current release.

### **COD:**

Select

MC - Capacity on Demand (COD)

By using COD we can enable the Inactive/no license devices (Processor, Memory etc). but before enable COD devices, we need get the license from IBM.

### **Managed System States:**

**Power off** --> When all Managed System down

**Initializing** --> When Managed System starting

**Standby** --> When Managed System activated fully and All Lpars will be not activate State.

**Operating** --> If we activate atleast one LPAR the MS state will come to Operating State.

### **Problems:**

no connection --> When FSP connection lost between HMC and MS.

recovery mode -> When MS having issues.

### **Minimum Hardware required to create LPAR:**

**Processor:** 1 (if dedicated) or 0.1 (if shared) -- P5, P6 but in P7 if shared minimum 0.05 required.

**Memory:** 128 MB (aix servers), 256MB (VIO servers)

**I/O Device:** One hard disk (5 to 10GB) to install OS. and If we need network also then we need one ethernet adapter also.

### **Granularity:**

after assigned minimum processor unit (0.1 or 0.05), we can assign smallest processor unit (0.01) minimum. this is called granularity.

Granularity value is 0.01

### **LPAR ACTIVITIES:**

**LPAR:** Stands for Logical partition, it is individual server/system/partition/ in MS. The lpar we can create in Managed Systems by adding available resources (Processor, Memory & I/O) in Frame. each LPAR having separate virtual console,date & time and OS etc.

Same Features are: In a MS all LPARs having Serial Number, Model Number etc

### **1) Properties:**

Select LPAR -> click on "Properties"

In properties we can LPAR information such as.

a) General -> Lpar name and ID

b) Hardware -> Processor/Memory/I/O Devices) assigned Values.

c) Virtual Adapters -> Created Virtual adapters

## **2) Operations:**

a) Shutdown/Restart/Activate/Delete: To shutdown/restart/activate and Delete the lpar,

**Shutdown:** We can shutdown with immediate or Delayed

**Restart:** immediate or Dump (if dump file required).

**Activate:** with required profile we will activate.

**Delete:** if we need to delete the LPAR but LPAR should be down before delete.

**3) Configuration:** it is very important option,

a) Manage Profiles

Select LPAR -> Configuration -> Manage Profile

In Manage Profile we can create new profiles, Delete/copy and edit the profiles.

Edit -> To modify (add/remove) the LPAR configuration (Memory/CPU and I/O) information by using edit option.

## **Edit Profile:**

Select LPAR -> configuration -> Manage Profile -> Select check box of Profile -> Actions -> Edit.

## **Note:**

1) Once modified the profile configuration we have to shutdown and activate LPAR to effect the modified changes. If we reboot the LPAR change will not effect the LPAR will take the last profile settings.

2) When ever we have modified profile we have to shutdown & activate to effect. without shutdown if we need to add we can change configuration by using DLPAR operations.

## **CONSOLE WINDOW:**

a) Open Console window --> To open LPAR console/Virtual Monitor. at a time one person only can access the console.

b) Close Console window --> To close/terminate the console window session if someone accessing.

## **DLPAR Operations:**

It is an dynamic operation to add/remove and move resources (memory/CPU and I/O) to LPARs without bring down the LPAR/Server.

By using DLPAR operations we can add/remove the resources memory/CPU and I/O devices without bring down the server.

### **DLPAR Points:**

- 1) In DLPAR max we can add/Increase value upto Maximum value.
- 2) In DLPAR Max we can remove/Decrease the CPU/Memory value upto Minimum value.
- 3) Before adding resources we have to check the availability of resources in Frame.
- 4) All DLPAR supported demons should be active on HMC side and LPAR side to implement DLPAR operations successfully.
- 5) Once DLPAR operation performed, if we need same settings permanently we need to modify the profile.
- 6) DLPAR operations will not remove after reboot the server. but if we shutdown & activate DLPAR operations will be remove.

### **Steps to DLPAR:**

#### **Pre-checks:**

- a) get info How much extra resource need to add
- b) check Max value of LPAR,
- c) Check free resources availability In Frame.

#### **Implementation:**

Select LPAR -> Dynamic Logical Partitioning -> Select resource (Processors/Memory/Physical adapter/Virtual Adapters) --> Select

#### **Add or Remove or Move:**

**Add:** To Increase the Desired value.

**Remove:** To Decrease the Desired value.

**Move:** To move resources from one lpar to another lpar.

### **DLPAR Demons:**

#### **LPAR Side Demons:**

##### **1) ctrmc**

startsrc -s ctrmc

stopsrc -s ctrmc

##### **2) rsct:**

/usr/sbin/rsct/bin/rmcctrl -z -> To stop the rsct Demons

/usr/sbin/rsct/bin/rmcctrl -A -> To start the rsct Demons

/usr/sbib/rsct/bin/rmcctrl -P -> To Start the rsct demons

lssrc -a |grep -i rsct --> To verify the RSCT demons status

### **HMC Side:**

RMC Demon/connection if not running in HMC.

Need to reboot HMC to start RMC demon hmc side.

Q) When we performing DLPAR operations its fail what is the reason?

1) May be LPAR side ctrmc or rsct demons not active

2) May be RMC connection not active in HMC side

3) May be which is resource trying to add to the LPAR, the resource not available in Frame.

### **Recent Activity**

### **Recent Troubleshooting**

### **Recent critical activity**

## **POWER HYPERVISOR:**

POWER Hypervisor is a firmware, it handle/manage all resources between LPARs as per users configuration. Power hypervisor will provide the virtual console and maintain the security and isolation between the lpars.

### **HMC Management:**

To restart/shutdown HMC

Login HMC --> Click on HMC Management -> select restart option --> Click ok --> To restart HMC

>>>hmcshutdown -t now -r --> To restart HMC

>>>hmcshutdown -t now 0 --> To shutdown HMC

### **HMC Data Backup:**

Click on HMC management -> Click on "Backup HMC Data" -> Select any one of the option below.

a) Back up to media on local system --> CD/DVD/Pendrive

once select "a" option -> Click next button --> Select below listed local media(CD/DVD rom) --> Finally click ok to take HMC data backup into DVD/CD/USB.

b) Back up to Mounted remove systems --> NFS

if we select "b" option --> Click next button --> fill the below field

**Remote Server:** 192.168.1.200 (NIM Server)

**Resource:** /export/mksysb/hmc71

**Description:** HMC data backup of 7.1 version on Aug 29th 2016

once filled above steps click ok button to take HMC 7.1 version backup into nimserver on /export/mksysb/hmc71 location.

**How to restore HMC data Backup:**

Click on HMC management --> Restore HMC Data -->

**IF NFS/NIM Server follow below steps to restore:**

- a) Select "restore from remote NFS server"
- b) Click Next button and fill the below steps:  
Remote Server : 192.168.1.200  
Resource: /export/mksysb/hmc71
- c) click ok button to restore HMC backup data from 192.168.1.200 NIM Server from /export/mksysb/hmc71/ location

**IF need to restore from the CD/DVD/USB follow below steps**

- a) Select "restore from removable media" option
- b) Select below listed DVD rom
- c) Click ok button to restore HMC backup from DVD

**PROCESSORS TYPES:**

- 1) Physical Processors
- 2) Virtual Processors
- 3) Logical Processors
- 4) Shared Processors/ Micro-Partitioning
- 5) Dedicated Processors
- 6) Inactive Processors
- 7) De-configured Processors

**LPAR States:**

- 1) Not available -> If MS down
- 2) Not Activated -> MS up but LPAR not activated
- 3) Open Firmware -> If we have booting related issues
- 4) Running -> Once activate the LPAR, if everything is fine, LPAR will be running state.

Q) System/LPAR Not booting or not activation reason?

**ANS:**

- a) May be bootlist order changed, if OS disk not 1st boot device
- b) May be OS corrupted
- c) May be Memory/CPU 100% usage in System

**To add I/O Devices to LPARs**

Click on MS --> Select LPAR --> Configuration -> Manage Profiles -> Select Profile --> Actions --> Edit --> Click on I/O devices --> Select I/O device which is need to be add --> Select add as required/desired.

**To Create Virtual Adapters:**

Click on MS --> Select LPAR --> Configuration --> Manage Profiles --> Select Profile --> Actions --> Edit --> Click on Virtual adapters --> Actions --> Create --> then we can create below type of adapters

- a) Virtual Ethernet Adapters
- b) Virtual SCSI Adapters
- c) Virtual Fiber Channel adapters

### **Ethernet Adapters:**

Ethernet adapters are two types we can

- 1) Physical Ethernet adapter
- 2) Virtual Ethernet adapter

### **VLAN:**

VLAN stands for Virtual Local Area Network, by using vlans we can make communication between the LPARs Internally.

VLAN IDs range will be 0 to 4095 = 4096 total

0, 4095 --> Reserved

4094 we can use.

1002 - 1005 --> sometimes these VLANs will be reserved for token ring adapters.

### **Inter-partition communication:**

If LPAR are communicating with each other by using virtual adapters via VLANs internally within Managed system this is called Inter-partition communication.

### **FIRMWARE UPGRADE:**

#### **Pre-Works:**

- 1) Take the mksysb backup of all LPARS in the managed system which is we are going to upgrade
- 2) Take the all important command output.  
lsmcode, lsmcode -Ar, ifconfig -a, netstat -nr, mount, df -g, lsfs, lsvg, lsvg -o etc.
- 3) If packages are not available in NIM server we have to download from the IBM Fix central and copy into NIM server (192.168.1.100)

**Location:** /software/firmware/power6

open fix central -> select product -> power -> select managed system model -> select the required firmware level

Ex:

**power 5 servers**

EL350\_075

EL350\_110

EL340\_100

EL360\_203

**Power 6 Servers:**

**Implementation:**

) Login to the HMC

2) Select the Managed System which is we have to upgrade

AL720\_096

AL730\_110

**Power 7 Servers**

AM720\_100

AM730\_118

## **VIO SERVER**

VIO stands for Virtualizaion of Input output devices, By using VIO server we can share/virtualize the physical resouces between the LPARs.

**VIO Advantages:**

- 1) In VIO we can share the Physical ethernet adapters between LPARs by implementiong the Shared Ethernet Adapters (SEA).
- 2) In VIO we can share the Physical fiber channel adapters by using vscsi disk mapping or NPIV disk mapping.
- 3) By using VIO we can configure MPIO (multipath input/output)
- 4) VIO server is an normal LPAR but its having virtualization capability.
- 5) VIO OS is combination of virtualization and Aix environment commands.
- 6) In Single managed System max we can configure 4 VIOS that to depends on Licence from IBM.
- 7) Default username padmin and password also padmin after installed OS.

**To add I/O Devices to LPARs**

Click on MS --> Select LPAR --> Configuration -> Manage Profiles -> Select Profile --> Actions --> Edit --> Click on I/O devices --> Select I/O device which is need to be add --> Select add as required/desired.

### **To Create Virtual Adapters:**

Click on MS --> Select LPAR --> Configuration --> Manage Profiles --> Select Profile --> Actions --> Edit --> Click on Virtual adapters --> Actions --> Create --> then we can create below type of adapters

- a) Virtual Ethernet Adapters
- b) Virtual SCSI Adapters
- c) Virtual Fiber Channel adapters

### **Ethernet Adapters:**

Ethernet adapters are two types we can

- 1) Physical Ethernet adapter
- 2) Virtual Ethernet adapter

**1) Physical Ethernet Adapter:** By using physical ethernet adapter we can configure IP address, it having capability to communicating another physical servers also.

**2) Virtual Ethernet Adapter:** By using virtual ethernet adapter we can configure IP address but this IP address can communicate only inter LPARs within the managed system this is also called inter partition communication. VEA adapter don't have capability of external network communication.

### **VLAN:**

VLAN stands for Virtual Local Area Network, by using VLAN we can make communication between the LPARs Internally.

VLAN IDs range will be 0 to 4095 = 4096 total

0, 4095 --> Reserved

4094 we can use.

1002 - 1005 --> sometimes these VLANs will be reserved for token ring adapters.

### **Inter-partition communication:**

If LPAR are communicating with each other by using virtual Ethernet adapters through VLAN IDs within the Managed system this is called Inter-partition communication.

\$oem\_setup\_env or \$r oem --> To move from VIO environment to AIX

### **SEA Configuration in Single VIO setup:**

1) Create one Virtual Ethernet adapter on VIO side and on each lpar with same VLAN ID, which are LPARs required external access

**Note:** while creating VEA on VIO side select "Access External Network" and if it is Dual VIO servers we have to select Trunk Priority.

2) Add the Physical adapters to VIO Server.

3) Shutdown & Activate the VIO and LPARS after created adapters

4) Create the SEA on VIO Server By using below command

**Note:** Which are adapters we are using to create SEA, on those adapters should not configured any IP address, should be free, once created SEA then we can create/add IP to SEA adapter.

**SYN:**`$mkvdev -sea <PEA> -vadapter <VEA> -default <VEA> -defaultid <VLANID>`

**EX:**`$mkvdev -sea ent1 -vadapter ent0 -default ent0 -defaultid 10`

PEA - Physical Ethernet Adapter

VEA - Virtual Ethernet Adapter

### **To Remove the single VIO setup:**

1) Login console on VIO server and Client server

2) Remove the IP both sides if configured

3) Remove the adapters (Vir & Phy) on OS level on both sides

4) Remove the adapters (vir & Phy) on profiles on both sides

5) Shutdown VIO and Client Server

6) Handover to next person

### **SEA (Shared Ethernet Adapter):**

SEA stands for Shared Ethernet Adapter, it one of the advance technology in vitulization, SEA adapter we can create by using physical and virtual adapters. Actually to create sea minimum one physical and one virtual adapter required.

To Provide external network communication by using less (one or more) Physical adapters to all/more LPARs in the Frame we can implement the SEA adapter.

### **Dual VIO Setup SEA/SEA Failover Setup:**

1) Create two VIO servers in Frame and install same VIO os version on both servers.

`$ioslevel`

1.5 (or) 2.2.2.3

2) Create two virtual Ethernet adapters on each VIO server with different VLAN ids. one for to create SEA adapter which is communicating with client

servers and another one for control channel adapter to communicate with another VIO server.

3) Add minimum one physical ethernet adapter on each VIO server, if we need etherchannel minimum two physical ethernet adapters required on each VIO.

## **VIO1**

Ex: ent0 (to create SEA) VLAN = 40 - VEA

ent1 (Control channel adapter ) VLAN=99 - VEA

ent2 (Physical adapter) - PEA

**Note:** while creating ent0 adapter need to select access external network and trunk priority 1

## **VIO2**

Ex ent0 (To create SEA) VLAN = 40 - VEA

ent1 (Control channel adapter) VLAN=99 - VEA

ent2 (Physical adapter) -PEA

**Note:** while creating ent0 adapter need to select access external network and trunk priority 2

4) Activate (if down), Shutdown/Activate (if running), both VIO servers.

5) login to the VIO1

```
$lsdev -type adapter|grep ent
```

```
$entstat -all ent0 |grep -i vlan -> To check 40 VLAN id status
```

```
$entstat -all ent1 |grep -i vlan -> To check 99 VLAN id status
```

6) Create SEA adapter on VIO1 by using below command

**SYN:**`$mkvdev -sea <PEA> -vadapter <VEA> -default <VEA> -defaultid <VLAN ID> -attr ha_mode=auto ctl_chan=<CCVEA>-netaddr=<router IP>`

EX:`$mkvdev -sea ent2 -vadapter ent0 -default ent0 -defaultid 40 -attr ha_mode=auto ctl_chan=ent1 -netaddr=10.3.1.6`

**O/P:**ent3 available - SEA adapter

en3 available - SEA Interface

et3 available

7) Login to second VIO (VIO2)

```
$lsdev -type adapter|grep ent
```

```
$entstat -all ent0 |grep -i vlan -> To check 40 VLAN id status
```

```
$entstat -all ent1 |grep -i vlan -> To check 99 VLAN id status
```

8) Create SEA adapter on VIO1 by using below command

**SYN:**`$mkvdev -sea <PEA> -vadapter <VEA> -default <VEA> -defaultid <VLAN ID> -attr ha_mode=auto ctl_chan=<CCVEA>-netaddr=<router IP>`

EX:`$mkvdev -sea ent2 -vadapter ent0 -default ent0 -defaultid 40 -attr ha_mode=auto ctl_chan=ent1 -netaddr=10.3.1.6`

**O/P:**ent3 available - SEA adapter

en3 available - SEA Interface, If we need we can configure IP

et3 available

9) Create the VEA adapter on all LPARs with VLAN id 40, which are LPARs/Servers required external network.

10) After create VEA on LPARs login to each LPAR and configure the IP

### **SEA Commands**

\$mkvdev --> To create SEA.

\$lsdev -type adapter |grep -i shared --> To list SEA adapter

EX:\$lsdev -type adapter |grep -i shared

ent5 Available Shared Ethernet Adapter

\$entstat -d <SEA>|grep -i active --> To check the active VIO

If Active=True -> It is active VIO server

If Active=False -> It is Inactive VIO server

EX:\$entstat -all ent5 |grep -i active

Active channel: primary channel

Priority: 1 Active: True

\$entstat -all ent5 |grep -i priority --> To check the VIO priority.

\$chdev -dev <SEA> -attr ha\_mode=standby --> To move/switch to SEA service access from active VIO to standby VIO.

EX: chdev -dev ent5 -attr ha\_mode=standby

\$entstat -all ent5|grep -i active

Active channel: primary channel

Priority: 1 Active: False

### **Go to VIO2:**

\$entstat -all ent5 |grep -i active

Active channel: Secondary channel

Priority: 2 Active: True

### **Go to VIO1:**

\$chdev -dev ent5 -attr ha\_mode=auto --> To bring the the SEA service access from Active VIO to standby VIO.

### **DISK MAPPING:**

In Virtualization one of the advanced technology disk mapping. The disk mapping will be 2 types from VIO to client servers.

1) VSCSI Disk Mapping

2) NPIV Disk Mapping

### **1) VSCSI Disk Mapping**

In VSCSI disk mapping we can assign disks in two ways

a) Entire disk mapping to single client.

b) Single disk can be shared between multiple LPARs/Servers.

### **2) NPIV Disk Mapping**

### **1) VSCSI Disk Mapping Configuration Single VIO:**

a) Create Virtual SCSI adapters on VIO side (VIO1) one and client (nimserver) side one with correct Mapping IDs.

Ex:VIO1 Side vscsi details:

Adapter ID : 4

Type of adapter : Server

Selecte LPAR : nimserver

Client ID : 6

nimserver side vscsi details

Adapter ID : 6

Type of adapter : Client

Server Partition : VIO1

Server ID : 4

2) Once created VSCSI Adapters on VIO & client side, Shutdown/Activate the both servers if we are not created by using DLPAR operation.

**Note:** If we create VSCSI adapters via DLPAR just create the adapters in profile also.

3) Once both servers are up, login to the VIO1 Server, check the vscsi adapter slot ID (4) in vio server, which vhost adapter having slot ID4 by using below command

`$lsdev -slots`

EX:`$ lsdev -slots |grep vhost`

U9117.570.65B9DBE-V2-C4 Virtual I/O Slot vhost0

4) Find the free disks in VIO to add to the vhost adapter

`$lspv -free`

hdisk2

hdisk3

5) Add the free disk to vhost adapter by using below command.

**SYN:**`$mkvdev -vdev <freedisk/baking Device> -vadapter <vhostname> -dev <VTD Name>`

EX:`$mkvdev -vdev hdisk2 -vadapter vhost0 -dev aix1disk`

`$lsmmap -all ->` To list the mapping

`$lsmmap -vadapter vhostname-->` to list the particular vhost mappings.

EX:`$lsmmap -vadapter vhost0 -->` To list vhost0 adapter mappings

6) Go to the client side (aix1) and scan new disks.

`#lspv -->` Take lspv output

`#cfgmgr -->` Scan new disks

`#lspv -->` verify latest disk

`#bootinfo -s hdiskX -->` Verify the disk size.

**b) Single disk can be shared between multiple LPARs/Servers in disk mapping.**

1) Create VSCSI adapters on VIO & Client sides with correct mapping IDs.

- 2) Logging to the VIO server.
- 3) Find the correct slots number of vhosts, which is mapped to required lpar.
- 4) Find the free disk
- 5) Create VG on free disk  
\$mkvg -vg <vgname><pvname>
- 6) Create LV with required disk size  
\$mklv -lv <lvname><vgname><lvsize>  
EX:\$mklv -lv nimdisklv sapsharevg 10G
- 7) Map the LV to vhost adapter by using \$mkvdev command  
\$mkvdev -vdev <lvname> -vdaper <vhostname> -dev <vtname>  
\$mkvdev -vdev nimdisklv -vdapter vhost0 -dev nimdisk1
- 8) Go to client side and scan the new disk.  
#lspv  
#cfgmgr  
#lspv  
#bootinfo -s hdiskX

### **Unmapping/Remove the mappings:**

- 1) Remove VG/FS and disks from the clients.
- 3) Go to VIO side
- 4) list VTDS mapping by adapter wise \$lsmap -vadaper vhostX
- 5) remove the VTDs for all vhosts  
\$rmvdev -vtd <vtdname>
- 6) remove the vhost adapters from VIO
- 7) remove all VSCSI adapters in VIO and Clients profiles
- 8) Shutdown VIO server.

### **External Storage**

**External Storage Types:** Hitachi, IBM and EMC.

Actually in each Managed System disks capacity will be 8 disks or 12 disks. if we need more disks to the servers Ex: 50, 100, 500 or 1000 disks, we can buy the external storage and add the disks to Managed System by using Zoning.

- 1) To receive the disks to Managed System from storage, The managed System should have the fiber channel or SCSI adapter.
- 2) SCSI adapters old models, now we are not using most of the companies.

### **Fiber Channel SCSI (FCS) adapter or Host Bus adapter (HBA):**

Fiber channel adapter is advanced adapter to receive the disks from Storage to Managed System. FCS adapter also called HBA adapter. Fiber channel adapter speed will be 4gig or 8gig. We are having Physical fiber channel adapters & virtual fiber channel adapter.

**Each Physical Fiber channel adapter contains the two unique numbers.**

- 1) WWNN (World Wide Node Name or Number)
- 2) WWPN or WWN (World Wide Port Name or Number).

**Note:** WWPN is the main number to receive the disks from storage.

Each Virtual Fiber channel adapter contains two WWN numbers one WWN for to receive the disks from storage. another WWN for Live Partition Mobility (LPM).

```
#lsdev -Cc adapter |grep -i fcs
fcs0 - Model number
fcs1 - Virtual I/O adapter
#lscfg -vpl <FCS>|grep -i network --> To find the wwn or wwpn number of
FCS adapter
EX:#lscfg -vpl fcs0 |grep -i network
c0507654ac7ip8ya
```

**Note:** 4gig speed adapters will not support for NPIV configuration.

**To receive the disk from Storage:**

- 1) To Receive the disk from Storage we need submit the storage request with below details.

Server Name or IP : VIO1 or aix1  
FCS name : fcs0  
WWN of FCS : c0156789xb123uy4  
Size of New Disk : 100G or 200G or 500G  
Total Disks : 4

=====

- 2) SAN team add the 4 disks with 100GB size to requested fiber channel adapter as requested in request.

- 3) Once added SAN team will send mail us with 4 LUN ids:

Ex: 0D1256, 0D1257, 0D1258, 0D1259 LUN IDs. Login Unit Name.

- 4) Login VIO server

- 5) take old disks output

```
#lspv (or) lsdev -Cc disk
```

```

6) Scan new disks
#cfgmgr (or) $cfgdev (if VIO)
7) Verify new disks LUN Ids in Hitachi Storage
#/usr/DynamicLinkmanager/bin/dlnkmgr view -lu
(or)
#cd /usr/DynamicLinkmanager/bin/
./dlnkmgr view -lu -> To display all disks LUN ids
./dlnkmgr view -lu |grep -i 0D1257 and 58 and 59
0d1256 hdisk5
0d1257 hdisk6
0d1258 hdisk7
0d1259 hdisk8
8) Verify the disk size
#bootinfo -s hdisk5, hdisk6, hdisk7, hdisk8
100240
9) Add PVID to New disks
#lspv
hdisk5 none none
hdisk6 none none
hdisk7 none none
hdisk8 none none
#chdev -l hdisk5 -a pv=yes, same for hdisk6, hdisk7 hdisk8
10) Check reserve_policy should be no_reserve if not change it.
#lsattr -El hdisk5 |grep -i reserve

```

Q) Once receive the disk from Storage disk size is zero displayed. what is the problem and how to fix it?.

**ANS:**

Yes, this common problem sometimes will happen when we receive the disks from storage, I have also fixed many times this issue.

**To fix this problem we have to clear the reservation key of new LUN or disk by using below command.**

```
#/usr/DynamicLinkmanager/bin/dlnkmgr -k <newdiskname>
```

after cleared key if issue not fixed the problem not OS side (AIX). The problem will be SAN team side, we can inform to them check the zoning again, they may not added disk correctly to WWN number.

**DUAL VIO DISK MAPPING CONFIGURATION TO VIO CLIENT via VSCSI Adapters:**

**VIO1 Side Adapters:**

fcs0

fcs1

vhost0 (Server ID: 4, Client name: aix1 Client ID: 6)

**VIO2 Side Adapters:**

fcs0

fcs1

vhost0 (Server ID:10 Client name:aix1, Client ID: 11)

**aix1 Client Side Adapters:**

vscsi0 (Client ID: 6 Server name: VIO1, Server ID:4)

vscsi1 (Client ID: 11 Server name: VIO2, Server ID:10)

So, on each VIO server having two FCS adapters if we want receive the disks from storage we have to provide the 4 wwns numbers to Storage team.

Then Storage team side 4 wwn numbers they will create as group and add the requested disks to group (4wwns) by using Zoning.

Once disks added by storage team all disks will be visible both VIO sides

ex:vio1: hdisk2 0D1560 (LUN ID)

vio2: hdisk1 0D1560

**Note:** Once received disks from storage the disk names may be same or may not be same on both VIO sides, we can find the which disk is same on both VIO sides by listing/seeing LUN IDs. so which disk having same LUN ID on VIO1 and VIO2 the disk is same disk.

above example hdisk2 having same LUN ID on vio1, hdisk1 having same LUN ID on VIO2, the both disks are same disk

**Mapping the disk from VIOS to Client:**

1) Login VIO1

\$lspv -free

hdisk2 none none

**Note:** create PVID and reserve\_policy, if not created

\$mkvdev -vdev hdisk2 -vadapter vhost0 -dev aix1disk

\$lsmap -vadapter vhost0 --> To verify the mapping

2) Login VIO2

\$lspv -free

hdisk1 none none

**Note:** create PVID and reserve\_policy, if not created

\$mkvdev -vdev hdisk1 -vadapter vhost0 -dev aix1disk

\$lsmap -vadapter vhost0 --> To verify the mapping.

**Note:** Once created PVID for hdisk2, hdisk1 both vio sides, the PVID should be same.

3) Login to the client side (aix1)

#lspv -> Take output

#cfgmgr --> scan new disk

#lspv -> Verify the new disk

#Create PVID and reserve\_policy

#Then create VG/FS/LV if we need to store the data.

### **MPIO via VSCSI Mapping:**

To implement MPIO configuration successfully we have to change the below attributes on VIO and client side:

#### **VIO Server side:**

Change the FCS adapters and disk attributes on both VIO servers:

##### **VIO1:**

fcs0 (fscsi0) fcs1 (fscsi1)

hdisk2

FCS attributes changing

dynktrk=yes

fc\_err\_recov=fast\_fail

#chdev -l fscsi0 -a dynktrk=yes

#chdev -l fscsi0 -a fc\_err\_recov=fast\_fail

\$chdev -dev fscsi0 -attr dynktrk=yes

\$chdev -dev fscsi0 -attr fc\_err\_recov=fast\_fail

**Note:** Do same on fscsi1 in VIO1 server and VIO2 Server FCS adapters.

#### **Disk Attributes on both VIO sides:**

algorithm=load\_balance

reserve\_policy=no\_reserve

hcheck\_mode=nonactive

hcheck\_interval=60

\$chdev -dev hdisk2 -attr algorithm=failover

\$chdev -dev hdisk2 -attr reserve\_policy=no\_reserve

\$chdev -dev hdisk2 -attr hcheck\_mode=nonactive

\$chdev -dev hdisk2 -attr hcheck\_interval=60

#### **Client Side (aix1) attributes:**

Devices

vscs0

vscs1

hdiskX  
VSCSI adapters  
#chdev -l vscsi0 -a vscsi\_path\_to=30  
#chdev -l vscsi0 -a vscsi\_err\_recover=fast\_fail  
**Note:** same do for vscsi1 also

**Disk:**

algorithm=failover  
reserve\_policy=no\_reserve  
hcheck\_mode=nonactive  
hcheck\_interval=60

#chdev -l hdiskX -a reserve\_policy=no\_reserve  
#chdev -l hdiskX -a algorithm=failover  
#chdev -l hdiskX -a hcheck\_mode=nonactive  
#chdev -l hdiskX -a hcheck\_interval=60

**VIO attributes:**

FCS  
dynaktrk=yes  
fc\_err\_recover=fast\_fail

**Disk:**

algorithm=load\_balance  
reserve\_policy=no\_reserve  
hcheck\_mode=nonactive  
hcheck\_interval=60

vscsi\_path\_to=30  
vscsi\_err\_recover=fast\_fail  
algorithm=failover  
reserve\_policy=no\_reserve  
hcheck\_mode=nonactive  
hcheck\_interval=60

**VIO attributes:**

FCS  
dynaktrk=yes --> When active path is failed, dynaktrk attribute will change track to next active/available path.  
fc\_err\_recover=fast\_fail --> When active path FC adapter failed, this attribute will switch to next available FC adapter

**Disk:**

algorithm=load\_balance --> Both VIO sides disk I/O operations load should be balance by using this attribute.

reserve\_policy=no\_reserve --> It is one of important attribute, this attribute will tell the paths, the disk is not reserved for the any path. when we receive the disk we should change this attribute 100% as no\_reserve, if we are not changing no use of MPIO configuration.

```
hcheck_mode=nonactive
hcheck_interval=60
```

The above two are related attributes hcheck\_interval attribute will monitor the all paths every 60 seconds, hcheck\_mode attribute will monitor the non-active paths, if active path is non-active/inactive it will switch to next available path

### **Client Side:**

vscsi\_path\_to=30 --> VSCSI adapters paths monitor every 30 seconds by using this attribute.

vscsi\_err\_recove=fast\_fail --> this attribute role will be If active vscsi adapter fails, immediatly switch to next available vscsi adapter.

algorithm=failover -> when active path fail, failover to next available path.

```
reserve_policy=no_reserve
hcheck_mode=nonactive
hcheck_interval=60
```

### **What is MPIO:**

MPIO stands for Multiple Paths Input/Output devices(disks).

If any disk is coming by more than one path that is called MPIO.

### **In MPIO we have two types of configurations.**

- 1) MPIO via VSCSI Mapping.
- 2) MPIO via NPIV Mapping.

### **To list the Paths**

#### **VIO1 Side**

\$lspath --> To list all disks all paths.

```
Enabled hdisk0 fscsi0
Enabled hdisk0 fscsi1
Enabled hdisk1 fscsi0
Enabled hdisk1 fscsi1
```

#### **VIO2 Side**

\$lspath --> To list all disks all paths.

```
Enabled hdisk0 fscsi0
Enabled hdisk0 fscsi1
```

Enabled hdisk1 fscsi0  
Enabled hdisk1 fscsi1

### **Cient Side (aix1):**

```
#lspath
Enabled hdisk0 vscsi0
Enabled hdisk0 vscsi1
Enabled hdisk1 vscsi0
Enabled hdisk1 vscsi1
```

```
#lspath -l hdisk1 -> To list particular disk paths
Enabled hdisk1 vscsi0 - VIO1
Enabled hdisk1 vscsi1 - VIO2
```

### **To check path priority & Status**

```
#lsapath -AEH -l <DiskName> -p <VSCSI NAME>
```

```
#lspath -AEH -l hdisk0 -p vscsi0
attribute value description user_settable
priority 1 Priority True
#lsapth -AEH -l hdisk0 -p vscsi1
attribute value description user_settable
priority 1 Priority True
```

### **To change path priority**

```
#chpath -l <Diskname> -p <VSCSI> -a priority=value
#chpath -l hdisk0 -p vscsi -a priority=2
```

### **To disable the path**

```
#chpath -l <diskname> -p <VSCSI> -s disabled
```

### **To Enable the path**

```
#chpath -l <diskname> -p <VSCSI> -s enabled
(or)
#smitty mpio
```

### **Testing:**

Shutdown VIO server1/Automaticlly down

```
aix1
#lspath -l hdisk1
Failed hdisk1 vscsi0 - VIO1
Enabled hdisk1 vscsi1 - VIO2
```

```
#lspath |grep -i failed --> To list failed paths
#lspath |grep -i missing --> To list the missing paths.
```

## **Recent Activity/issue**

### **Critical activity:**

#### **To fix the failed/missing paths:**

##### **Problem reason:**

It may be fiber cables loose connectivity issue.

It may be VIO server1 or 2nd VIO server rebooted.

##### **Fix the Problem:**

1) If cables problem, Inform to DC technician to insert cables properly server side and storage side and enable the paths from OS level.

2) If any one of the VIO server rebooted, Just enable the paths OS level.

#### **To enable the failed/missing paths from OS level by using below command.**

#smitty mpio

MPIO Path Management

Enable the Paths

Enable selected or all paths

Final window

\*Paths to Enable [Select missing/failed paths] To enable.

we can select multiple missing paths by using F7 to enable.

**Note:** after above steps followed, if failed/missing paths not enabled the reasons will be:

- 1) VIO server may be down
- 2) VIO side FCS adapter may be down
- 3) VIO side FCS adapter cable connection may be not good
- 4) Client side VSCSI adapter may be removed.

#### **When VIO Server required?**

Suppose if Managed system having less Physical I/O Devices (Ethernet, FCS, and Disks) and Many LPARs VIO required to share physical devices between LPARs.

#### **When VIO not required?**

If Managed System having more I/O Devices less LPARs VIO server not required.

#### **NPIV (or) MPIO via NPIV:**

NPIV stands for Node Port ID Virtualization, NPIV is one of the advanced technology in disk mapping, By using NPIV configuration we can send disk from storage to clients directly without sending disk to VIO server.

**Single VIO NPIV Configuration:**

0) Add/Assign Physical Fiber Channel adapter (PFCS) to VIO server.

1) Create virtual fiber channel adapter (VFCS) on VIO side(vios1) and LPAR Side (nimserver).

**Note:** If VIO & LPARs are running create via DLPAR and update profile manually. if not running create directly in profile

**Create VFCS Adapter DLPAR.**

Select VIO/LPAR -> Dynamic Logical Partition -> select Virtual Adapters option

**Create VFCS Directly in Profile:**

Select VIO/LPAR -> Configuration -> Manage Profile -> Actions -> Create -> Select VFCS adapter

**Note:** if we create Directly VFCS adapter on VIO/LPAR without DLPAR operation we have to Shutdown & Activate VIO/LPAR to reflect on servers.

Mostly on VIO server we will create VFCS adapter via DLPAR because many servers (Lpars) depends on VIO server.

**Mapping IDs on VIOS1 Side in VFCS adapter:**

Adapter ID : 31

Client Server : nimserver

Clinet ID : 31

**Mapping IDs on nimserver side in VFCS adapter:**

Adapter ID : 31

VIOS Server : VIOS1

Server ID : 31

2) Login VIOS1 Server and Map the PFCS adapter to VFCS adapter by using below command.

`$lsdev -type adapter |grep -i vfchost* -> To list VFCS adapters`

vfchost0 Available Virtual FC Server Adapter

vfchost1 Available Virtual FC Server Adapter

vfchost2 Available Virtual FC Server Adapter

`$lsdev -slots -> To find the VFCS Adapter slot number (31) which is we have created.`

vfchost0 V2-C20

vfchost1 V2-C28

vfchost2 V2-C31

**To list the PFCS Adapters**

`$lsdev -type adapter |grep -i fcs*`

fcs2 Available 8Gb PCI Express Dual Port FC Adapter  
fcs3 Available 8Gb PCI Express Dual Port FC Adapter

### **To Verify the PFCS Adapters Ports status**

\$lsnports

3) To map the NPIV configuration (or) To map PFCS adapter to VFCS adapter by using below command.

**SYN:**\$vfcsmap -vadapter <VFCS adapter> -fcp <PFCS>

**EX:**\$vfcsmap -vadapter vfcs host2 -fcp fcs2

4) Go to client side (nimserver)

#lsdev -Cc adapter |grep -i fcs

fcs0 Available 30-T1 Virtual Fibre Channel Client Adapter

#lscfg -l fcs0

V4-C31

To list WWN of VFCS adapter on client side, This WWN number we have to provide to SAN team if we want disks on client server (nimserver)

#lscfg -vpl fcs0 |grep -i network

Network Address.....C0507603A272008B

Q) Submit SAN request to receive two disks with 500GB on client server (nimserver) in NPIV mapping.

**Ans:**

### **Dual VIO NPIV Configuration:**

Servers: VIO1, VIO2 and aix1 (client)

1) Add the minimum two physical fiber channel adapters on each VIO Server.

2) Create one virtual fiber channel adapter on both VIO servers and client server with correct mapping details.

**Note:** IDs mapping we will give like vscsi mapping.

3) Shutdown & Activate the VIO Servers and aix1 client server if adapters are not created via DLPAR operation.

4) Login to **VIO1** Server list the fiber channel adapters.

\$lsdev -type adapter |grep -i fcs

fcs0 (PFCA)

fcs1 (PFCA)

vfcvhost0 (VFCA)

\$lsnports

tlsrv808

Name physloc

fabric tports aports

```
fcs0 U5802.001.008A824-P1-C9-T1 1 64 64
fcs1 U5802.001.008A824-P1-C9-T2 0 64 64
```

fabric = 1 - FCS Adapter connected SAN switch, NPIV configuration will be support (fcs1)  
fabric = 0 - FCS adapter not connected SAN switch, NPIV configuration will not support (fcs0)  
tports - Total/Max ports to connect clients VFCS adapters  
aports - Available ports list to connect clients.

### **To map the physical fiber channel adapter to virtual fiber channel adapter.**

```
$lsmap -all -npiv --> To list all NPIV mappings.
$lsmap -vadapter vfchost0 -npiv --> To list particular VFCA adapter NPIV mappings.
Status : Not_logged_in
Flags : <Not_Mapped,Not_Connect>
$vfcmmap -vadapter <VFCA> -fcp <PFCA>
$vfcmmap -vadapter vfchost0 -fcp fcs0 --> to map the vfchost0 to fcs0 PFCA.
```

**Note:** once mapped VFCA (vfchost0) will get the physical FCA capabilities to receive the disks from storage to client directly.

```
$lsmap -all -npiv
```

```
Status : logged_in
Flags : Mapped
```

### **VIO2 side:**

Login VIO2 and do the mapping.

```
$lsdev -type adapter |grep -i fcs
fcs0 (PFCA)
fcs1 (PFCA)
vfchost0 (VFCA)
$lsnports
Name physloc fabric tports aports
fcs2 U5802.001.008A824-P1-C9-T1 1 64 40
fcs1 U5802.001.008A824-P1-C9-T2 1 64 64
```

### **To map the physical fiber channel adapter to virtual fiber channel adapter.**

```
$vfcmmap -vadapter vfchost0 -fcp fcs1
$lsmap -all -npiv --> To list all NPIV mappings.
$lsmap -vadapter vfchost0 -npiv --> To list particular (vfchost0) VFCA adapter NPIV mappings.
```

### **To Receive the disks from Storage to Client (aix1):**

- 1) Login client aix1
- 2) list fiber channel adapters  
`#lsdev -Cc adapter |grep -i fcs`  
fcs0 (VFCA)  
fcs1 (VFCA)
- 3) list wwn numbers of FCS adapters  
`#lscfg -vpl fcs0 |grep -i network`  
070987260000006A -WWN  
`#lscfg -vpl fcs1 |grep -i network`  
070987260000006C -WWN
- 4) Give the WWNs numbers to SAN team and get the required disks with required size.
- 5) Once SAN team added disks, they will provide us LUN IDs scan them and find the LUN IDs disks.
- 6) Verify the disk size, create PVID, check reserve\_policy.
- 7) Create VG/LV/FS etc... to store the data.

### **To Unmap NPIV:**

- 1) Login client server remove the VG/FS/LV and disks.
- 2) Login to VIO servers then run below command  
`$vfcmap -vadapter vfchost0 -fcp ->` To unmap VFCS adapter with PFCS adapter

### **NPIV Important Points:**

- 1) Fabric=1 NPIV support, Fabric=0 NPIV will not support
- 2) FCS capacity/speed should be 8Gb speed, 4GB will not be support.
- 3) Each PFCA will support/connect/Map Max 64 Lpars or 64 VFCA.
- 4) Before adding PFCA to VFCA need to check available ports
- 5) VFCA adapter once created client side, each adapter having two WWN numbers, one wwn number for received the disks from storage and another one for Live Partition Mobility (LPM).

### **Identify below Disks receiving technology**

```
nimserver:/# lspath -l hdisk44
Enabled hdisk44 vscsi0
Enabled hdisk44 vscsi1
```

```
aix1:/# lspath -l hdisk30
Enabled hdisk30 scsi0
Enabled hdisk30 scsi1
```

```
aix2:/# lspath -l hdisk13
```

```
Enabled hdisk13 fscsi0
Enabled hdisk13 fscsi1
```

```
#lsdev -Cc adapter
fcs0 Available 31-T1 Virtual Fiber Channel Client Adapter
fcs1 Available 32-T1 Virtual Fiber Channel Client Adapter
```

```
aix3:/# lspath -l hdisk26
Enabled hdisk26 fscsi0
Enabled hdisk26 fscsi1
```

```
#lsdev -Cc adapter |grep fcs
fcs0 U78A0.001.DNWK4W9-P1-C3-T1 1 64 52 2048 1988
fcs1 U78A0.001.DNWK4W9-P1-C3-T2 1 64 52 2048 1988
```

```
aix4:/# lspath -l hdisk13
Enabled hdisk13 fscsi0
Enabled hdisk13 fscsi1
#lsdev -Cc adapter
fcs1 Available 31-T1 Virtual Fiber Channel Client Adapter
fcs2 Available 32-T1 Virtual Fiber Channel Client Adapter
```

### **To Receive the disk from Storage to VIO Server or LPAR:**

If we want receive the disk directly from storage to VIO or LPAR server. we should have the physical fiber channel adapter on VIO server or LPAR.

1) Check Fiber Channel adapters

```
#lsdev -Cc adapter |grep -i fcs
fcs0
```

2) To get the WWN number of FCS adapter and provide to the WWN number to Storage team.

```
#lscfg -vpl fcs0 |grep -i network
```

3) Then Storage team will add the disk to WWN and send mail us with new disk IDs (LUN ID) Ex: 0D16B1

4) Login to the VIO Server follow below steps.

a)

```
#lspv --> Take output
```

```
#cfgmgr --> To scan new disk or LUNs
```

```
#lspv --> To verify the new disks
```

```
#verify the LUN ID of new disk
```

```
#!/usr/DynamicLinkmanager/bin/dlnkmgr view -lu |grep -i 0D16B1
```

```
O/P:0D16B1 hdisk8 (0D16B1 is LUN ID)
```

```
Ex: hdisk8 (newdisk)
```

b) Verify the disk new disk size

```
#bootinfo -s <newdisk>
```

c) Add the PVID to new disk

```
#chdev -l <diskname> -a pv=yes
d) Check/Change the reserve_policy attribute as no_reserve
#lsattr -El hdisk8 |grep -i reserve --> To check reserve policy
O/P: reserve_policy=singlepath
#chdev -l hdisk8 -a reserve_policy=no_reserve
Note: Reserve policy always should be no_reserve.
```

```
root@AIX: / # echo "cvai" | kdb | grep vscsi
vscsi0 0x000007 0x00000000001 0x0 VIO5->vhost1
vscsi1 0x000007 0x00000000003 0x0 VIO6->vhost1
```

### **Check NPIV adapter mapping**

To check the connection between your VFCHOST and FC physical port in the VIOS level (show vfchost <-> fcs mappings).

### **Run on LPAR Client (not on VIO Server):**

```
root@AIX: / # echo "vfcs" | kdb
NAME ADDRESS STATE HOST HOST_ADAP OPENED NUM_ACTIVE
fcs0 0xF1000A000033C123 0x0008 VIO1 vfchost2 0x01 0x0000
fcs1 0xF1000A000033D542 0x0008 VIO2 vfchost3 0x01 0x0000
```

### **To Identify the Disk mappings Adapters:**

We can identify the VSCSI disk mapping 3 ways between VIO and client servers:

1) Login to client server run below command:

```
#hostname
nimserver
echo "cvai" | kdb | grep -i VIO
O/P:
```

```
vscsi0 - VIO1->vhost0
```

```
vscsi1 - VIO2->vhost0
```

2) Go to manage profiles VIO side or client to find the mapping adapters, IDs. Servers etc.

(or)

3) Go to VIO side run lsmap command

EX:\$lsmap -all --> To list all mappings (or)

\$lsmap -vadapter vhostX -> To list specific adapter mapping.

Here we can Xexa decimal number of LAR ID.

### **Control Channel Ethernet Adapter (CCEA)**

## **What is control Channel:**

In Dual VIO setup we can create one CCEA on each VIO server. by CCEA both VIO servers communication with each other, if any problems/failures happen to active VIO CCEA adapter immediately will inform to next available VIO to server network communication to the LPARs.

(OR)

By using CCEA adapter both VIO servers communicating with each other in Dual VIO SEA setup.

```
oixvp001:/# echo "cvai"|kdb|grep -i vio
vscsi0 0x000007 0x000000000000 0x0 DREAMS5-vio1->vhost0
vscsi1 0x000007 0x000000000000 0x0 DREAMS5-vio2->vhost0
```

## **HACMP**

HACMP stands for High Availability cluster multi processing,

### **Hacmp Theam/Advantages**

- 1) By using hacmp we can provide the continuous service to the client's applications.
- 2) By using hacmp we can eliminate the Single point of failure (SPOF).

2) By using hacmp we can eliminate downtime of the servers.

Generally, Downtimes will be two types

- 1) Planned Downtime
- 2) Unplanned Downtime

**1) Planned Downtime:**

- Hardware upgrades
- Repairs
- Software updates/upgrades
- Testing (periodic testing is required for cluster validation.)
- Development

**2) Unplanned Downtime:**

- Administrator errors
- Application failures
- Hardware failures
- Environmental disasters

**HACMP TWO NODE CLUSTER CONFIGURATION:**

**HARDWARE REQUIREMENTS:**

- 1) TWO Servers
- 2) TWO Switches
- 3) 5 LAN Cables
- 4) 2 Ethernet adapters on each node, (4 both nodes)
- 5) TWO FCS/HBA Adapters on each node to receive the external storage
- 6) 4 Fiber Cables
- 7) Minimum one NON-IP Network (rs232 or diskhb)

**SOFTWARE REQUIREMENTS:**

- 1) Install the Aix OS on both nodes with same version, TL and SP level.
- 2) Install HACMP Software on both nodes (5.3, 5.4, 5.5, 6.1 and 7.1)
  - a) First install the HACMP Pre-requisites
  - b) Install the HACMP software

**HACMP Pre-requisites:**

```
rsct.compat.basic.hacmp
rsct.compat.basic.rte
rsct.compat.basic.sp
rsct.compat.clients.hacmp
rsct.compat.clients.rte
rsct.compat.clients.sp
```

rsct.msg.en\_US.basic.rte  
bos.adt.lib  
bos.adt.libm  
bos.adt.syscalls  
bos.net.tcp.client  
bos.net.tcp.server  
bos.rte.SRC  
bos.rte.libc  
bos.rte.libcfg  
bos.rte.libcu

### **HACMP Software:**

cluster.es.cfs.rte  
cluster.es.client.lib  
cluster.es.client.rte  
bos.clvm.enh  
cluster.es.client.nfs  
cluster.es.client.utils  
cluster.es.clvm.rte  
cluster.es.cspoc.cmds  
cluster.es.cspoc.dsh  
cluster.es.cspoc.rte  
cluster.es.server.diag  
cluster.es.server.events  
cluster.es.server.rte  
cluster.es.server.utils  
cluster.es.worksheets  
cluster.license  
cluster.man.en\_US.es.data

### **TWO Node Cluster Configuration STEPS:**

- 1) Configure the bootips in Aix level by using smitty mkctpip
- 2) Update the hacmp IP address in /etc/hosts file on both nodes
- 3) Configure the cluster name
- 4) Configure the cluster nodes (cl\_node1, cl\_node2)
- 5) Configure the HACMP networks
- 6) Configure the HACMP communication Interfaces (IP Networks) & Devices (Non-IP networks)
- 7) Configure the Persistent IP address
- 8) Configure the Service IP
- 9) Configure the Application Server
- 10) Configure the Resource Group
- 11) Perform the discover if required

- 12) Add the resources (Service IP, Application server, VGs and Filesystems) to Resources group
- 13) Perform the cluster verification & Synchronization
- 14) Start the cluster services

=====

- 1) Configure the bootips in Aix level by using smitty mktcpip

**Login node1 run below commands:**

```
#smitty mktcpip
*Hostname [node1_bootip1
*IP Address [192.168.1.10]
*Interface [en0]
#smitty mktcpip
*Hostname [node1_bootip2
*IP Address [192.168.2.10]
*Interface [en1]
```

**Login Node2 run below command to create bootips in Aix level**

```
#smitty mktcpip
*Hostname [node2_bootip1]
*IP Address [192.168.1.20]
*Interface [en0]
#smitty mktcpip
*Hostname [node2_bootip2]
*IP Address [192.168.2.20]
*Interface [en1]
```

- 2) Update the hacmp IP address in /etc/hosts file on both nodes

```
vi /etc/hosts
node1_bootip1 192.168.1.10
node1_bootip2 192.168.2.10
node1_per 192.168.3.10
node2_bootip1 192.168.1.20
node2_bootip2 192.168.2.20
node2_per 192.168.3.20
icici_serv 192.168.4.10
:wq
```

- 3) Configure the cluster name

```
#smitty hacmp
-> Extended configuration
```

- > Extended Topology configuration
- > Configure an HACMP Cluster
- > Add/Change/Show HACMP Cluster

Final Window

\*Cluster Name [IBM\_Cluster]

**Note:** Cluster Name is our choice, we can give which is customer preferred, or generally we can give application names (Ex: ICICI\_Cluster) etc.

4) Configure the cluster nodes (node1 , node2)

### **Node1 Configuration:**

#smitty hacmp

- > Extended configuration
- > Extended Topology configuration
- > Configure HACMP Nodes
- > Add a Node to the HACMP Cluster

Final Window

\*Node Name [cl\_node1 ]

Communication path to node [node1\_bootip1] +

### **Node2 Configuration**

#smitty hacmp

- > Extended configuration
- > Extended Topology configuration
- > Configure HACMP Nodes
- > Add a Node to the HACMP Cluster

Final Window

\*Node Name [cl\_node2 ]

Communication path to node [node2\_bootip1] +

### **Note:**

- 1) Node name is our choice which is we want we can give,
- 2) Communication path we have to select any one bootip for the each node

5) Configure the HACMP networks

**Note:** In HACMP we are having two types of networks one is IP-Networks (ether), another one is NON-IP Networks (diskhb or rs232)

### **To configure "ether" network**

#smitty hacmp

- > Extended configuration
- > Extended Topology configuration

- > Configure HACMP Networks
- > Add a Network to the HACMP Cluster
- > Select IP-Network Name/Type like "ether"

Final Window

```
*Network Name [net_ether_01]
*Network Type [ether]
*Netmask [255.255.255.0]
*Enable IP-Address Takeover via IP Aliases [yes]
```

**Note:**

- a) Network name will provide the server, if we need to change we can change
- b) Netmask value we have to give which is given by the network team
- c) IP-Address Takeover field is very important field, we can decide here which IP mechanism we are using IPAT via aliasing or IPAT via replacement.

if we select "yes" we are using IPAT via aliasing, if we select "No" we are using IPAT via replacement mechanism, mostly cluster setups we will using IPAT via aliasing concept

**To Configure the NON-Networks (diskhb/rs232):**

**To Configure "diskhb" NON-IP Network**

**Note:** Before configure diskhb we have to follow below steps.

- a) Take one external disk (LUN) from the storage (Size 1GB or 5GB )
- b) Create heartbeat VG (hbvg) on LUN,
- c) Follow the below steps to configure diskhb in hacmp.

```
#smitty hacmp
```

- > Extended configuration
- > Extended Topology configuration
- > Configure HACMP Networks
- > Add a Network to the HACMP Cluster
- > Select "diskhb" network

Final window

```
*Network Name [net_diskhb_01]
*Network Type [diskhb]
```

**To Configure "rs232" NON-IP Network**

```
#smitty hacmp
```

- > Extended configuration
- > Extended Topology configuration
- > Configure HACMP Networks
- > Add a Network to the HACMP Cluster
- > Select "rs232" network

Final Window

```
Network Name [net_rs232_01]
Network Type [rs232]
```

6) Configure the HACMP communication Interfaces (IP Networks) & Devices (Non-IP networks)

**a) Configure the Communication Interfaces (each two, Total 4 bootips)**

```
node1_bootip1 192.168.1.20
node1_bootip2 192.168.2.20
node2_bootip1 192.168.1.30
node2_bootip2 192.168.2.30
```

```
#smitty hacmp
-> Extended configuration
-> Extended Topology configuration
-> Configure HACMP Communication Interfaces/Devices
-> Add Communication Interfaces/Devices
-> Add Pre-defined Communication Interfaces/Devices
-> Communication Interfaces
-> Select IP-Network "net_ether_01"
Final Window
*IP Label/Address [node1_bootip1]
*Network Type [ether]
*Network Name [net_ether_01]
*Node Name [cl_node1]
```

```
#smitty hacmp
-> Extended configuration
-> Extended Topology configuration
-> Configure HACMP Communication Interfaces/Devices
-> Add Communication Interfaces/Devices
-> Add Pre-defined Communication Interfaces/Devices
-> Communication Interfaces
-> Select IP-Network "net_ether_01"
Final Window
*IP Label/Address [node1_bootip2]
*Network Type [ether]
*Network Name [net_ether_01]
*Node Name [cl_node1]
```

```
#smitty hacmp
-> Extended configuration
-> Extended Topology configuration
-> Configure HACMP Communication Interfaces/Devices
-> Add Communication Interfaces/Devices
-> Add Pre-defined Communication Interfaces/Devices
-> Communication Interfaces
-> Select IP-Network "net_ether_01"
Final Window
```

```
*IP Label/Address [node2_bootip1]
*Network Type [ether]
*Network Name [net_ether_01]
*Node Name [cl_node2]
```

```
#smitty hacmp
-> Extended configuration
-> Extended Topology configuration
-> Configure HACMP Communication Interfaces/Devices
-> Add Communication Interfaces/Devices
-> Add Pre-defined Communication Interfaces/Devices
-> Communication Interfaces
-> Select IP-Network "net_ether_01"
```

Final Window

```
*IP Label/Address [node2_bootip2]
*Network Type [ether]
*Network Name [net_ether_01]
*Node Name [cl_node2]
```

### **To Configure the Communication Devices (Non-IP Networks) diskhp configuration for node1:**

```
#smitty hacmp
-> Extended configuration
-> Extended Topology configuration
-> Configure HACMP Communication Interfaces/Devices
-> Add Communication Interfaces/Devices
-> Add Pre-defined Communication Interfaces/Devices
-> Communication Devices
-> select Non-network "net_diskhb_01"
```

Final window

```
*Device Name [node1_diskhb]
*Network Type [diskhb]
*Network Name [net_diskhb_01]
*Device Path [/dev/hdisk4]
*Node Name [cl_node1]
```

### **diskhb configuration for node2:**

```
#smitty hacmp
-> Extended configuration
-> Extended Topology configuration
-> Configure HACMP Communication Interfaces/Devices
-> Add Communication Interfaces/Devices
-> Add Pre-defined Communication Interfaces/Devices
-> Communication Devices
-> select Non-network "net_diskhb_01"
```

```
Final window
*Device Name [node2_diskhb]
*Network Type [diskhb]
*Network Name [net_diskhb_01]
*Device Path [/dev/hdisk4]
*Node Name [cl_node2]
```

### **rs232 configuration for node1:**

```
#smitty hacmp
-> Extended configuration
-> Extended Topology configuration
-> Configure HACMP Communication Interfaces/Devices
-> Add Communication Interfaces/Devices
-> Add Pre-defined Communication Interfaces/Devices
-> Communication Devices
-> select Non-network "net_rs232_01"
Final window
*Device Name [node1_rs232]
*Network Type [rs232]
*Network Name [net_rs232_01]
*Device Path [/dev/tty0]
*Node Name [cl_node1]
```

### **rs232 configuration for node2:**

```
#smitty hacmp
-> Extended configuration
-> Extended Topology configuration
-> Configure HACMP Communication Interfaces/Devices
-> Add Communication Interfaces/Devices
-> Add Pre-defined Communication Interfaces/Devices
-> Communication Devices
-> select Non-network "net_rs232_01"
Final window
*Device Name [node2_rs232]
*Network Type [rs232]
*Network Name [net_rs232_01]
*Device Path [/dev/tty0]
*Node Name [cl_node2]
```

7) Configure the Persistent IP address for both nodes

For node1 also do like this

**For Node2 Steps:**

```
#smitty hacmp
```

```
-> Extended configuration
```

```
-> Extended Topology configuration
```

```
-> Configure HACMP Persistent Node IP Label/Addresses
```

```
-> Add a Persistent Node IP Label/Address
```

```
-> Select "cl_node2"
```

Final Window

```
*Node Name [cl_node2]
```

```
*Network Name [net_ether_01]
```

```
*Node IP Label/Address [node2_per]
```

8) Configure the Service IP

```
#smitty hacmp
```

```
-> Extended configuration
```

```
-> Extended Resource Configuration
```

```
-> HACMP Extended Resource Configuration
```

```
-> Configure HACMP Service IP Labels/Address
```

```
-> Add a service IP Labels/Address
```

```
-> Configurable on Multiple Nodes
```

```
-> Select "net_ether_01" Network
```

Final Window

```
* IP Label/Address [ICICI_Serviceip]
```

```
* Network Name [Net_ether_01]
```

9) Configure the Application Server

**Note:**Before configure application server, we have to get the start/stop applications scripts from application team with correct locations.

Ex:Start script location [/var/app/icici\_startapp]

Stop script location [/var/app/icici\_stopapp]

**Now follow the below steps to configure application server.**

```
#smitty hacmp
```

```
-> Extended Configuration
```

```
-> Extended Resource Configuration
```

```
-> HACMP Extended Resource Configuration
```

```
-> Configure HACMP Application Servers
```

```
-> Add an Application Server
```

Final Window:

```
* Server Name [icici_appserver]
* Start Script [/var/app/icici_startapp]
* Stop Script [/var/app/icici_stopapp]
```

#### 10) Configure the Resource Group

```
#smitty hacmp
-> Extended Configuration
-> Extended Resource Configuration
-> HACMP Extended Resource Group Configuration
-> Add a Resource Group
Final Window
* Resource Group Name [ICICI_RG]
* Participating Nodes [cl_node1,cl_node2]
Startup Policy [Online on Home Node Only]
Fallover Policy [Fallover to Next Priority Node]
Fallback Policy [Never Fall Back]
```

**Note:** Resource group policies are total 9, Start policy having 4 policies, Fallover Policy having 3 Policies, Fall back policies are 2, We have to Select which is required.

#### 11) Perform the discover if required

```
#smitty hacmp
-> Extended Configuration
-> Discover HACMP-Related Information From Configured Nodes
```

**Note:** Discover option only available on HACMP 5.3,5.4, 5.5 Version. From 6.1, 7.1 Discover not required.

**Discover Why required:** When we have added new things (VGs, File systems, Disks and any resources etc) into the cluster. Sometimes HACMP will not be aware about newly added things so then we have to perform discover to inform to the hacmp software.

(or)

By using discover option we can inform to HACMP about the newly configured resources.

**Note:** Before adding resources into Resource Group we have to perform discover to get newly added resources.

#### 12) Add the resources (Service IP, Application server, VGs and Filesystems) to Resources group

```
#smitty hacmp
-> Extended configuration
```

- > Extended Resource Configuration
- > HACMP Extended Resource Group Configuration
- > Chang/Show Resources and Attributes for a Resource Group
- > Select Resources group "ICICI\_RG"
- > Add the resources into Resource Group follow the below steps.

Service IP Labels/Addresses [ICICI\_Serviceip]  
 Application Servers [ICICI\_appserver]  
 Volume Groups [icicivg,sapvg]  
 File Systems [/sapfs10]

### 13) Perform the cluster verification & Synchronization

- ```
#smitty hacmp
```
- > Extended Configuration
 - > Extended Verification & Synchronization

Final Window

- * Verify, Synchronize or Both [Both] +
- * Automatically Correct Errors found during verification [No]
- * Force Synchronization if verification fails [No]
- * Verify Changes Only [No]
- * Logging [Standard]

Verification & Synchronization:

If we added/removed/modified in the cluster topologies in HACMP or Resource Group we have to perform verification & synchronization

Verification:

verification will verify the cluster components and it will give the results whether cluster configuration is fine or not, if verification is fails we have to fix the problems first before run synchronizaion.

Synchronization:

Synchronization will update the added/modified cluster components on all the cluster nodes.

Note:

- 1) If we add/remove/modified anything on RG synchronization required,
- 2) For LVM operations Synchronization not required.

14) Start the cluster services

- ```
#smitty hacmp (or) #smitty cl_start (Fast Path)
```
- > System Management (C-SPÖC)

- > Manage HACMP Services
- > Start Cluster Services

Final Window

```
* Start Now, On system restart or Both [Both] +
Start Cluster Services on these nodes [cl_node1,cl_node2] +
BROADCAST Message at startup [True]+
Startup Cluster Information Daemon [True]+
Ignore Verification errors [False]+
Automatically correct errors found during cluster startup [Interactively] +
```

**Note:** Once cluster services are started, HACMP will take care all resources and it will monitor the total cluster topologies etc.

### **BOOT IPs:**

Which IP Address is we are configuring first time for NIC/EA, that IP address we can call bootip, also we can call base IP, first IP address or Communication Interfaces in hacmp. each node should have two boot IPs for the redundant Purpose, In two node cluster minimum 4 bootips we should configure. Bootips we can configure Aix Level first by using `mktcpip` and we have to configure hacmp level also by using `#smitty hacmp`

### **IP Label:**

IP Label nothing but its like hostname of IP Address.

### **Persistent IP:**

It is one of the IP Address in HACMP, its an alias IP Address in HACMP, We will configure persistent IP top of the bootip it is always bound the single node.

Admins will use persistent IP only to access the cluster servers if total network is down on server/node persistent IP will be down. also persistent IP only jump from one interface to another interface within the node. It will not be jump from one node to another node.

**Note:** Each node should have atleast one persistent IP.

### **Service IP:**

It is one of the important IP address in HACMP Configuration, The main use of service IP, clients/customers will access the applications via service IP address only. This is high availability IP address in HACMP. It is alias IP address. it will be configure top of bootip. service IP will jump to

from one interface to another interface also its having capability of to jump from one node to another node. Service IP always bound on the multiple nodes.

**Note:**

- 1) Each Resource group should have one service IP.
- 2) If total HACMP nodes or Total Bootips are down on Cluster/HACMP Service IP will be down.

**HACMP Key Points:**

- 1) To provide continuous applications service to clients we need redundant components (Servers, Networks, Switches, Routers, NICs, HBAs etc)
- 2) All HACMP Nodes should have same Hardware/Software configuration
- 3) On Each cluster setup Maximum we can configure 32 nodes.
- 4) On Each cluster setup Maximum we can create 64 Resource Groups
- 5) On Each cluster setup Maximum we can configure 256 Networks.

**Downtime:**

The downtime is the time frame when an application is not available to serve its clients. We can classify the downtime as:

**Planned Down Time:**

- Hardware upgrades
- Repairs
- Software updates/upgrades
- Testing (periodic testing is required for cluster validation.)
- Development

**Unplanned Downtime:**

- Administrator errors
- Application failures
- Hardware failures
- Environmental disasters

**For a high availability solution you need:**

Redundant servers  
Redundant networks  
Redundant network adapters  
Monitoring  
Failure detection  
Failure diagnosis  
Automated fallover

Automated reintegration

### **Cluster**

Loosely-coupled collection of independent systems (nodes) or LPARs organized into a network for the purpose of sharing resources and communicating with each other.

### **Cluster topology:**

Contains basic cluster components nodes, networks, communication interfaces, communication devices, and communication adapters.

### **Resource:**

Resource is logical component it will be configured in Resource Group.

In HACMP resources are:

- 1) Volume Groups
- 2) File systems
- 3) Service IPs
- 4) Application Servers
- 5) RAW Disks

Resources will be move from one node to another node along with Resource Group

### **Resource Group (RG):**

Resource group is logical component its contains the resources (VGs, FSs, Service IP and application Server). RG will move from one to another node when active node failed. also we can move RG manually from one node to another node when required

### **Fallover:**

To move the Resource Group from Active node to Passive node is called Fallover. Fallover will be two types,

#### **1) Automatic Fallover:**

When Active Node Total network down or If Active Node is down RG will automatically fallover into the next available node by HACMP

#### **2) Manually Fallover:**

If we scheduled any activity (TL Upgrade, Reboot etc..) on HACMP Active node. we can move the Resource group from Active node to Passive Node manually by using below command.

```
#clRGmove -g <RGName> -n <passivenode> -m
```

(or)

Represents the movement of a resource group from one active node to another node (backup node) in response to a failure on that active node.

**Fallback:**

If RG is moving from Secondary node to Primary Node is called Fallback.

(or)

Represents the movement of a resource group back from the backup node to the previous node, when it becomes available. This movement is typically in response to the reintegration of the previously failed node.

**Split Brain:**

Generally, if we don't have the NON-IP networks in HACMP setup split brain problems will come.

Suppose if we are having only IP-Networks on HACMP setup, when total IP-Networks are down, communication will stop between both nodes then both nodes will try to activate the VGs. This situation VG's data will be corrupted/lose this concept called split brain. So to eliminate the split brain problems we have to configure the NON-IP networks on HACMP Setup

**Communication Devices/NON-IP Networks:**

Communication Devices also we can call NON-IP networks, its alternate network to communicate hacmp nodes with each other. we can eliminate the split brain problems by configuring communication devices.

NON-IP Networks: diskhb, rs232, tmscsi etc...

**HACMP C-SPOC Operations:**

C-SPOC stands for Cluster Single Point Of Control

By using C-SPOC we can implement the few operations on all cluster nodes at a time.

**C-SPOC Operations are:**

- > Manage HACMP Services
- > HACMP Communication Interfaces Management
- > HACMP Resources Group and Application Management (VVIMP)
- > HACMP Log viewing
- > HACMP Security and Users Management
- > HACMP Logical Volume Management (VVIMP)
- > HACMP Concurrent Logical Volume Management

-> Configure GPFS

## **To Enter C-SPOC**

```
#smitty hacmp
System Management (C-SPOC)
(or)
#smitty cl_admin (or) #smitty cspoc
```

## **Manage HACMP Services:**

By using this option we can start/stop/show the hacmp services at a time on all node in cluster setup

```
#smitty cl_start -> To start the cluster services
```

**Note:** While starting cluster services applications will be start automatically by using application server start script.

```
#smitty cl_stop -> To Stop the cluster services
```

**Note:** While stoping cluster services applications will stop automatically by using application server stop script.

## **HACMP services we can stop 3 ways:**

- 1) Gracefull
- 2) Gracefull with Takeover
- 3) Forced

### **1) Gracefull:**

While stopping cluster services if we select Gracefull option, Cluster services will be stop & RG will be release from the current node but RG will not come online on another node if we need RG online on another node we have to bring RG online manually by using below command.

```
#clRGmove -g <RG_Name> -n <Nodename> -u -> To bring resource group on required node.
```

### **2) Gracefull with Takeover**

While stopping cluster services if we select Graceful with Takeover option, Cluster Services will stop & RG will release from current node then RG will come online on next available node automatically.

### **3) Forced:**

While stopping cluster services if we select Forced option, Cluster services will stop & RG will not release from current node

**Note:** in Forced option if we tried bring RG online manually on another node it will not work.

### **HACMP Logical Volume Management:**

In HACMP we have 3 types of VGs Such as:

- 1) NON-Concurrent VGs
- 2) Concurrent VGs
- 3) Enhanced Concurrent VGs

**1) NON-Concurrent VGs:** This type of VGs presented on all cluster nodes but VG & File systems will be active on active node. On stand by node VGs & File systems will offline/varyoff state

**Note:** When Active node fails, on active node file systems will be umount & VG will be varyoff then on Standby node VG will be active & file systems will mounted by using HACMP.

**2) Concurrent VG:** This type of VG presented on all cluster nodes, VG will be active on all cluster nodes but file systems will be active on only active node and on standby node file systems will umount state.

**Note:** When Active node fails, on active node file systems will be umount & VG will be varyoff then on Standby node only file systems will mounted by using HACMP because VG already active.

**3) Enhanced Concurrent VG:** This type of VGs presented on all cluster nodes, on active node VGs & File systems active & standby also VGs & File systems will active but on standby node the VG having only read permissions (LVM listing command only work)

**Note:** When Active node fails, on active node file systems will be umount & VG will be varyoff then on Standby node just HACMP will enable the write permissions.

This concept also we can call "fast disk takeover"  
To implement ECVGs we need diskhb configuration.

### **Active & Passive Cluster Setup:**

If we are having only one RG configuration in cluster setup. The RG is active on primary node & secondary node RG will be offline. This setup we can call active & passive cluster setup.

### **Active/Active or Passive/Passive Cluster Setup:**

Suppose if we are having morethan one RG or Two RGs in cluster setup one RG will be online on one node & another RG will be online on another node this concept we can call Active/Active & Passive/Passive cluster setup.

**Note:** In office environment we will be having both kind of setups

50 servers one cluster servers,

### **HACMP LVM:**

#### **To create VG on 7.1 Version:**

- 1) Get the new luns from the storage
- 2) Once received disks from the storage we have to scan, verify lunids, size and create PVID on both nodes.
- 3) Create VG by using below steps

```
#smitty hacmp
```

```
-> System Management (c-spoc)
```

```
-> Storage
```

```
-> Volume Groups
```

```
-> Create Volume Groups
```

```
-> Select the both cluster nodes by using F7
```

```
-> Select the New Disks to create VG using F7
```

```
-> Select the type of VG
```

```
Final Window
```

```
Type or select values in entry fields.
```

```
Press Enter AFTER making all desired changes.
```

```
[TOP] [Entry Fields]
```

```
Node Names tlsrv229per,tlsrv228per
```

```
Resource Group Name [ICICI_RG] +
```

```
PVID <00c3cfc74f076e21 00c3cfc>
```

```
VOLUME GROUP name [sapvg]
```

```
Physical partition SIZE in megabytes 4 + Volume group MAJOR
NUMBER [50]#
```

```
Enable Fast Disk Takeover or Concurrent Access Fast Disk Takeover
+
```

```
Volume Group Type Scalable
```

```
CRITICAL volume group? no +
```

```
Max PPs per VG in units of 1024 32 +
```

```
Max Logical Volumes 256 +
```

|                            |    |   |
|----------------------------|----|---|
| Enable Strict Mirror Pools | No | + |
| Mirror Pool name           | [] |   |

## Major Number:

Major number should be same on both nodes for the VGs  
 #lvlstmajor -> To display the major number

## To add the new disks to existing VGs:

1) Once received disks from the storage, scan, verify lun ids, size, reserve\_policy and create PVID on both nodes.

Then follow the below steps

```
#smitty hacmp
-> System Management (c-spoc)
-> Storage
-> Volume Groups
-> Set Characteristics of a Volume Group
-> Add a volume to volume group
-> Next it will display the VG with respected RG name & cluster nodes,
select the correct VG.
-> Next it will display the free PVs select them
-> enter
```

Final Window:

|                     |                          |              |
|---------------------|--------------------------|--------------|
| VOLUME GROUP name   | bprsapdataX_vg           |              |
| Resource Group Name | bpr_rg1                  |              |
| Node List           | tlsrv228per, tlsrv229per | VOLUME names |
| hdisk68 hdisk69     |                          |              |
| Physical Volume IDs | 00c3cfc74f0c7605         |              |

## To create the LV:

```
#smitty hacmp
-> System Management (c-spoc)
-> Storage
-> Logical Volume
-> Add a Logical Volume
-> Select the VG to create LV
-> Select the PV
```

Final Window

```
*Logical Partitions [10]
Logical volume Name [saplv]
Type of Logical volume [jfs2]
```

## Create the File systems:

```
#smitty hacmp
-> System Management (c-spoc)
-> Storage
-> File systems
-> Add a File system
-> Select the VG
-> Select the type of file system
Enhanced Journal File system
Standard Journal file system
```

Final Window:

|                           | [Entry Fields]           |   |   |
|---------------------------|--------------------------|---|---|
| Resource Group            | bpr_rg1                  |   |   |
| * Node Names              | tlsrv228per, tlsrv229per |   |   |
| Volume group name         | bprsapdataX_vg           |   |   |
| SIZE of file system       |                          |   |   |
| Unit Size                 | Gigabytes                | + |   |
| * Number of units         | [10]                     | # |   |
| * MOUNT POINT             | [/spafs]                 | / |   |
| PERMISSIONS               | read/write               | + |   |
| Mount OPTIONS             | []                       | + |   |
| Block Size (bytes)        | 4096                     | + |   |
| Inline Log?               | no                       | + |   |
| Inline Log size (MBytes)  | []                       | # |   |
| Logical Volume for Log    |                          | + |   |
| Extended Attribute Format | Version 1                |   | + |
| ENABLE Quota Management?  | no                       |   | + |
| Enable EFS?               | no                       | + |   |

### To Increase/Decrease File system on HACMP:

- 1) Get the new luns from storage, if we don't have free space in VG
- 2) Add the disks to new luns to VG
- 3) Take the #df -g output
- 4) Increase file system by using below steps

```
#smitty hacmp
-> System Management (c-spoc)
-> Storage
-> File systems
-> Change / Show Characteristics of a File System
-> Select the respected file system
```

Final Window:

## Change/Show Characteristics of a Enhanced Journal File System

Type or select values in entry fields.

Press Enter AFTER making all desired changes.

| [TOP]                                  | [Entry Fields]          |
|----------------------------------------|-------------------------|
| Volume group name                      | bprsapdataX_vg          |
| Resource Group Name                    | bpr_rg1                 |
| * Node Names                           | tlsrv228per,tlsrv229per |
| * File system name                     | /db2/BPR/sapdata1       |
| NEW mount point                        | [/db2/BPR/sapdata1] /   |
| SIZE of file system                    |                         |
| Unit Size                              | Gigabytes +             |
| Number of Units                        | [600] #                 |
| Mount GROUP                            | []                      |
| Mount AUTOMATICALLY at system restart? | no +                    |
| PERMISSIONS                            | read/write +            |
| Mount OPTIONS                          | [] +                    |
| Start Disk Accounting?                 | no +                    |
| Block Size (bytes)                     | 4096                    |
| Inline Log?                            | yes                     |
| Inline Log size (MBytes)               | [675] #                 |

[MORE...5]

### To Convert into GBs to 512 byte blocks:

Ex: /sapfs

current size: 10G

Need to add: 5G

$10 \times 1024 = 10240$  - MBs

$10240 \times 1024 = 10485760$  - KBs

$10485760 \times 1024 = 10737418240$  Bytes Value

$10737418240 / 512 = 20971520$  512 byte blocks value

$5 \times 1024 = 5120$

$5120 \times 1024 = 5242880$

$5242880 \times 1024 = 5368709120$

$5368709120 / 512 = 10485760$

20971520 + 10485760 = 31457280

### **To convert from 512 byte blocks to GBs:**

Ex: 31457280 512 bytes value

31457280\*512 = 16106127360 Bytes value

16106127360/1024 = 15728640 KBs value

15728640/1024 = 15360 MBs value

15360/1024 = 15 GB

**Note:** 2 512 byte blocks equal to one KB

### **Cluster health checking commands:**

tlsrv022per(VPR-PRD):/

# lssrc -ls clstrmgrES -> To check the cluster status

Current state: ST\_STABLE

#/usr/es/sbin/cluster/utilities/clRGinfo or clfindres => To check the RG status

#/usr/es/sbin/cluster/utilities/clRGinfo -m -> To display RG status with monitoring scripts.

### **HACMP OPERATIONS:**

#### **HACMP Log Files**

/tmp/hacmp.out

tail -f /tmp/hacmp.out

/tmp/cspoc.log

/var/adm/cluster/hacmp.out

/usr/es/adm/cluster.log

/var/hacmp/clverify/clverify.log

/usr/es/sbin/cluster/history/cluster.mmddyyyy

#### **Cluster Health checks:**

1) lssrc -ls clstrmgrES -> To verify cluster state

2) #/usr/es/sbin/cluster/clstat or cldump -> To see cluster topology statistics

3) /usr/es/sbin/cluster/utilities/clRGinfo -m or clfindres

4) lssrc -ls topsvc -> To verify the missing heart beats and Dead Man Switch (DMS) status (Enabled/Disabled).

5) lssrc -s clcomdES

### **Cluster configuration C-SPOC activities (LVM Management)**

=====

Manage HACMP Services

HACMP Communication Interfaces Management

HACMP Resources Group and Application Management (VVIMP)

HACMP Log viewing

HACMP Security and Users Management  
HACMP Logical Volume Management (VVIMP)  
HACMP Concurrent Logical Volume Management  
Configure GPFS

## **Cluster Health checks, How to test IP and Non-IP networks, Cluster Daemons**

=====

clcomdES  
clinfoES  
clstrmgrES  
ctrmc  
topsvcs  
gslvmd  
Cluster log files  
DARE Operations  
RG Policies  
Service IP  
IPAT via Aliasing  
IPAT via Replacement  
Lazy update

## **TICKETS**

Tickets will be two types

- 1) Incidents
- 2) CSR (Customer Service Request)

Tickets will be generating into ticketing tool,

**Ticketing Tools:** Service Now, HPSM and BMC Remedy tool

**Monitoring Tools:** IBM Tivoli Manager (ITM), Control Minder (CM).

**Ticketing Tool:** In ticketing tool we can manage the Incident, CSR and Changes etc.

Ex: Service Now

(Incidents, RITM/Task, Changes)

HPSM ( Incidents, CSR, Changes)

BMC Remedy (Incidents, SR, Changes)

### **Incidents:**

Incidents will be two types

1) Automatic incidents

2) User created/Manually incidents.

### **Automatic incidents:**

Automatic incidents will be generate by the monitoring tool (ITM) in the ticketing tool (SNOW). if anything wrong/errors/problems occurred on the server the monitoring tool will create incident into SNOW

**sample:** INC012345, INC012346 etc

### **Automatic Tickets will be:**

#### **P1:**

Server rebooted

Servers down, Server hung state

#### **P2:**

Applications down, not running, hung

Database down, not running, hung

### **P3,P4,P5:**

#### **1)Backup related problems:**

a)Backup server not able to take the backup

b)No communication between backup and client server,

c) Backup services may be not running properly

#### **2) File systems related tickets:**

file system is full, file system unmounted,server not pinging,Cpu usage, Memory usage and network related tickets,Hardware failures, call home alerts, LED alerts.

### **Manually/User created Tickets:**

if users are facing some problems on servers the users will be open Incidents.

#### **P1, P2:**

if clients/users need reboot the server immediately

Application/DB not responding to check the system performance. network login issues

### **p3,p4,p5:**

Password reset, unlock user account, network login issues, backup failed etc.

**CSR/TASK/SR:**

Create/Increase/Decrease/remove VGs, FSs, LVs  
Create/remove users,  
Enable/Disable sudo Access for the users,  
Install softwares  
Add/Remove Memory, CPU  
Update the configuration files /etc/hosts, /etc/resolv.conf,  
/etc/mail/sendmail.cf etc  
Copy the files from one server to another server  
Enable/disable the permissions/ownership etc

**Note:** For Task/CSR/SR implementation we have to create the Change and implement the plan, except user, sudo related activities.

**Type of Change activities:**

Create/Increase/Decrease/remove VGs, FSs, LVs  
Install softwares/OS/TL Levels/SP etc  
Add/Remove Memory, CPU  
Update the configuration files /etc/hosts, /etc/resolv.conf,  
/etc/mail/sendmail.cf etc  
Faulty disk/adapters replacements, sysplanner replacements  
Power supply cables replacements/ VRM Replacement, DIMM slots  
(memory) replacements etc...

**Changes:**

Actually we have 3 types changes in Service Now,

**SNOW:** OSA Change, Comprehensive Change, E-Change

**HPSM:** NoN-Standard, Standard, E-Change

**OSA change/Non Standard:** this type of change will create for Test/Dev servers only. also if any online activity on production/DR servers we can create anyday anytime.

**Comprehensive/Standard Change:** This type of change only we will create production/DR servers mostly. this change always we will schedule on weekends (Saturday, Sunday).

**E-Change:**

If we have to perform any offline activities on production/DR Servers on weekdays so then we will submit Emergency Changes (E-Change). This change also we can call contingency change.

## **P5,P6 AND P7 SERVERS**

Difference Between P5, P6, P7

### **P5 features:**

Virtual Adapters introduced (VEA, VSA etc)

DLPAR operations support

SMT introduced

It will support only one Shared Processor Pool

VSCSI Disk Mapping

VIO Features

Dual VIO Support

255 LPARs we can create depends on resource availability.

### **P6 Servers:**

LPM Support,

WPAR Support,

64 Shared processor pools we can configure,

Memory pools configuration,

HEA adapter (LHEAs) support,

NPIV Support

### **Power 7 Servers:**

AME (Active Memory Sharing)

NPIV Support

We can configure 32 bit kernel and 64 bit kernel.

While installing OS we can select JFS/JFS2 file system in rootvg.

## OTHER TOPICS

### **Sudo:**

By using sudo we can give the specific or all commands access to the specific users,

### **sudo install:**

sudo we can install by using installp (if bff software) or we can install by using rpm command (if it is rpm packages).

To provies sudo access for users.

```
1) take backup sudoers file
cp /etc/sudoers /etc/sudoers_date
2) visudo or vi /etc/sudors
ravi ALL=NOPASSWD /usr/sbin/mksuer, /user/bin/lspv
meerza ALL=PASSWD /usr/sbin/sar, /user/sbin/netstat,
venkat ALL=ALL ALL -> to give full access as root.
```

### **SAN REQUEST:**

\*Please increase Production TL sapdata fs space on priority\*. Increase by 125 gb each fs. Details to follow:

```
server: tlsrv228per / tlsrv229per
Database: Prod BPR
filesystems: /db2/BPR/sapdata(1 thru 10)
fs current size (each fs): 550 gb
increase by (each fs): 125 gb
fs new size (each fs): 675 gb
total space requested (all fs): 1,250
copy services required?: yes
```

Hi Team,

As per SCTASK1312080 request please provide the 5 disks on tlsrv228per / tlsrv229per servers to increase the /db2/BPR/sapdata1 to 10 file systems, each disk size should be 250GB.

Kindly find server & WWN details below:

Servers: tlsrv228per / tlsrv229per

FCS Adapters/WWNs:

```
tlsrv228per(BPR-PRD):/# lscfg -vpl fcs0 |grep -i network
 Network Address.....10000000C9CFDD14
tlsrv228per(BPR-PRD):/# lscfg -vpl fcs2 |grep -i network
 Network Address.....10000000C9CFE932
tlsrv228per(BPR-PRD):/#
tlsrv229per(BPR-PRD):/# lscfg -vpl fcs0 |grep -i network
 Network Address.....10000090FA0A15E8
tlsrv229per(BPR-PRD):/# lscfg -vpl fcs2 |grep -i network
 Network Address.....10000090FA0A45A2
```

## **PROCESS CLASSES:**

### **PROCESS CLASSES/REAL TIME OFFICE ENVIRONMENT:**

**Tell me about your self:**

#### **Company/client Environment:**

In my environment we are having P5, P6, P7 servers also,  
Total servers we are managing.

100 (or)  
200 (or)  
300 (or)  
400 (or)  
500 (or)  
1000 (or)

Ex: 400 servers (LPARs) we are managing.  
Physical servers/Frames : 40 or 50 servers

**OSs:** Aix, Linux, Solaris & HP-UX

**Note:** if we dont know any OS don't tell in interview say only we are having  
Aix Servers.

#### **Unix Team Persons:**

Total: 10 People

SDM -> Manager -> Team Lead -> Colleagues (L3, L2 and L1) and onsite.

**your Level:** my current level is L2 and also I can perform L3 level activities.

## **Total Teams in Office environment:**

- 1) Unix Team
- 2) Storage Team
- 3) Backup Team
- 4) Network Team
- 5) Command Center / Operations Team/ Monitoring Team
- 6) Application Team
- 7) Database Team

**1) Unix Team:** Aix, Linux, Solaris & HP-UX

**2) Storage Team:** To allocate/Deallocate disks LUNs to the servers, verify the Fiber cable connections etc....

**3) Backup Team:** TSM, Legato, Netbackup

To take back up requested file systems from the Unix servers & Restore the file systems data into Unix servers incase lose.

**4) Network Team:**

This team will work on network related tasks/problems. if there is no network problem on Unix servers but still not able to login into the servers by using putty then we can contact the Network team to fix the issue.

**5) Command Center/Monitoring Team:**

This team will monitor the tickets, if any high priority tickets (P1, P2) are coming these guys will push us to fix the issue immediatly, also they will open Bridge/Conference call will all teams if required.

**Bridge calls:** B1, B2, B3, B4

**6) Application Team:**

**Applications:** Apache/tomcat, sas, pega, java, oracle etc..

We can request apps team stop/start when rebooting/shutdown the application servers.

**7) Database Team:**

We can request database team stop/start when rebooting/shutdown the database servers.

**Databases:** DB2, Oracle

## **SERVERS ENVIRONMENTS:**

- 1) Production Servers
- 2) DR (Disaster Recovery) Servers
- 3) Test Servers
- 4) Development Servers
- 5) Build/Staging/Pre-Production Servers

## **Aix Teams.**

- 1) Build/Development Team
- 2) Supporting Team
- 3) Monitoring Team.

## **Data Centers:**

**USA:** wilson, charlotte, newyork, orlando, kings mountain

**AUS:** sidney, melborne etc.,  
Canada, Dubai.

**Domestic:** Chennai, HYD, Mumbai, Pune, Delhi etc.

## **Daily Activities/Day to Day Activities:**

- 1) I will login into the system.
  - 2) I will read my mails, if any important mails I will reply to them (outlook)
  - 3) I will check my queue tickets, if I dont have any tickets I will look on the teammates tickets
  - 4) Also if any changes are scheduled I will implement them
  - 5) If any follow ups received from the previous shift persons we will work on them.
- these are my daily activities

## **Shift Timings:**

We are having three types of shifts

- 1) Morning Shift (6:30 AM to 4:00 PM)
- 2) After noon Shift (2:30 PM to 12:00 AM)
- 3) Night Shift (10:30 PM to 8:00 AM)